

Middle East Technical University

Northern Cyprus Campus

EEE 202 Design Project Report

Project 6: Water Level Indicator

Conductive-Probe-Based Discrete Analog Water Level Indicator Circuit

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Abstract

This report presents the design and LTspice simulation of a conductive-probe-based discrete analog water level indicator circuit. The implemented design, referred to as Version 3, operates from a $12\,\mathrm{V}$ DC supply and uses six conductive sensing levels, BC547B BJT switching stages, LED indicators, base-node stabilization networks, reverse-polarity protection, and an astable multivibrator alarm stage.

The first five water levels are indicated by steady LEDs, while the sixth and highest level activates a blinking LED alarm. In the LTspice model, water contact is represented by voltage-controlled switches and resistive water paths. These switches are activated sequentially to imitate a rising water level. The simulation results verify that the first five LEDs turn on in order and remain steady, while the final LED blinks after the highest level is reached. The simulated supply current and power consumption also remain within the expected low-current and low-power operating range.

The report also discusses component selection, datasheet-based verification, and practical limitations of the design. A MOSFET-based Version 4 is proposed as an improvement direction for reducing sensing current through the water path, and an AC-based sensing method is identified as a possible future improvement for reducing electrolysis and probe corrosion. Overall, the implemented Version 3 circuit satisfies the intended discrete analog water-level indication behavior in simulation, while physical testing is required to evaluate real water conductivity, probe contact stability, component tolerance, and long-term corrosion effects.

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1 Objective

The objective of this design project is to design, simulate, and physically implement a discrete analog water level indicator circuit using conductive probes and LED indicators. The circuit is intended to detect different water levels inside a container and provide a clear visual indication of the reached level.

The implemented design aims to satisfy the given project requirements while remaining simple, buildable, and suitable for laboratory implementation. In addition to the basic level indication function, the design also includes a blinking indication for the highest water level by using an astable multivibrator stage.

Another objective of the project is to evaluate the design from an engineering perspective rather than only demonstrating that the circuit works. For this reason, the report discusses the design evolution from the initial BJT-based circuit to the implemented Version 3 circuit, and also considers a MOSFET-based Version 4 as a possible improvement for reducing sensing current and probe corrosion risk.

Finally, the project aims to compare the simulated behavior of the circuit with its expected physical implementation, while identifying practical limitations such as water conductivity variation, component tolerances, probe corrosion, and the effects of using DC conductive sensing.

2 Problem Statement and Design Requirements

The design problem requires a circuit that can detect the water level inside a container and indicate the reached level using LEDs. The circuit must operate as a water level indicator, where different water heights correspond to different visual output states. Instead of using a mechanical float, ultrasonic sensor, microcontroller, or integrated sensing module, the sensing mechanism is based on conductive probes placed at different heights inside the container.

The given design requirements can be summarized as follows:

Design Parameter	**Requirement**
Supply voltage	$5\,\text{V} \leq V_S \leq 12\,\text{V}$
Minimum number of detected levels	At least 3 levels
Output indicator	LEDs
Sensing method	Conductive probes
Circuit type	Discrete analog implementation

Table 2.1: Given design requirements for the water level indicator circuit.

In this project, the implemented laboratory design uses a $12\,\text{V}$ DC supply, which is within the allowed supply voltage range. This value was selected because the circuit will be tested using a laboratory DC power supply, making $12\,\text{V}$ a stable and practical choice. The design also exceeds the minimum level requirement by using six sensing levels instead of only three.

The first five levels are used as normal visual level indicators, where each reached probe turns on its corresponding LED. The sixth level is treated as the highest or critical level. Instead of only turning on a steady LED, this level activates an astable multivibrator stage, causing the final LED to blink. This makes the highest level more noticeable and allows it to act as a simple alarm indication.

Therefore, the design problem is not only to make LEDs turn on when water reaches different probes, but also to create a practical, stable, and buildable discrete analog circuit. The circuit should detect multiple water levels, avoid floating transistor inputs, reduce false triggering caused by water movement, protect the supply input against reverse polarity, and provide a distinct warning behavior at the highest water level.

3 Design Summary

The implemented design is a six-level conductive-probe-based discrete analog water level indicator circuit. Although the design problem requires at least three detected levels, six levels were selected to provide a more detailed visual indication of the water height. The circuit is designed to operate from a $12\,\text{V}$ DC laboratory power supply, which is within the allowed $5\,\text{V}$ to $12\,\text{V}$ supply range.

The laboratory implementation is based on Version 3 of the design. In this version, the first five water levels are displayed by steady LEDs, while the sixth and highest level is used as a critical-level alarm. When the water reaches one of the first five probes, the corresponding BJT switching stage drives its LED. When the water reaches the sixth probe, an astable multivibrator stage is enabled and the final LED blinks instead of remaining steadily on.

The implemented Version 3 circuit contains five main functional roles: conductive probes, BJT-based switching stages, steady LED indicators, an astable multivibrator alarm stage, and practical protection/filtering elements. The practical improvements include base pull-down resistors to prevent floating inputs, debounce capacitors to reduce unstable switching caused by water movement, and a series 1N5819 Schottky diode for reverse-polarity supply protection. The detailed operation of these blocks is discussed later in the operation section, while their numerical design calculations are given in the component selection and calculation section.

Figure 3.1 shows the simplified functional structure of the implemented Version 3 design. The diagram is not intended to show every circuit connection; instead, it summarizes the main signal flow of the design. The water level is detected by conductive probes, the first five probe signals drive BJT switching stages and steady LEDs, and the sixth probe enables the astable multivibrator that drives the blinking alarm LED.

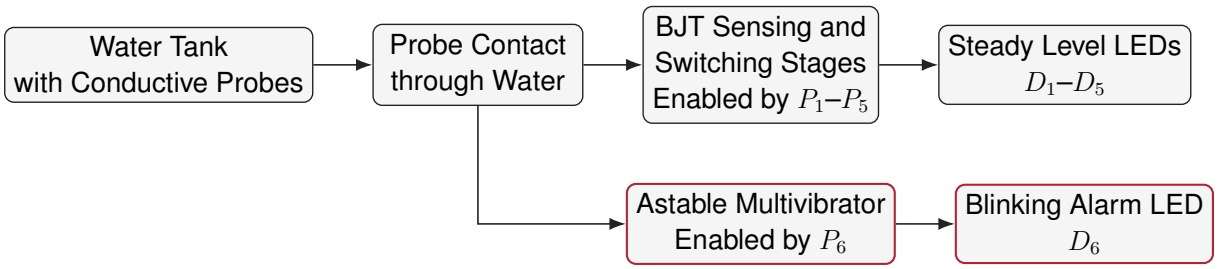


Figure 3.1: Simplified functional overview of the implemented Version 3 water level indicator design.

Version 3 is the physically implemented laboratory design in this report. Later sections also discuss a MOSFET-based Version 4 and a possible AC-sensing Version 5 as improvements, but these alternatives are not the main implemented circuit. Therefore, the main design, simulation, and physical testing focus remains on the Version 3 BJT-based circuit.

4 Design Approach

The design approach was based on treating the water level indicator as a modular discrete analog system. Instead of designing the circuit as one large undivided structure, the problem was separated into smaller functional tasks: water detection, signal conversion, LED switching, input stabilization, and highest-level alarm indication.

This modular approach was chosen because a water level indicator naturally consists of repeated level-detection channels. Once one sensing and switching channel is designed, the same basic structure can be repeated for additional water levels. Therefore, the circuit can be expanded or reduced depending on the required number of detected levels. In this project, six levels were used even though the minimum requirement was three levels, because six levels provide a clearer visual representation of the water height.

The design was kept discrete and analog. No microcontroller, comparator IC, digital sensor module, or ready-made liquid-level sensor was used. This decision keeps the design close to the educational purpose of the project: showing how probes, resistors, capacitors, diodes, BJTs, LEDs, and an astable multivibrator can be combined to form a complete sensing and indication system.

A key design decision was to separate normal level indication from critical-level indication. The lower water levels only need to show the approximate height of the water, so steady LEDs are sufficient for those levels. The highest water level, however, represents a more critical condition. For this reason, the sixth level was assigned to an alarm function instead of being treated as another ordinary steady indicator, leading to a fully analog blinking stage.

Practical buildability was also considered as part of the design approach. A circuit that works only under ideal simulation conditions is not sufficient for laboratory implementation. Real water contact can be unstable, transistor input nodes can float, supply connections can be reversed accidentally, and component values can vary because of tolerance. Therefore, the design included practical stability and protection features from the beginning.

The LTspice simulation was planned as a verification tool rather than as a perfect physical model of water. The simulation does not model the full chemical and physical behavior of water, probe corrosion, or electrolysis. Instead, water contact is represented electrically as a controlled conductive path, which is sufficient for verifying sequential LED activation, highest-level alarm triggering, and blinking operation.

Overall, the design approach follows three main principles: modularity, analog discreteness, and practical buildability. Modularity makes the design scalable, analog discreteness keeps the circuit based on fundamental components, and practical buildability ensures that the circuit can be assembled and tested with available laboratory components.

5 Design Flow and Design Evolution

The implemented circuit was not designed in a single step. Instead, it was developed through several versions. Each version either added a new function or solved a practical limitation observed in the previous version. This design flow is important because the project is not only a circuit construction task, but also a design problem. Therefore, the evolution of the circuit shows how the final implemented Version 3 design was reached.

The version sequence can be summarized as follows. Version 1 established the basic conductive-probe sensing and BJT switching principle. Version 2 added a sixth level and a blinking alarm stage. Version 3 added practical input filtering and reverse-polarity protection, and this version was selected as the implemented laboratory circuit. Version 4 was developed as a proposed MOSFET-based improvement, while Version 5 is only considered as a future AC-sensing direction.

5.1 Version 1: Basic Five-Level BJT Indicator

The first version was a basic five-level water level indicator circuit. It used five conductive level probes and five BJT-based LED switching stages. When the water reached a probe, a conductive path was formed between the common supply probe and that level probe. This allowed base current to flow into the corresponding BJT, turning the transistor on and lighting the related LED.

The purpose of Version 1 was to verify the basic sensing and switching principle of the design:

$$\text{Water reaches probe} \rightarrow \text{base current flows} \rightarrow \text{BJT turns ON} \rightarrow \text{LED turns ON}.$$

This version satisfied the minimum requirement of detecting more than three water levels. However, it only provided steady LED indication and did not include a special warning behavior for the highest water level. It also did not include input filtering or supply protection.

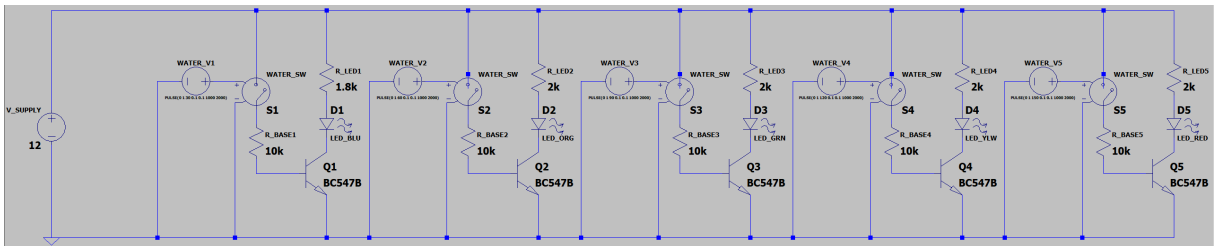


Figure 5.1: Version 1 schematic: basic five-level BJT-based water level indicator.

5.2 Version 2: Sixth Level and Blinking Alarm

Version 2 added a sixth sensing level. This level was assigned to the highest or critical water level. Instead of using the sixth level as another ordinary steady LED indicator, it was connected to an astable multivibrator stage.

When the water reaches P_6, the astable multivibrator is enabled and the final LED blinks. Therefore, the first five levels indicate the water height with steady LEDs, while the sixth level produces a distinct alarm indication.

The main improvement of Version 2 can be summarized as

$$P_6 \text{ reached} \rightarrow \text{astable multivibrator enabled} \rightarrow \text{final LED blinks.}$$

This made the highest water level more noticeable and allowed it to function as a simple overflow warning.

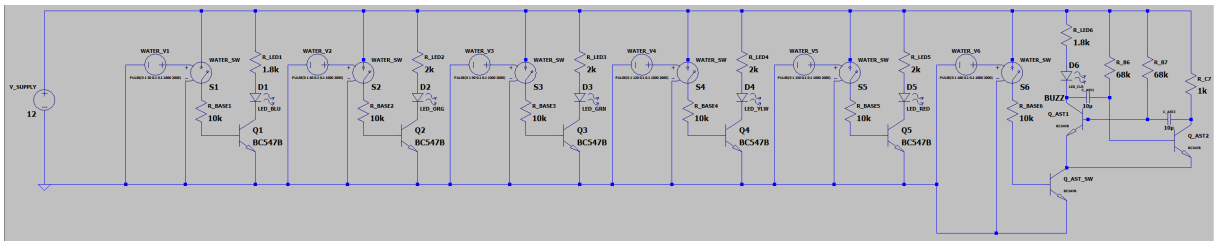


Figure 5.2: Version 2 schematic: six-level BJT-based circuit with an astable multivibrator alarm stage.

5.3 Version 3: Input Stabilization and Supply Protection

Version 3 introduced the practical improvements needed for laboratory implementation. A pull-down resistor and a debounce capacitor were added to each BJT base node. The pull-down resistor prevents the base from floating when the probe is dry, while the debounce capacitor reduces unstable switching caused by small water movements around the probe.

Together, the pull-down resistor and debounce capacitor form an input-stabilization network at each sensing node. This network should not be interpreted as filtering the DC supply voltage itself. The supply is DC and remains approximately constant. Instead, the filtering behavior appears when the water contact near a probe changes rapidly because of surface motion, vibration, or partial wetting. These rapid contact changes produce rapid changes in the base-node voltage, and the capacitor slows those changes.

Therefore, the practical role of the network is

$$\text{rapid water-contact variation} \rightarrow \text{rapid base-voltage variation} \rightarrow \underset{\text{input stabilization}}{R_{\mathrm{PD}}\text{–}C_{\mathrm{DB}}}.$$

In addition, a 1N5819 Schottky diode was added in series with the supply input for reverse-polarity protection. If the power supply is accidentally connected in reverse, the diode blocks current and helps protect the circuit.

Version 3 is the implemented laboratory design in this report. It preserves the discrete analog structure of the earlier versions while adding the stability and protection features needed for a more practical physical circuit.

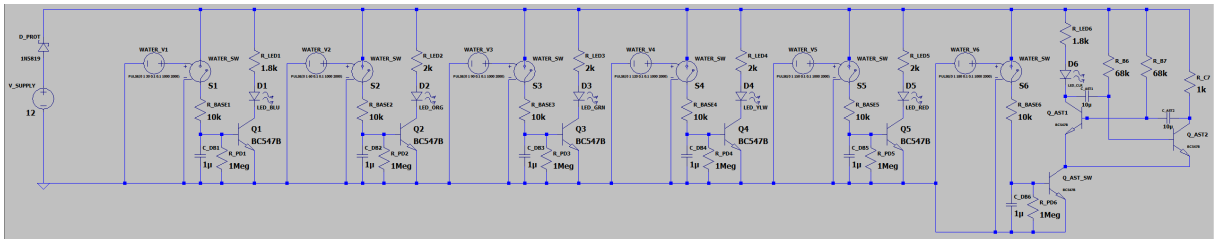


Figure 5.3: Version 3 schematic: implemented BJT-based circuit with low-pass filtering and reverse-polarity protection.

5.4 Version 4: Proposed MOSFET-Based Improvement

Version 4 was developed as a proposed improvement, but it is not physically implemented in the laboratory due to MOSFET component availability. In this version, the BJT switching stages are replaced with BS170 N-channel MOSFETs.

The motivation for this change is that a BJT requires base current to turn on, while a MOSFET is controlled mainly by gate-source voltage. Therefore, a MOSFET-based sensing stage can reduce the current passing through the water path. This is expected to reduce electrolysis and probe corrosion compared with the implemented BJT-based DC sensing design.

Because MOSFET gates are sensitive to excessive gate-source voltage, reverse-connected zener diodes are included for gate protection. The proposed MOSFET version is discussed in more detail later in the report.

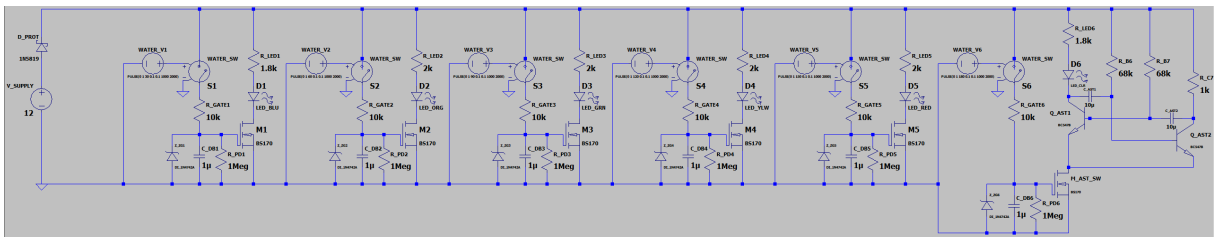


Figure 5.4: Version 4 schematic: proposed MOSFET-based improvement with BS170 switching stages and zener gate protection.

5.5 Possible Version 5: AC-Based Conductive Sensing

A possible Version 5 is considered only as a future improvement. Both the implemented BJT design and the proposed MOSFET design still use DC conductive sensing. Therefore, long-term use may still cause electrolysis and corrosion at the probes.

A more durable version could use AC-based probe excitation instead of DC sensing. This would reduce the net DC current through the water and therefore reduce electrolysis effects. This version is not designed in detail in this report; it is only mentioned as a possible future direction.

Version	Main Feature	Design Role
V1	Five-level BJT indicator	Establishes the basic conductive-probe sensing and LED switching principle.
V2	Sixth level with astable alarm	Adds a blinking warning behavior for the highest water level.
V3	Filtering and supply protection	Becomes the implemented laboratory design by adding practical stability and protection features.
V4	MOSFET-based proposed improvement	Reduces sensing current and corrosion risk, but is not physically implemented due to component availability.
V5	Possible AC-based sensing	Future direction for reducing net DC probe current and improving long-term durability.

Table 5.1: Design evolution of the water level indicator circuit.

This version-based design flow shows why Version 3 was selected as the implemented circuit. Version 1 established the basic sensing idea, Version 2 added the alarm behavior, and Version 3 added the practical features needed for physical implementation. Version 4 and Version 5 are kept as improvement directions rather than as the main implemented design.

6 Implemented Circuit Overview

This section presents the implemented laboratory circuit, namely Version 3 of the design, from a circuit-level modular perspective. Instead of introducing the complete schematic immediately, the implemented circuit is first decomposed into its main circuit modules. This makes the structure easier to read and helps the reader understand how the full circuit is formed from smaller functional subcircuits. After the modules are introduced individually, the complete Version 3 schematic is presented.

The implemented Version 3 design is a six-level discrete analog water level indicator powered from a $12\,\mathrm{V}$ DC supply. It uses conductive probes for sensing, BC547B NPN BJTs for switching, LEDs for visual indication, RC input stabilization at each sensing node, and an astable multivibrator for the highest-level blinking alarm. A 1N5819 Schottky diode is also used at the supply input for reverse-polarity protection.

6.1 Circuit-Level Modules of the Implemented Design

The implemented circuit may be divided into three main module groups:

1. reverse-polarity supply protection,
2. repeated sensing and indication stages for P_1 to P_5,
3. the combined P_6 sensing trigger and astable multivibrator alarm stage.

This grouping avoids repeating the same circuit explanation several times. The first five level-detection channels have the same basic structure, so one representative sensing-and-indication module is shown and interpreted as a repeated stage. The sixth level is different because it does not simply drive a steady LED; instead, it enables the astable alarm stage.

6.1.1 Reverse-Polarity Protection Module

Figure 6.1 shows the reverse-polarity protection stage of the implemented circuit. A 1N5819 Schottky diode is connected in series with the positive supply input. Under correct polarity, the diode conducts and establishes the protected supply rail V_{CC}. Under reverse polarity, the diode blocks current and helps protect the rest of the circuit.

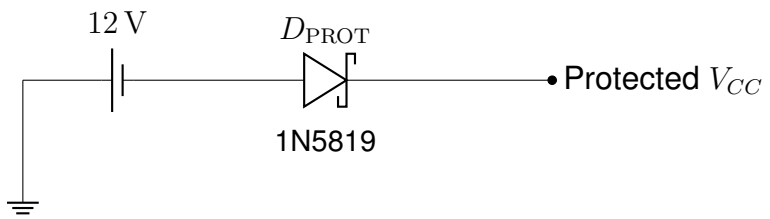


Figure 6.1: Reverse-polarity protection stage using a series 1N5819 Schottky diode.

6.1.2 Representative Sensing and Indication Module for P_1 to P_5

Figure 6.2 shows one representative sensing and indication stage used for the first five water levels. The same basic structure is repeated for P_1 to P_5. The practical differences between the repeated stages are mainly the LED color and the corresponding LED-series resistor value.

In the drawing, R_{water} is shown as a conceptual resistance between the common supply probe and the level probe. It represents the conductive path that is formed when water touches both probes. When this path exists, current can flow through R_{BASE} into the BJT base. The transistor then turns on and completes the LED current path.

The base pull-down resistor R_{PD} keeps the base node from floating when the probe is dry. The capacitor C_{DB} acts as a debounce or low-pass element that reduces unstable triggering caused by small water movements near the probe.

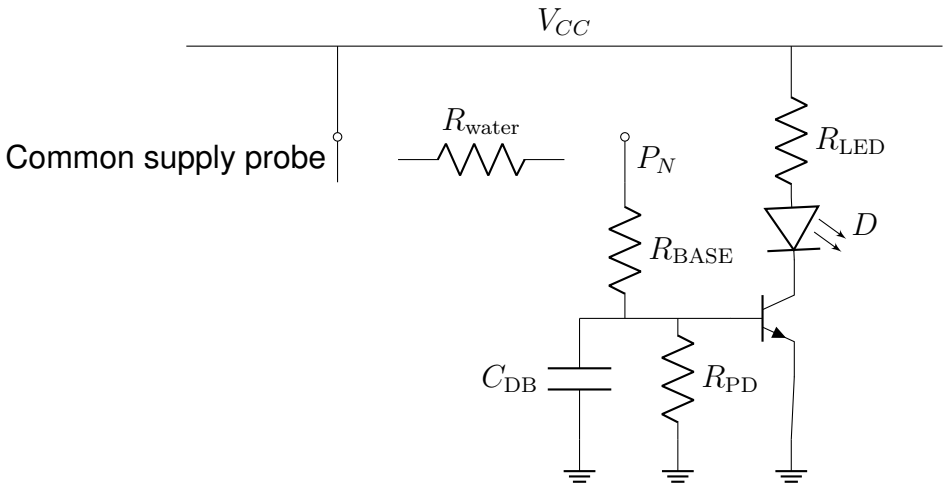


Figure 6.2: Representative sensing and indication stage repeated for P_1 to P_5. The base-related elements share the same electrical node but are drawn along a horizontal base rail for clarity.

6.1.3 Combined P_6 Trigger and Astable Alarm Module

Figure 6.3 shows the combined P_6 sensing trigger and astable multivibrator alarm module. The left side of the module represents the sixth-level conductive sensing stage. This part is similar to the earlier sensing stages, but instead of driving an ordinary steady LED directly, it controls $Q_{\text{AST_SW}}$, which enables the astable alarm circuit.

The right side of the module represents the astable multivibrator. Once the P_6 trigger enables the astable stage, the circuit oscillates and drives the final LED as a blinking warning indicator. This makes the highest water level visually distinct from the lower levels.

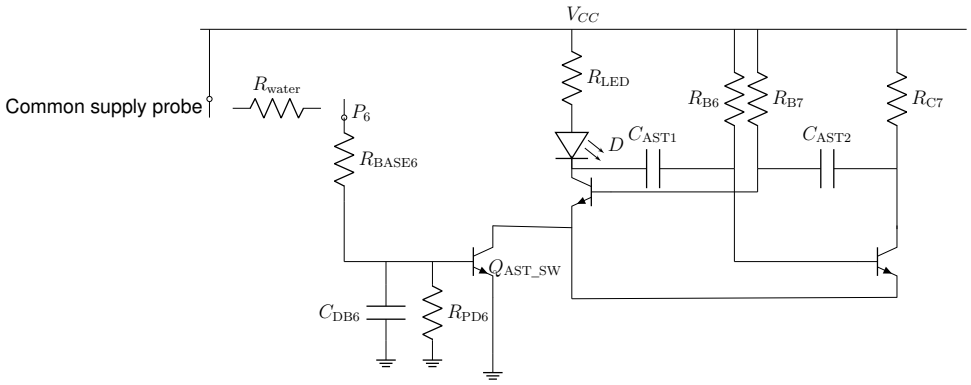


Figure 6.3: Combined P_6 sensing trigger and astable multivibrator alarm module used to drive the blinking highest-level LED.

6.2 Complete Implemented Version 3 Schematic

After introducing the implemented design at the module level, the complete Version 3 circuit is shown in Figure 6.4. This is the full schematic used as the basis for LTspice simulation and physical implementation.

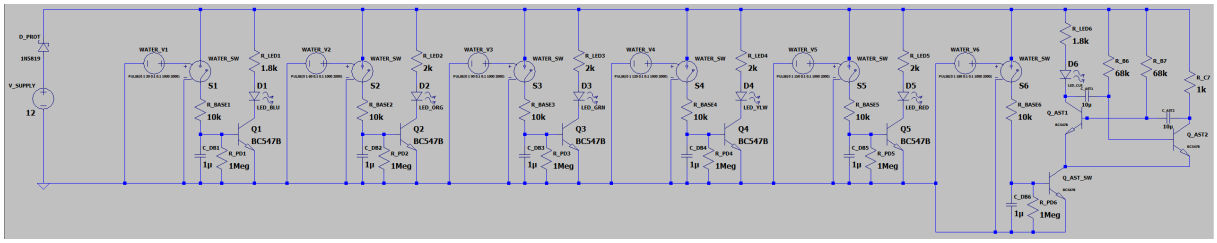


Figure 6.4: Complete schematic of the implemented Version 3 water level indicator circuit.

The complete schematic should be read together with the module drawings above. The module drawings explain the repeated local structures, while the full schematic shows how these structures are interconnected in the complete Version 3 circuit. The left side of the full schematic contains the protected supply input, the central portion contains the repeated BJT sensing and LED indication stages, and the right side contains the combined highest-level trigger and astable alarm circuitry.

7 Operation of the Implemented BJT-Based Circuit

The implemented Version 3 circuit operates by converting water contact into electrical switching action. The circuit uses one common supply probe and six level probes, P_1 to P_6. The common supply probe is connected to the protected positive supply rail, while the level probes are placed at different heights inside the container. As the water level rises, water reaches the probes one by one and creates conductive paths between the common supply probe and the reached level probes.

This section explains the operating sequence of the circuit. The purpose here is not to repeat the numerical design calculations, but to describe how the circuit behaves during operation. The detailed calculations for supply voltage, LED current, BJT base current, RC time constants, astable frequency, and power dissipation are given in the Section 8.

7.1 Initial State: No Water Contact

When the water level is below all level probes, none of the sensing probes P_1 to P_6 is electrically connected to the common supply probe through water. Therefore, no sensing current path is formed.

In this condition, the base nodes of the BJT stages are held close to ground by their pull-down resistors. Since the base-emitter junctions are not forward biased, the BJTs remain off. As a result, the collector-emitter paths of the BJTs are open, and no LED current path is completed.

Therefore, when the water level is below P_1,

$$D_1, D_2, D_3, D_4, D_5, D_6 \quad \text{are all OFF.}$$

7.2 Protected Supply Operation

The circuit receives power through the reverse-polarity protection stage. Under correct supply polarity, the 1N5819 Schottky diode conducts and the protected supply rail V_{CC} is established. The rest of the circuit is then supplied through this protected rail.

If the external supply is accidentally connected in reverse, the Schottky diode becomes reverse biased. In that case, it blocks the current path from the supply to the circuit. The circuit does not operate under reverse polarity, but the protection diode helps prevent reverse current from damaging the components.

Thus, the protection stage has a simple operating role:

$$\text{correct polarity} \rightarrow \text{diode conducts} \rightarrow \text{circuit operates},$$

reverse polarity $\rightarrow$ diode blocks $\rightarrow$ circuit protected.

7.3 Conductive Probe Sensing Principle

The sensing mechanism is based on the conductivity of water. When water touches both the common supply probe and a level probe P_n, it creates a conductive path between them. This path is not an ideal wire. Real water has finite resistance, so the water contact behaves as a resistive path rather than a perfect short circuit.

For a reached probe P_n, the sensing path can be represented conceptually as

$$V_{CC} \rightarrow \text{common supply probe} \rightarrow \text{water path} \rightarrow P_n \rightarrow R_{\mathrm{BASE},n} \rightarrow \text{BJT base}.$$

This path allows current to enter the base of the corresponding BJT. Once sufficient base drive is produced, the transistor turns on and allows current to flow through the LED branch.

When a probe is dry, this water path does not exist. Therefore, the corresponding transistor remains off.

7.4 Operation of the First Five Level Stages

The first five probes, P_1 to P_5, are used for normal water level indication. Each of these probes controls one repeated BJT sensing and indication stage.

When water reaches a probe P_n, where

$$n = 1, 2, 3, 4, 5,$$

the water forms a conductive path from the common supply probe to P_n. This allows current to flow through the base resistor into the base of the corresponding BC547B transistor. The base-emitter junction becomes forward biased, and the transistor turns on.

Once the transistor is on, it completes the LED current path:

$$V_{CC} \rightarrow R_{\mathrm{LED},n} \rightarrow D_n \rightarrow \text{BJT collector} \rightarrow \text{BJT emitter} \rightarrow \text{ground}.$$

Therefore, the corresponding LED turns on.

For example, when water reaches P_1, transistor Q_1 turns on and LED D_1 lights. When water reaches P_2, the second transistor turns on and D_2 lights. The same logic continues for P_3, P_4, and P_5.

As the water level rises, the circuit therefore gives a cumulative visual indication of the water height:

$$P_1 \text{ reached} \rightarrow D_1 \text{ ON},$$

$$P_2 \text{ reached} \rightarrow D_1, D_2 \text{ ON},$$

$$P_3 \text{ reached} \rightarrow D_1, D_2, D_3 \text{ ON},$$

and so on.

7.5 Role of the Pull-Down Resistor and Debounce Capacitor

Each BJT base node includes a pull-down resistor and a debounce capacitor. These elements do not change the basic conductive sensing principle, but they make the input behavior more stable during practical operation.

The pull-down resistor connects the base node to ground. Its purpose is to define the base voltage when the probe is dry. Without this resistor, the base node could float and respond unpredictably to leakage current, electrical noise, or accidental touch. Therefore, when there is no water contact, the pull-down resistor keeps the base close to 0 V.

The debounce capacitor is connected from the base node to ground. Its purpose is not to filter the DC supply voltage. The circuit is supplied by a DC source, and the intended input condition is also a threshold-like water-contact event. The capacitor becomes important because real water contact is not perfectly steady. If the water surface moves near a probe, the probe may rapidly make and lose contact with the water. This produces rapid changes in the base-node voltage.

The capacitor resists sudden voltage changes at the base node. Therefore, fast contact disturbances are smoothed, while slower changes corresponding to an actual change in water level can still activate the transistor. In this practical sense, the pull-down resistor and debounce capacitor produce a low-pass-like input stabilization behavior for water-contact-induced voltage fluctuations.

This behavior can be summarized as

$$\begin{array}{c}\text{fast water-surface}\\ \text{motion near probe}\end{array} \rightarrow \begin{array}{c}\text{rapid probe contact}\\ \text{changes}\end{array} \rightarrow \begin{array}{c}\text{rapid base-voltage}\\ \text{changes}\end{array} \rightarrow \begin{array}{c}\text{smoothed by}\\ R_{\mathrm{PD}} \text{ and } C_{\mathrm{DB}}\end{array}.$$

Thus, the network is better described as a debounce or input-stabilization network

with low-pass-like behavior, rather than as a filter applied to the DC supply itself. The detailed turn-on and turn-off time constant calculations are given in Section 8 and Appendix C.

7.6 Operation of the Sixth Level

The sixth probe, P_6, is treated differently from P_1 to P_5. The first five probes are used for ordinary level indication, but P_6 represents the highest or critical water level. Therefore, P_6 is used to activate an alarm stage rather than simply lighting another steady LED.

When water reaches P_6, the sixth sensing path is completed. This turns on the transistor associated with the P_6 trigger stage. The purpose of this transistor is to enable the astable multivibrator alarm circuit.

Thus, the sixth-level action can be summarized as

$$P_6 \text{ reached} \rightarrow Q_{\mathrm{AST_SW}} \text{ turns ON} \rightarrow \text{astable multivibrator enabled}.$$

Once enabled, the astable multivibrator begins oscillating. The output of this oscillation drives the final LED, D_6, so that it blinks instead of remaining steadily on.

7.7 Astable Multivibrator Alarm Behavior

The astable multivibrator is a two-transistor oscillator. It does not have one stable resting state. Instead, the two transistor stages alternately switch on and off due to the cross-coupled capacitor network.

Operationally, the behavior can be understood as follows. When one transistor conducts, the other transistor is driven toward cutoff. Then the timing capacitor voltage changes until the circuit switches state. After the switch, the previously off transistor turns on and the previously on transistor turns off. This process repeats continuously as long as the astable stage is enabled.

The blinking LED is connected to this oscillating stage. Therefore, when the highest water level is reached, the final LED repeatedly turns on and off. This makes the highest level visually distinct from the lower levels.

The purpose of this behavior is not only to show that water reached P_6, but also to make the highest level appear as an alarm condition:

$$P_6 \text{ reached} \rightarrow \text{oscillation} \rightarrow D_6 \text{ blinking alarm}.$$

The detailed frequency calculation of the astable multivibrator is given in the Section 8.

7.8 Sequential Operation Summary

The complete operation of the circuit can be summarized as a sequence of water-level states. When the water level is below all probes, all LEDs are off. As the water rises and reaches each probe, the corresponding indication stage is activated.

Water Level Condition	**Activated Sensing Path**	**Circuit Response**
Below P_1	No probe contact	All LEDs are off.
Water reaches P_1	P_1 connected through water	D_1 turns on.
Water reaches P_2	P_1 and P_2 connected through water	D_1 and D_2 are on.
Water reaches P_3	P_1 to P_3 connected through water	D_1, D_2, and D_3 are on.
Water reaches P_4	P_1 to P_4 connected through water	D_1 to D_4 are on.
Water reaches P_5	P_1 to P_5 connected through water	D_1 to D_5 are on steadily.
Water reaches P_6	Highest-level trigger path completed	Astable multivibrator is enabled and D_6 blinks.

Table 7.1: Sequential operation of the implemented Version 3 water level indicator.

Therefore, the implemented circuit provides two types of visual information. The first five LEDs give a steady indication of the approximate water level. The sixth LED gives a blinking alarm indication when the highest level is reached. This separation between normal indication and alarm behavior is the main operational feature of the implemented Version 3 circuit.

8 Component Selection and Design Calculations

This section explains the component selection and design justification for the implemented Version 3 circuit. The purpose of this section is not to show every arithmetic step in the main report, but to explain which component values were selected, why they were selected, how datasheet values were used, and what the calculated results mean for the physical circuit.

The detailed step-by-step calculations are provided in the appendices. This keeps the main report focused on engineering interpretation while still preserving the full calculation process for verification. The detailed LED resistor calculations are given in Appendix A, the BJT base-current and gain calculations are given in Appendix B, the RC time-constant derivations are given in Appendix C, the astable frequency calculation is given in Appendix D, and the power dissipation calculations are given in Appendix E.

The relevant datasheet excerpts are included in Appendix F. The BC547B datasheet excerpt is shown in Figure F.1, the 1N5819 Schottky diode datasheet excerpt is shown in Figure F.2, and the Agilent E3631A power supply datasheet excerpt is shown in Figure F.3.

8.1 Selected Components and Their Roles

The implemented Version 3 circuit is built from discrete analog components. The main selected components are listed in Table 8.1. Each component has a specific role in the sensing, switching, indication, filtering, protection, or alarm behavior of the circuit.

Component	Role in the Circuit
Carbon film resistors, 5% tolerance	Used as LED current-limiting resistors, BJT base resistors, pull-down resistors, and astable timing resistors.
$1\,\mu$F, 50 V electrolytic capacitors	Used as debounce capacitors at the BJT base nodes to reduce unstable switching caused by water movement.
$10\,\mu$F non-polarized capacitors	Used as timing and coupling capacitors in the astable multivibrator stage.
Blue, orange, green, yellow, red, and clear LEDs	Used as visual indicators for the detected water levels.
1N5819 Schottky diode	Used as a series reverse-polarity protection diode at the supply input.
BC547B NPN BJT transistors	Used as low-side switching transistors for the LED branches and the astable enabling stage.
Agilent E3631A DC power supply	Used as the laboratory 12 V DC source.

Table 8.1: Main components used in the implemented Version 3 circuit.

8.2 How Datasheet Values Are Used

Datasheets are used in this report as engineering references for checking whether the selected components can safely operate in the designed circuit. A datasheet value should not be copied without interpretation; it must be understood according to what type of value it is and under which test condition it is specified.

The most important distinction is between maximum ratings and operating characteristics. Maximum ratings define limits that should not be exceeded, such as maximum collector-emitter voltage, maximum collector current, or maximum reverse voltage. These values are used for safety verification. Operating characteristics describe expected behavior under specified conditions, such as transistor gain, saturation voltage, diode forward voltage, and pin configuration. These values are used to estimate how the component will behave in the circuit.

Therefore, the datasheet-based verification process used in this report follows the structure

identify the relevant datasheet quantity

↓

interpret what the quantity means

↓

compare it with the circuit requirement

↓

decide whether the component is suitable

For example, the BC547B collector current rating is compared with the LED branch current, the 1N5819 reverse-voltage rating is compared with the $12\,\mathrm{V}$ supply, and the Agilent E3631A output capability is compared with the expected current demand of the circuit. The relevant datasheet excerpts are provided in Appendix F.

8.3 Datasheet-Based Verification of Main Components

Table 8.2 summarizes the main datasheet values used in this design and explains how they are used to verify component suitability.

Component	Datasheet Quantity	Meaning	Use in This Design
BC547B	$V_{CEO} = 45\,\text{V}$	Maximum collector-emitter voltage with base open.	Compared with the $12\,\text{V}$ supply to verify voltage safety.
BC547B	$I_C = 100\,\text{mA}$	Maximum continuous collector current.	Compared with the approximately $5\,\text{mA}$ LED branch current.
BC547B	h_{FE}	DC current gain under specified test conditions.	Used to judge whether the available base current can switch the LED load.
BC547B	$V_{CE(sat)}$	Collector-emitter voltage when the transistor is on and saturated.	Used in LED resistor selection and BJT power estimation.
1N5819	$V_{RRM} = 40\,\text{V}$	Maximum repetitive reverse voltage.	Compared with the $12\,\text{V}$ supply for reverse-polarity protection.
1N5819	$I_F(AV) = 1.0\,\text{A}$	Average forward current rating.	Compared with the estimated $30\,\text{mA}$ to $40\,\text{mA}$ total circuit current.
Agilent E3631A	0 to $+25\,\text{V}$, 0 to $1\,\text{A}$ output	Available laboratory supply range and current capability.	Confirms that $12\,\text{V}$ operation and the expected current demand are within the supply capability.

Table 8.2: Datasheet values used to verify the suitability of the selected components.

The datasheet checks show that the selected components are operated far below their main voltage and current limits. The BC547B transistors are used at low collector current, the 1N5819 diode has sufficient reverse-voltage and current margin, and the Agilent E3631A supply can easily provide the required voltage and current.

8.4 Main Design Assumptions

Some circuit quantities are not perfectly fixed. LED forward voltage depends on LED color and manufacturer, water resistance depends on water conductivity and probe geometry, and capacitor values may vary due to tolerance. Therefore, the calculations use the design assumptions summarized in Table 8.3.

Quantity	Assumed Value	Design Meaning
Input supply voltage	$V_S = 12\,\text{V}$	Laboratory supply value selected within the allowed $5\,\text{V}$ to $12\,\text{V}$ range.
1N5819 forward voltage drop	$V_D \approx 0.3\,\text{V}$	Approximate Schottky diode voltage loss at low current.
Effective circuit supply	$V_{CC,\text{eff}} \approx 11.7\,\text{V}$	Approximate supply voltage available after the protection diode.
BJT base-emitter voltage	$V_{BE} \approx 0.7\,\text{V}$	Approximate voltage required to forward-bias the base-emitter junction.
BJT saturation voltage	$V_{CE(sat)} \approx 0.2\,\text{V}$	Approximate collector-emitter voltage when the BJT is on.
Modeled water resistance	$R_{\text{water}} = 100\,\text{k}\Omega$	LTspice model for the non-ideal conductive path through water.
Target LED current	$I_{\text{LED}} \approx 5\,\text{mA}$	Selected as a visible but low-current LED operating point.
Resistor tolerance	$\pm 5\%$	Used for carbon film resistor tolerance considerations.
Debounce capacitor	$C_{\text{DB}} = 1\,\mu\text{F}$	Used to reduce unstable switching at the base node.
Astable timing capacitor	$C_{\text{AST}} = 10\,\mu\text{F}$	Used in the astable multivibrator timing network.

Table 8.3: Main assumptions used for component selection and design calculations.

These assumptions are used to obtain the main design results in the following subsections. The detailed arithmetic steps are provided in the appendices.

8.5 Supply Voltage and Reverse-Polarity Protection

The circuit is supplied from a $12\,\text{V}$ DC source. Since the 1N5819 Schottky diode is placed in series with the positive supply input, the rest of the circuit receives slightly less than the input voltage. Using the low-current Schottky diode approximation,

$$V_{CC,\text{eff}} \approx 12\,\text{V} - 0.3\,\text{V} = 11.7\,\text{V}.$$

This effective supply voltage is still high enough for the LED branches, BJT switching stages, and astable multivibrator. The purpose of using a Schottky diode instead of a standard silicon diode is to reduce the lost voltage across the protection element. A standard silicon diode would typically drop around $0.7\,\text{V}$, while the Schottky diode drop is closer to $0.3\,\text{V}$ at low current. Therefore, the Schottky diode preserves more usable supply voltage.

The 1N5819 is suitable for this role because its reverse-voltage and current ratings are much larger than the expected requirements of the circuit. The detailed supply-voltage and protection-diode calculations are given in Appendix E.

8.6 LED Indicator Design

The LED current was targeted around

$$I_{\mathrm{LED}} \approx 5\,\mathrm{mA}.$$

This value was selected as a compromise between visible brightness and low total current consumption. Since several LEDs may be on at the same time, a moderate LED current keeps the total current low while still providing visible indication.

For each LED branch, the current-limiting resistor is selected using

$$R_{\mathrm{LED}} \approx \frac{V_{CC,\mathrm{eff}} - V_F - V_{CE(sat)}}{I_{\mathrm{LED}}}.$$

Using $V_{CC,\mathrm{eff}} \approx 11.7\,\mathrm{V}$, $V_{CE(sat)} \approx 0.2\,\mathrm{V}$, and typical LED forward voltages, $1.8\,\mathrm{k\Omega}$ was selected for the blue and clear LEDs, while $2.0\,\mathrm{k\Omega}$ was selected for the orange, green, yellow, and red LEDs.

The resulting LED currents are summarized in Table 8.4. All values remain close to the $5\,\mathrm{mA}$ design target.

LED Color	**Assumed V_F**	**Selected R_{LED}**	**Estimated I_{LED}**
Blue	$3.2\,\mathrm{V}$	$1.8\,\mathrm{k\Omega}$	$4.61\,\mathrm{mA}$
Orange	$2.0\,\mathrm{V}$	$2.0\,\mathrm{k\Omega}$	$4.75\,\mathrm{mA}$
Green	$2.1\,\mathrm{V}$	$2.0\,\mathrm{k\Omega}$	$4.70\,\mathrm{mA}$
Yellow	$2.0\,\mathrm{V}$	$2.0\,\mathrm{k\Omega}$	$4.75\,\mathrm{mA}$
Red	$1.9\,\mathrm{V}$	$2.0\,\mathrm{k\Omega}$	$4.80\,\mathrm{mA}$
Clear	$3.2\,\mathrm{V}$	$1.8\,\mathrm{k\Omega}$	$4.61\,\mathrm{mA}$

Table 8.4: Selected LED resistors and estimated LED currents.

The exact brightness of the LEDs may not be identical, because brightness depends on LED color, device efficiency, and manufacturer. Equal current does not guarantee equal perceived brightness. However, the selected currents are sufficiently close for a visual water level indicator. It should be noted that the LED resistor calculation uses $V_{CE(sat)} \approx 0.2\,\mathrm{V}$ as a design approximation for the BJT on-state voltage. This does not mean that the BJT is guaranteed to be deeply saturated under every possible water-conductivity condition. The actual collector-emitter voltage depends on the available

base current, the water-path resistance, and the transistor operating condition. Therefore, the calculated LED currents should be interpreted as approximate design currents under the assumed switching condition, not as exact guaranteed currents for all possible water samples. The detailed per-color LED resistor calculations are provided in Appendix A.

8.7 BJT Switching and Water-Path Sensitivity

The first five indication stages use BC547B NPN BJTs as low-side switching devices. In the LTspice model, the water path is represented by

$$R_{\text{water}} = 100\,\text{k}\Omega.$$

When water reaches a probe, the corresponding base-current path is formed through the modeled water resistance and the base resistor. Under the assumed water-path resistance, the available base current is approximately

$$I_B \approx 0.1\,\text{mA}.$$

Since the LED branch current is approximately

$$I_C \approx I_{\text{LED}} \approx 5\,\text{mA},$$

the required effective current gain is

$$\beta_{\text{required}} = \frac{I_C}{I_B} \approx \frac{5\,\text{mA}}{0.1\,\text{mA}} = 50.$$

This value is below the reference h_{FE} range of the BC547B device, so the transistor is expected to turn on and switch the LED branch under the modeled water-resistance condition. However, h_{FE} is not a fixed design constant. It depends on collector current, temperature, device variation, and transistor operating region. Therefore, this gain comparison should be interpreted as an approximate switching check rather than as a guarantee of identical behavior under all physical water conditions.

For a conservative saturated-switch design, engineers often use a forced gain such as

$$\beta_{\text{forced}} \approx 10.$$

With a collector current of approximately

$$I_C \approx 5\,\text{mA},$$

this conservative condition would require

$$I_{B,\text{forced}} = \frac{I_C}{\beta_{\text{forced}}} = \frac{5\,\text{mA}}{10} = 0.5\,\text{mA}.$$

The available base current through the modeled water path is only approximately

$$I_B \approx 0.1\,\text{mA}.$$

Therefore,

$$I_B < I_{B,\text{forced}}.$$

This means that the BJT stages should not be described as strongly overdriven saturated switches under all possible water conditions. The LED resistor and power calculations use

$$V_{CE(sat)} \approx 0.2\,\text{V}$$

as an approximate on-state design assumption, but the actual collector-emitter voltage depends on the available base current and the real water-path resistance.

This approximation is supported by the LTspice collector-voltage verification in Section 10.3. In that simulation, the collector voltages of the active low-side BJT stages drop close to ground under the modeled $R_{\text{water}} = 100\,\text{k}\Omega$ condition. Since the emitters are connected to ground, this indicates a low V_{CE} during the active state. Therefore, the saturated-switch approximation is reasonable for the simulated design condition.

However, this does not guarantee deep saturation for all real water samples. If the water is less conductive, R_{water} increases, the base current decreases, and the BJT may move away from strong saturation. In that case, V_{CE} may increase, the LED current may decrease, and the LED may become dimmer or less reliable.

Thus, the implemented BJT circuit is expected to work under the modeled water-resistance condition and is verified functionally by LTspice simulation, but its switching strength depends on water conductivity. This dependence is one of the main motivations for considering the MOSFET-based Version 4 design.

The detailed base-current, gain, and forced-β calculations are given in Appendix B.

8.8 Input Stabilization Network

Each sensing node uses a $1\,\text{M}\Omega$ pull-down resistor and a $1\,\mu$F debounce capacitor. The pull-down resistor keeps the BJT base node near ground when the probe is dry. This prevents the base from floating and reduces false triggering caused by leakage

current, electrical noise, or accidental touch.

The debounce capacitor is included because real water contact can be unstable. When the water surface is close to a probe, small waves, vibration, filling motion, or partial wetting can cause the electrical contact between the water and the probe to appear and disappear rapidly. These rapid contact changes produce rapid changes in the base-node voltage.

The capacitor slows sudden base-voltage changes. Therefore, it suppresses fast water-contact-induced fluctuations, while slower changes corresponding to an actual water-level transition can still turn the transistor on. For this reason, the network can be described as an input stabilization or debounce network with low-pass-like behavior.

It is important to note that this filtering description does not mean that the circuit filters the DC supply voltage. The supply is DC and approximately constant. The filtered quantity is the base-node voltage variation caused by unstable probe-water contact.

For the implemented values, the approximate turn-on and turn-off time constants are

$$\tau_{\text{on}} \approx 0.11\,\text{s}, \qquad \tau_{\text{off}} \approx 1.0\,\text{s}.$$

The shorter turn-on time allows the circuit to respond to real water-level changes, while the longer turn-off time helps prevent brief contact interruptions from immediately turning the LED off. Therefore, the input stabilization network improves the practical stability of the indicator without changing its basic sensing principle.

The $1\,\mu$F, 50 V electrolytic capacitors should be connected with correct polarity. Since the base node is positive with respect to ground during normal operation, the positive terminal of the capacitor should be connected to the base node and the negative terminal should be connected to ground. The detailed RC time-constant derivations are provided in Appendix C.

8.9 Astable Alarm Design

The sixth water level is treated as a critical level. Instead of driving another steady LED, P_6 enables an astable multivibrator stage. The astable multivibrator makes the final LED blink, which makes the highest water level more visually noticeable.

The astable stage uses timing resistors and capacitors to determine the blinking rate. For a symmetric BJT astable multivibrator, the approximate frequency is

$$f \approx \frac{1}{1.386RC}.$$

Using the selected timing values,

$$R = 68\,\mathrm{k}\Omega, \qquad C = 10\,\mu\mathrm{F},$$

the expected blinking frequency is approximately

$$f \approx 1.06\,\mathrm{Hz}.$$

This means that the alarm LED is expected to blink roughly once per second, which is suitable for a human-visible warning indication. Due to resistor and capacitor tolerances, the real blinking frequency may deviate from this value. With typical tolerance assumptions, the blinking frequency may reasonably fall in the range

$$0.84\,\mathrm{Hz} \lesssim f \lesssim 1.40\,\mathrm{Hz}.$$

This range is acceptable because the LED remains visibly blinking. The detailed frequency and tolerance calculations are given in Appendix D.

8.10 Current and Power Budget

The estimated total current consumption of the circuit is approximately

$$I_{\mathrm{total}} \approx 30\,\mathrm{mA} - 40\,\mathrm{mA}.$$

This current is much lower than the $1.0\,\mathrm{A}$ average forward current rating of the 1N5819 diode and the $1\,\mathrm{A}$ output capability of the Agilent E3631A $+25\,\mathrm{V}$ output. Therefore, the current demand of the circuit is safe for both the protection diode and the laboratory supply.

Using

$$V_{CC,\mathrm{eff}} \approx 11.7\,\mathrm{V},$$

the total power drawn by the circuit is estimated as

$$0.35\,\mathrm{W} \lesssim P_{\mathrm{total}} \lesssim 0.47\,\mathrm{W}.$$

A component-by-component summation gives approximately

$$P_{\mathrm{sum}} \approx 0.39\,\mathrm{W}.$$

Since P_{sum} lies inside the supply-side estimate range, the two power estimates are consistent. This confirms that the circuit is a low-power design and that heating is not

expected to be a major issue.

The highest individual resistor powers remain below typical quarter-watt resistor ratings. The most important long-term issue is not thermal power dissipation, but the presence of DC current through the water path. Even if the water-path power is thermally small, DC current through metal probes and water can contribute to electrolysis and corrosion over time. This limitation is discussed again in the practical implementation and limitations sections. The detailed power calculations are given in Appendix E.

8.11 Summary of Selected Component Values

The selected component values are summarized in Table 8.5. These values are used in the implemented Version 3 circuit and form the basis for the LTspice simulation and physical laboratory implementation.

Component / Quantity	**Selected Value**	**Purpose**
Supply voltage	$12\,\text{V}$ DC	Main laboratory supply voltage.
Effective circuit supply	$V_{CC,\text{eff}} \approx 11.7\,\text{V}$	Approximate supply after the 1N5819 protection diode.
Protection diode	1N5819 Schottky diode	Reverse-polarity protection with low forward voltage drop.
Switching transistor	BC547B NPN BJT	Low-side switching for LED branches and alarm enabling.
Base resistor	$10\,\text{k}\Omega$	Limits BJT base current.
Modeled water resistance	$100\,\text{k}\Omega$	LTspice model for the conductive water path.
Pull-down resistor	$1\,\text{M}\Omega$	Prevents floating BJT base nodes.
Debounce capacitor	$1\,\mu\text{F}$, $50\,\text{V}$ electrolytic	Smooths unstable probe contact.
LED resistors	$1.8\,\text{k}\Omega$, $2.0\,\text{k}\Omega$	Set LED currents close to $5\,\text{mA}$.
Astable timing resistor	$68\,\text{k}\Omega$	Sets blinking period with the timing capacitors.
Astable timing capacitor	$10\,\mu\text{F}$, non-polarized	Used in the astable multivibrator timing network.
Expected blink frequency	$\approx 1.06\,\text{Hz}$	Produces approximately once-per-second alarm blinking.
Estimated total current	$30\,\text{mA} - 40\,\text{mA}$	Confirms low current demand.
Estimated total power	$0.35\,\text{W} - 0.47\,\text{W}$	Confirms low-power operation.

Table 8.5: Summary of selected component values and design results for Version 3.

Overall, the selected components satisfy the voltage, current, and power requirements of the implemented circuit. The LED branches operate at low current, the BC547B transistors are used far below their maximum voltage and current ratings, the 1N5819 diode has sufficient reverse-voltage and current margin, and the Agilent E3631A power supply can easily provide the required supply voltage and current. The main practical limitation is the dependence of the BJT sensing stages on water conductivity and the long-term effect of DC current through the probes.

9 LTspice Simulation Method

The purpose of the LTspice simulation is to verify the operating behavior of the implemented Version 3 circuit before physical construction and testing. The simulation is not intended to model the full chemical or fluid behavior of water. Instead, it models the electrical effect of water contact between probes. In this way, the simulation can verify whether the circuit produces the required electrical responses: sequential LED activation for the first five levels and blinking alarm operation at the sixth level.

The simulated circuit is based on the implemented Version 3 schematic. Therefore, the simulation includes the $12\,\text{V}$ DC supply, the 1N5819 reverse-polarity protection diode, the repeated BJT sensing and LED indication stages, the base pull-down and debounce networks, and the astable multivibrator alarm stage. The MOSFET-based Version 4 is not used as the main simulated circuit because Version 3 is the design selected for physical implementation.

9.1 Simulation Objective

The LTspice simulation was used to check three main behaviors of the implemented circuit. First, each LED should turn on when its corresponding water-level probe is reached. Second, the first five LEDs should remain steady after activation. Third, when the highest probe P_6 is reached, the astable multivibrator should be enabled and the final LED should blink.

Therefore, the simulation verifies the functional sequence

$$\begin{gathered} P_1 \to D_1, \quad P_2 \to D_2, \quad P_3 \to D_3 \\ P_4 \to D_4, \quad P_5 \to D_5, \quad P_6 \to D_6 \text{ blinking} \end{gathered}.$$

This means that the main purpose of the simulation is functional verification rather than exact real-world prediction. The real circuit may behave differently because of water conductivity variation, probe surface condition, component tolerance, breadboard contact resistance, and capacitor tolerance.

9.2 Modeling Water Contact

In the physical circuit, water creates a conductive path between the common supply probe and a level probe when the water reaches that height. Since water is not an ideal conductor, this path should not be modeled as a perfect short circuit. In the LTspice simulation, the water path is represented as a controlled resistive connection.

Each probe contact is modeled using a voltage-controlled switch. The switch is placed in series with a resistance representing the non-ideal conductivity of the water path.

When the control signal of the switch is low, the switch remains open and the corresponding probe is treated as dry. When the control signal becomes high, the switch closes and the water path becomes electrically active.

Conceptually, each simulated sensing path behaves as

$$\text{dry probe} \rightarrow \text{open switch} \rightarrow \text{no base current},$$

and

$$\text{wet probe} \rightarrow \text{closed switch} \rightarrow \text{base current flows}.$$

Thus, the switch does not represent a mechanical switch in the real circuit. It is only a simulation tool used to represent the moment when water reaches a probe.

9.3 Sequential Probe Activation

To represent a rising water level, the six voltage-controlled switches are not activated at the same time. Instead, their control inputs are driven by pulse voltage sources with different delay times. Each pulse source turns on one switch at a specific time during the transient simulation.

This creates a simulated rising water sequence:

$$P_1 \text{ becomes wet first},$$

$$P_2 \text{ becomes wet after } P_1,$$

$$P_3 \text{ becomes wet after } P_2,$$

and so on until P_6 is reached.

This method allows the complete water-level sequence to be observed in a single transient simulation. The LEDs should turn on in the same order as the probes are activated. When P_6 is activated, the simulation should show the transition from normal level indication to alarm behavior.

9.4 Transient Analysis Setup

A transient analysis was used because the circuit behavior changes with time. The simulated water level rises step by step, and the astable multivibrator produces a time-varying blinking output. Therefore, a DC operating point analysis alone is not sufficient

for verifying this circuit.

The transient simulation command used in LTspice was

```
.tran 0 190 0 0.1
```

This command runs the simulation from $t = 0\,\text{s}$ to $t = 190\,\text{s}$. The simulation data is saved from the beginning of the simulation, and the maximum time step is limited to $0.1\,\text{s}$.

The total simulation time of $190\,\text{s}$ was selected so that all six probe levels could be activated sequentially and the final blinking behavior could be observed after the highest level is reached. Since the water-level contacts are modeled by delayed control pulses, the simulation must be long enough to include the activation of P_1, P_2, P_3, P_4, P_5, and P_6.

The maximum time step of $0.1\,\text{s}$ was used to keep the plotted waveforms sufficiently detailed for observing LED switching and the blinking alarm behavior. Since the expected blinking period is close to one second, this time-step limit gives several simulation points within each blinking cycle. Therefore, the transient setup is suitable for functional verification of the sequential LED activation and the astable alarm response.

The purpose of this transient simulation is not to obtain an exact electrochemical model of water contact, but to verify the time-domain electrical behavior of the implemented Version 3 circuit under the assumed LTspice water-contact model.

9.5 Observed Simulation Quantities

The main quantities observed in the simulation are the LED branch currents, selected transistor node voltages, and the output waveform of the astable multivibrator. The LED currents are especially useful because they directly show whether each LED is off, steadily on, or blinking.

For the first five levels, the expected LED current behavior is approximately

$$I_{D_1}, I_{D_2}, I_{D_3}, I_{D_4}, I_{D_5} \rightarrow \text{step-like steady currents after activation.}$$

For the sixth level, the expected current behavior is different:

$$I_{D_6} \rightarrow \text{periodic blinking current after } P_6 \text{ activation.}$$

Therefore, the current waveform of D_6 is used to confirm that the astable alarm stage is functioning correctly.

The base-node voltages can also be observed to check the effect of the pull-down resistors and debounce capacitors. When a probe is dry, the corresponding base node

should remain close to ground. When the probe is activated, the base voltage should rise rather than float unpredictably.

9.6 Simulation Assumptions and Limitations

The LTspice simulation uses idealized models for several physical effects. The water path is modeled as a controlled resistive path, but real water conductivity depends on dissolved ions, temperature, probe spacing, probe material, and surface contamination. Therefore, the selected water resistance in LTspice should be understood as a design assumption rather than a universal physical value.

The voltage-controlled switches also idealize the moment of water contact. In the real circuit, contact may be unstable because of waves, splashing, bubbles, or partial wetting of the probe. The debounce capacitors are included to reduce this kind of unstable switching, but LTspice does not fully reproduce the fluid motion inside the container.

In addition, the simulation does not model long-term electrolysis, corrosion, or chemical degradation of the probes. These effects are important for practical use, but they require physical testing or a more specialized electrochemical model. For this reason, the simulation results should be interpreted as functional circuit verification, not as complete proof of long-term real-world reliability.

9.7 Summary of the Simulation Method

The LTspice simulation method can be summarized as follows:

1. The implemented Version 3 circuit was recreated in LTspice.
2. Water contact was modeled using voltage-controlled switches and resistive water paths.
3. The switch control signals were delayed sequentially to represent a rising water level.
4. A transient simulation was used to observe the time-domain operation of the circuit.
5. LED currents and selected node voltages were observed to verify correct circuit behavior.
6. The first five LEDs were expected to turn on steadily, while the sixth LED was expected to blink after the astable alarm stage was enabled.

This simulation method provides a controlled way to test the functional behavior of the circuit before physical implementation. It verifies whether the designed circuit topology can produce the required indication sequence under the assumed electrical model of water contact.

10 Simulation Results of the Implemented Circuit

The LTspice simulation results were examined to verify whether the implemented Version 3 circuit satisfies the intended operating behavior. The main points checked in simulation are the sequential activation of the sensing stages, the response of the BJT base nodes, the LED current behavior, the blinking alarm operation, and the supply-side current and power behavior.

10.1 Sequential Water-Level Triggering

The first simulation result verifies how the rising water level is represented in LTspice. Since the physical motion of water is not directly modeled, each water contact is represented by a voltage-controlled switch. The control voltages of these switches are activated one after another to imitate the water reaching the probes sequentially.

Figure 10.1 shows the six switch-control signals used to model the rising water level.

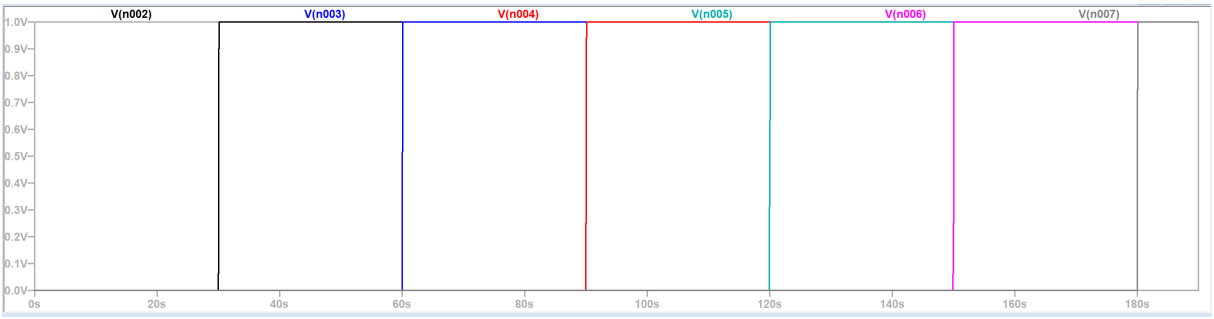


Figure 10.1: Sequential activation of the voltage-controlled switch inputs used to model the rising water level.

In this plot, the LTspice node voltages $V(n002)$, $V(n003)$, $V(n004)$, $V(n005)$, $V(n006)$, and $V(n007)$ correspond to the control signals of the six water-contact switches. Each signal represents one simulated probe contact. When a control signal is low, the corresponding switch is open and the probe is treated as dry. When the control signal becomes high, the corresponding switch closes and the probe is treated as reached by water.

The signals become active at approximately

$$30\,\text{s}, \quad 60\,\text{s}, \quad 90\,\text{s}, \quad 120\,\text{s}, \quad 150\,\text{s}, \quad 180\,\text{s}.$$

Therefore, the simulation represents the water level reaching the probes in the order

$$P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4 \rightarrow P_5 \rightarrow P_6.$$

This is important because the later LED current results must be interpreted with respect to this triggering sequence. If the switch-control signals were not sequential, then the LED activations could not be interpreted as the response of the circuit to a rising water level. Since Figure 10.1 confirms the intended triggering order, the following simulation results can be read as the circuit response to a progressively increasing water height.

The control signals rise from approximately $0\,\mathrm{V}$ to $1\,\mathrm{V}$. In the LTspice switch model, this voltage is not itself the water-level signal seen by the BJT base. Instead, it is the artificial control signal that determines whether the simulated water-contact switch is open or closed. Thus, Figure 10.1 should be interpreted as the simulation input sequence rather than as a physical voltage that would appear in the real water tank.

10.2 Input Triggering and RC Delay Measurement

After verifying the switch-control sequence, the next step is to examine how the BJT base nodes respond to these simulated water-contact events. The base-node voltage is important because the BJT turns on only when the base-emitter junction is forward biased. Therefore, the base voltage gives direct information about whether the sensing stage is being properly activated.

Figure 10.2 shows the base-node voltages of the switching transistors during the full transient simulation.

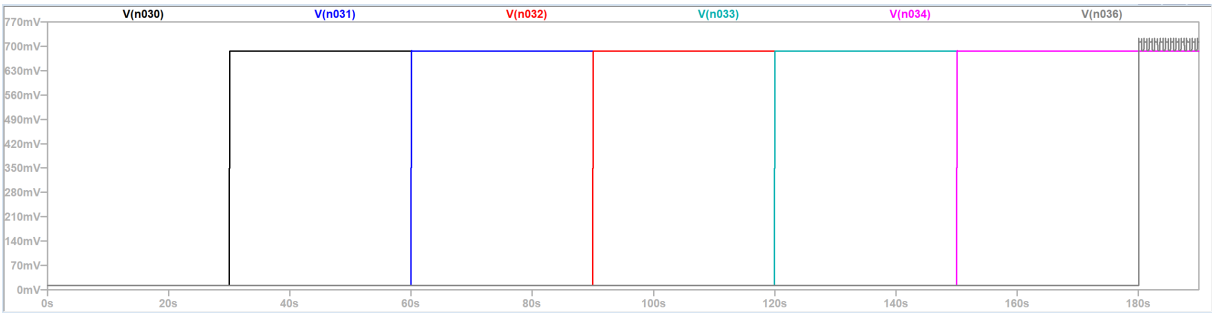


Figure 10.2: Simulated base-node voltages of the switching transistors during sequential water-level activation.

In this plot, $V(n030)$, $V(n031)$, $V(n032)$, $V(n033)$, and $V(n034)$ correspond to the base nodes of the first five BJT sensing stages. These nodes rise one after another as the corresponding water-contact switches are activated. After each transition, the base voltage settles close to

$$V_B \approx 0.69\,\mathrm{V}.$$

This value is physically meaningful because the base-emitter junction of a silicon BJT begins to conduct at approximately $0.7\,\mathrm{V}$. Therefore, the simulated base-node voltage confirms that the water path and base resistor provide enough drive to forward bias the BJT base-emitter junction.

The first five base-node voltages remain nearly constant after activation. This agrees with the intended behavior of the first five stages: each stage should turn on its transistor and keep the corresponding LED steadily on after the modeled water level reaches that probe.

The sixth base-node voltage, $V(n036)$, behaves differently after approximately $180\,\mathrm{s}$. This is expected because the sixth stage is not an ordinary steady LED stage. Instead, it activates the alarm stage and becomes coupled to the astable multivibrator behavior. The small oscillatory variation seen after the sixth trigger indicates that the highest-level stage is interacting with the blinking alarm circuit rather than remaining a purely steady switching stage.

The full-scale base-node plot verifies the correct order of switching, but it does not clearly show the triggering delay at one transition. For this reason, a zoomed view of the first triggering event is shown in Figure 10.3.

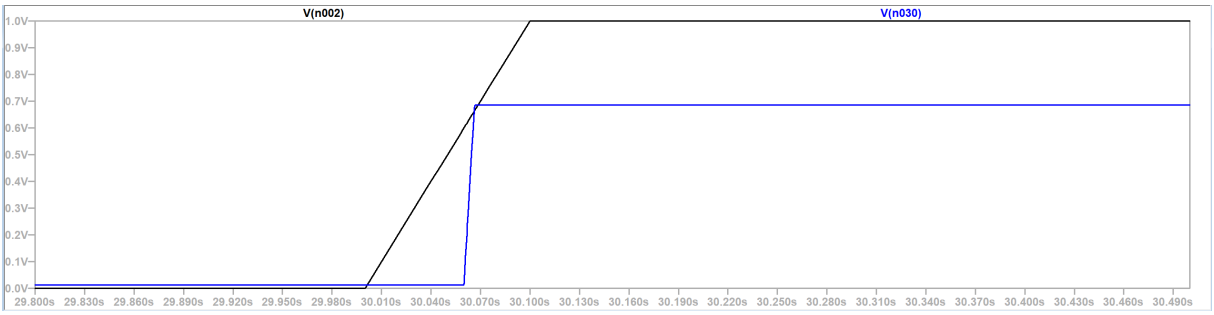


Figure 10.3: Zoomed comparison of the first switch-control signal $V(n002)$ and the first BJT base-node voltage $V(n030)$.

In Figure 10.3, $V(n002)$ is the control signal of the first simulated water-contact switch, while $V(n030)$ is the base-node voltage of the first BJT stage. The control signal begins to rise near

$$t \approx 30.0\,\mathrm{s}.$$

The base node remains close to ground before the switch turns on. This confirms the role of the pull-down resistor: when the probe is dry, the base is not left floating but is held near $0\,\mathrm{V}$.

As the control voltage rises, the voltage-controlled switch reaches its switching condition and the simulated water path becomes conductive. After this point, the base-node voltage rises quickly toward approximately

$$0.68\text{–}0.69\,\mathrm{V}.$$

This is the expected forward-bias voltage range for the BJT base-emitter junction. Once the base reaches this level, the transistor turns on and the corresponding LED branch

can conduct current.

This zoomed plot also shows why the input network should not be interpreted as an ideal instantaneous digital input. The switch control voltage, the voltage-controlled switch threshold, the water-path resistance, the base resistor, the pull-down resistor, and the debounce capacitor all affect the observed transition. Therefore, the apparent delay in the plot includes both the artificial switch-control behavior and the analog response of the base-node network.

The debounce capacitor is intended to reduce unstable switching caused by brief water-contact disturbances. In the theoretical design discussion, the approximate turn-on time constant was estimated as

$$\tau_{\mathrm{on}} \approx 0.11\,\mathrm{s}.$$

The simulation waveform is consistent with the idea that the base node does not behave as an uncontrolled floating node; it changes only when the water-contact switch is activated and then settles to the BJT forward-bias region. However, because the simulation uses an idealized voltage-controlled switch and a finite transient time step, this plot should be used mainly as a qualitative verification of input triggering rather than as a highly precise measurement of the physical RC time constant.

Therefore, Figures 10.2 and 10.3 together confirm that the input side of the circuit operates correctly. The probes are activated in order, the corresponding BJT base nodes rise to physically reasonable forward-bias voltages, and the pull-down/debounce network prevents the base nodes from remaining undefined when the probes are dry.

10.3 BJT Switching Stage Verification

After examining the input trigger signals and the base-node responses, the next step is to verify whether the BJT stages actually behave as switching stages. This is important because the LED current calculations use a low on-state collector-emitter voltage approximation. Therefore, the collector voltages of the switching transistors were inspected in the LTspice simulation.

Since the emitters of the low-side BJT switching stages are connected to ground, the collector voltage of each transistor directly represents its collector-emitter voltage:

$$V_{CE} = V_C - V_E.$$

Because

$$V_E = 0,$$

we obtain

$$V_{CE} = V_C.$$

Therefore, when a collector node drops close to ground, the corresponding BJT is operating with a low collector-emitter voltage and is behaving like a closed low-side switch.

Figure 10.4 shows the collector-voltage behavior of the BJT switching stages. The plotted collector nodes correspond to the transistors associated with the water-level stages. Each collector voltage is high while its transistor is off and drops close to ground when the corresponding stage is activated.

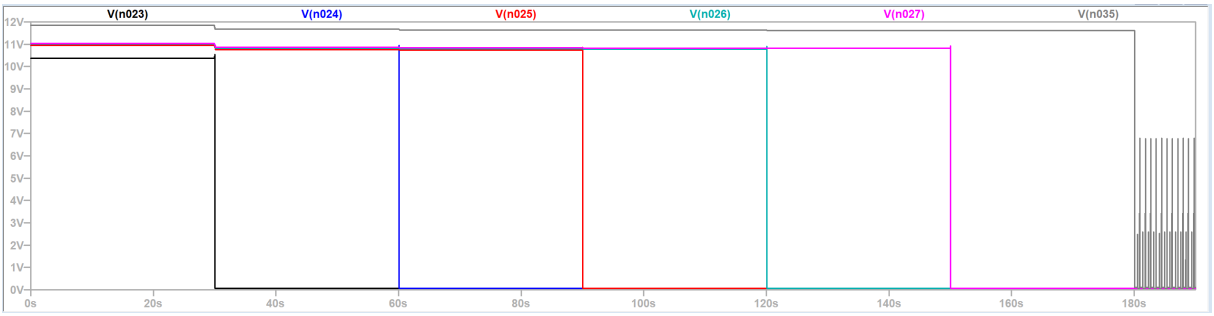


Figure 10.4: Collector-voltage behavior of the BJT switching stages. Since the emitters of the low-side switching transistors are connected to ground, each collector voltage represents the corresponding collector-emitter voltage V_{CE}.

For the first five level-indication stages, the collector voltages drop sequentially as the modeled water level reaches P_1, P_2, P_3, P_4, and P_5. Before activation, each collector node is pulled to a high voltage through its LED branch. This means the BJT is off and no LED current path is completed. After activation, the collector voltage falls close to ground. This means the BJT has turned on and now provides a low-side conduction path for the corresponding LED branch.

This result supports the intended switching sequence:

$$\begin{array}{c}\text{probe contact}\\ \text{is modeled}\end{array} \rightarrow \begin{array}{c}\text{base node}\\ \text{is driven}\end{array} \rightarrow \begin{array}{c}\text{BJT collector}\\ \text{drops low}\end{array} \rightarrow \begin{array}{c}\text{LED current path}\\ \text{is completed}\end{array}.$$

The low collector voltages also support the use of the approximate on-state assumption

$$V_{CE(sat)} \approx 0.2\,\text{V}$$

in the LED current and power calculations for the modeled LTspice condition. In other words, under the assumed water resistance and component models, the BJT switching stages behave close to low-voltage on-state switches when their corresponding probes are activated.

However, this conclusion must be interpreted carefully. The simulation result verifies the behavior under the modeled water resistance,

$$R_{\text{water}} = 100\,\text{k}\Omega.$$

It does not prove that the BJTs will be deeply saturated for every possible real water sample. In the physical circuit, the available base current depends on water conductivity, probe spacing, probe condition, and contact quality. If the real water resistance is much higher than the modeled value, the base current may decrease and the actual collector-emitter voltage may increase. Therefore, the saturated-switch approximation is reasonable for the simulated design condition, but it should not be treated as a universal guarantee.

The sixth switching stage shows time-varying collector behavior after the highest level is reached. This is expected because the sixth stage is connected to the alarm path rather than to a simple steady LED branch. Once the highest level activates the alarm section, the astable multivibrator begins oscillating, and the collector behavior of the alarm-related stage becomes periodic. Therefore, the sixth-stage waveform should be interpreted together with the alarm LED current and astable multivibrator waveforms discussed later.

Overall, this verification confirms that the BJT network performs the intended switching action in the LTspice model. The first five stages operate as steady low-side switches after activation, while the final stage participates in the time-varying alarm behavior.

10.4 LED Currents and Sequential Level Indication

After the input triggering and base-node behavior were verified, the next important result is the LED current response. The LED currents are the most direct output variables of the circuit, because they show whether each water level is being indicated correctly.

Figure 10.5 shows the simulated currents of the six LEDs during the full transient simulation.

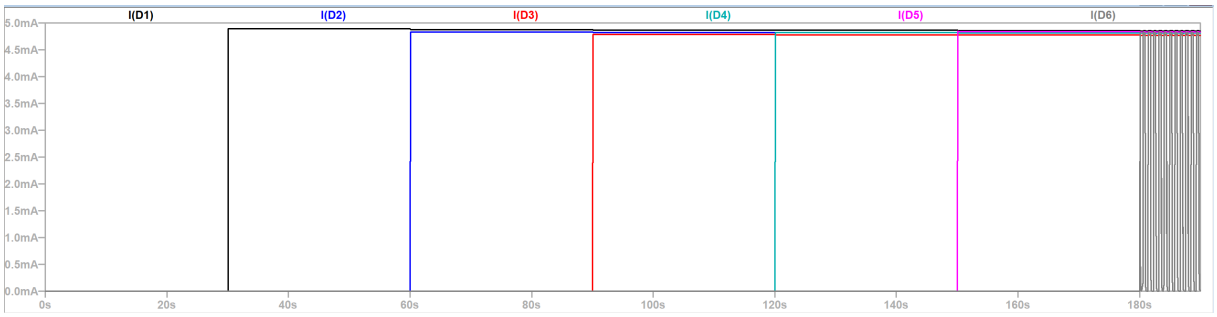


Figure 10.5: Simulated LED currents showing sequential activation of the first five LEDs and blinking operation of the sixth LED.

The first five LED currents, $I(D_1)$ to $I(D_5)$, turn on sequentially as the simulated water level reaches the corresponding probes. The first LED current rises near $t = 30\,\text{s}$, the second near $t = 60\,\text{s}$, the third near $t = 90\,\text{s}$, the fourth near $t = 120\,\text{s}$, and the fifth near $t = 150\,\text{s}$. This timing matches the switch-control sequence shown earlier, so the LED response follows the intended water-level order.

After each LED turns on, its current remains approximately constant. This confirms that the first five LEDs behave as steady level indicators rather than blinking or unstable outputs. The simulated LED currents are close to the intended design range of approximately $5\,\text{mA}$. The exact values are not identical because the LEDs have different forward-voltage models and because different current-limiting resistors are used. However, all first-five LED currents remain in the expected low-current indicator range.

This result verifies the main function of the lower five sensing stages:

$$P_n \text{ reached} \rightarrow Q_n \text{ turns on} \rightarrow I(D_n) \text{ becomes nonzero}, \qquad n = 1, 2, 3, 4, 5.$$

Therefore, the simulation confirms that the repeated BJT sensing and indication stages operate correctly. Each stage converts the modeled water contact into a steady LED current.

The sixth LED current, $I(D_6)$, behaves differently from the first five LED currents. It begins only after the sixth probe is activated near $t = 180\,\text{s}$, but it does not remain steady. Instead, it oscillates rapidly between an on-state current and a near-zero current. In the full-range plot, this appears as a dense vertical band near the end of the simulation. This indicates that the sixth level is not acting as an ordinary steady indication stage; it is activating the astable alarm stage.

Thus, Figure 10.5 verifies the intended separation between normal indication and alarm indication. The lower levels produce steady LED outputs, while the highest level produces a blinking LED output.

10.5 Alarm LED Blinking Measurement

The blinking behavior of the sixth LED is more clearly visible in a zoomed view. Figure 10.6 shows the simulated current of D_6 between approximately $180\,\text{s}$ and $190\,\text{s}$, after the highest water level has activated the alarm stage.

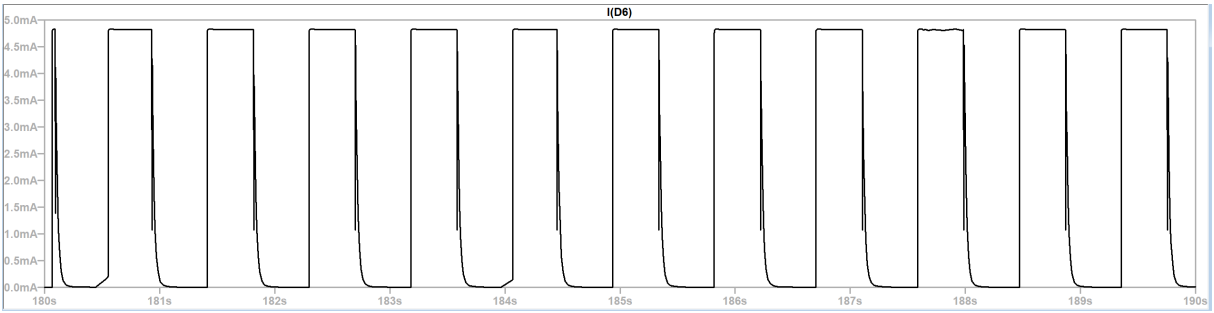


Figure 10.6: Zoomed simulation result of the alarm LED current $I(D_6)$ after activation of the highest water level.

The waveform shows that D_6 repeatedly switches between a near-zero current state and an on-state current of approximately $4.8\,\text{mA}$. This confirms that the sixth LED is not continuously on. Instead, it operates as a blinking alarm indicator.

The on-state current of D_6 is close to the design current used for the other LED branches. This is important because the alarm LED should be visually noticeable but should not draw excessive current. The near-zero portions of the waveform represent the off intervals of the blinking cycle.

The period of the blinking waveform is approximately on the order of one second. This agrees with the expected astable multivibrator behavior obtained from the selected $68\,\text{k}\Omega$ timing resistors and $10\,\mu\text{F}$ timing capacitors. Since the purpose of the alarm LED is to provide a clearly visible warning, a blinking rate close to once per second is suitable for human observation.

The waveform is not a purely ideal square wave. The falling transitions include a short decay region instead of an infinitely sharp drop. This is expected because the astable multivibrator contains capacitors that charge and discharge through resistive paths, and the transistor switching is not perfectly instantaneous. Therefore, the non-ideal transition shape is not a failure of the circuit; it is a natural result of the analog astable oscillator.

Figure 10.6 also supports the interpretation of the full LED current plot. In the full simulation result, the D_6 current appears as a dense oscillatory region near the end of the time axis. The zoomed plot shows that this region is actually a regular blinking waveform. Therefore, the simulation verifies that the highest-level sensing stage successfully enables the astable multivibrator and produces the required blinking alarm indication.

10.6 Astable Multivibrator Operation

After verifying that the sixth level activates the alarm LED, it is also necessary to confirm that the internal astable multivibrator itself operates correctly. The blinking behavior of the alarm LED is a consequence of the alternating switching action inside this astable stage. Therefore, the internal node voltages of the two cross-coupled transistor stages

were examined.

Figure 10.7 shows the voltages of the two main astable multivibrator nodes during the interval in which the highest water level has already activated the alarm stage.

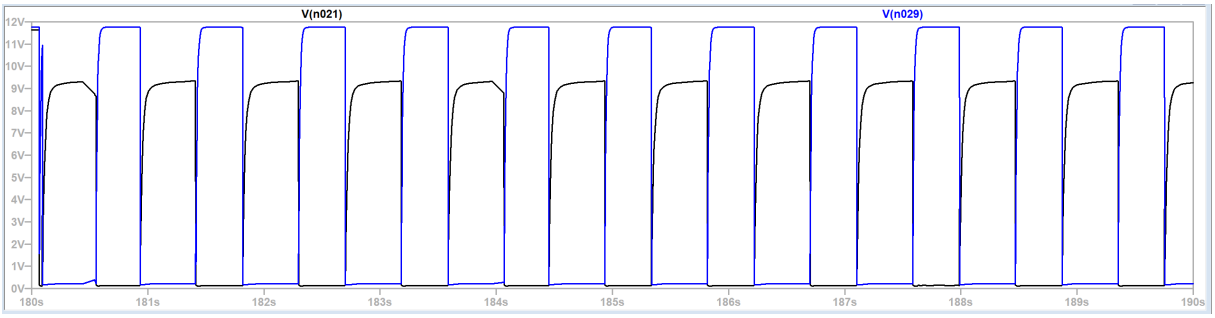


Figure 10.7: Simulated voltages of the two main astable multivibrator nodes after the highest-level alarm stage is enabled.

The waveform clearly shows that the two astable node voltages alternate with each other. When one node is near the high-voltage state, the other node is near the low-voltage state. Then, after a finite time interval, their roles interchange. This is the characteristic behavior of an astable multivibrator.

In other words, the circuit does not settle to a single stable DC operating point. Instead, it continuously switches between two complementary states:

$$\text{State 1: } Q_{\text{AST1}} \text{ on, } Q_{\text{AST2}} \text{ off}$$

$$\text{State 2: } Q_{\text{AST1}} \text{ off, } Q_{\text{AST2}} \text{ on}$$

Because of the cross-coupled RC network, each transistor alternately drives the other one toward cutoff and conduction. As a result, the circuit produces a self-sustained oscillation without any external clock source.

The blue waveform in Figure 10.7 behaves very nearly like a square wave. It repeatedly switches between a value close to 0 V and a value close to the available supply voltage. This is the expected collector-type behavior of a transistor in switching operation: when the transistor is on, the corresponding node is pulled low; when the transistor is off, the node rises toward the supply through the collector resistor.

The black waveform also alternates between low and high states, but its transitions are less ideal. In particular, some portions of the waveform show rounded rising behavior rather than perfectly vertical transitions. This is physically reasonable, because the astable multivibrator depends on the charging and discharging of timing capacitors through resistors. Therefore, certain node voltages evolve exponentially rather than instantaneously.

This graph is important for two reasons. First, it confirms that the blinking action seen

previously in the alarm LED current is truly generated by the astable multivibrator itself. Second, it shows that the oscillation is not random. The waveform is periodic and repeatable, which means that the highest-level alarm stage behaves as a controlled analog oscillator.

The oscillation period is approximately on the order of one second, which is consistent with the chosen timing components of the astable stage. This is also appropriate for visual indication, since an alarm LED that blinks too fast may be difficult to distinguish, while one that blinks too slowly may be less noticeable. The selected astable design therefore produces a practically useful warning signal.

Overall, Figure 10.7 verifies that the highest-level stage does not simply turn the final LED on. Instead, it activates a genuine astable multivibrator whose alternating internal node voltages generate the required flashing alarm behavior.

10.7 Supply Voltage Behavior Under Loading

In addition to the signal and LED responses, it is also useful to examine the supply behavior of the implemented circuit. Since the circuit includes a series reverse-polarity protection diode and multiple LED-transistor branches that turn on progressively, the internal circuit supply voltage is expected to differ slightly from the ideal laboratory source voltage.

Figure 10.8 shows the source-side supply voltage and the internal circuit supply voltage V_{CC} throughout the simulation.

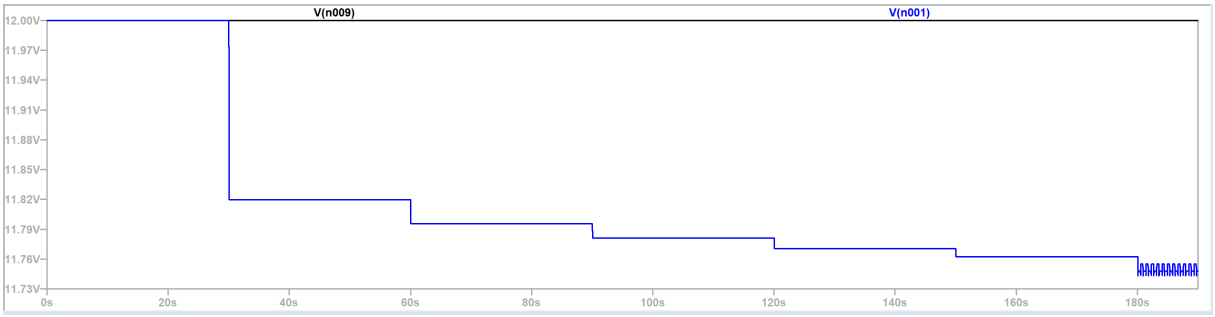


Figure 10.8: Simulated source voltage and internal circuit supply voltage V_{CC} as the water level rises and more loads become active.

The black trace remains at approximately $12\,\text{V}$, which represents the ideal DC supply voltage provided by the source. This is expected, since the source itself is modeled as a fixed voltage source.

The blue trace represents the internal circuit supply voltage after the protection diode. At the beginning of the simulation, when no water level is active and essentially no load current is drawn, this voltage is also close to $12\,\text{V}$. However, when the first water level is triggered near $t = 30\,\text{s}$, the internal supply voltage drops slightly, from about $12.0\,\text{V}$ to about $11.82\,\text{V}$.

This behavior is expected. Once the first LED branch turns on, current begins to flow through the series protection diode. Since the diode is not ideal, it develops a forward voltage drop. Therefore, the voltage available to the rest of the circuit becomes slightly smaller than the original source voltage.

As additional water levels are triggered, more LED branches turn on, so the total current drawn from the source increases. Consequently, the forward current through the protection diode also increases, and its voltage drop becomes slightly larger. This causes the internal supply voltage V_{CC} to decrease step by step as the water level rises.

The graph therefore provides a useful system-level interpretation:

$$\begin{gathered}\text{more active levels} \rightarrow \text{more supply current} \\ \rightarrow \text{larger diode drop} \rightarrow \text{slightly smaller } V_{CC}\end{gathered}.$$

This explains why the blue trace is not constant. Instead, it decreases in discrete steps at approximately $30\,\text{s}$, $60\,\text{s}$, $90\,\text{s}$, $120\,\text{s}$, and $150\,\text{s}$, exactly when the lower five steady-indication branches become active.

After the sixth level is reached near $t = 180\,\text{s}$, the behavior changes again. Instead of a single additional step, a small oscillatory ripple appears on V_{CC}. This is also expected. At the highest level, the astable alarm stage begins to switch on and off periodically. Therefore, the load current is no longer constant but time-varying. As the astable stage alternates between its states, the instantaneous supply current changes, and this causes a corresponding small fluctuation in the voltage drop across the series protection diode.

Thus, the ripple visible near the end of the graph is not a problem; it is a direct consequence of the blinking alarm stage. In fact, it provides further confirmation that the highest-level stage is actively oscillating and modulating the total circuit current.

Another important conclusion from Figure 10.8 is that the reduction in V_{CC} remains small. Even after multiple levels are active, the internal supply voltage remains close to the original $12\,\text{V}$ source value. Therefore, the use of the Schottky protection diode does not excessively reduce the available operating voltage. This supports the design choice of using a Schottky diode rather than an ordinary silicon diode, since the lower forward drop helps preserve more voltage for the sensing, switching, and LED branches.

Overall, this result shows that the power distribution of the circuit remains well behaved. The source voltage stays fixed, the internal supply voltage decreases only moderately as more loads are added, and the small ripple at the highest level is fully consistent with the intended alarm-flashing operation.

10.8 Supply Current and Power Dissipation Verification

After verifying the signal behavior of the sensing and alarm stages, the supply-side behavior of the circuit was also examined. This is important because the circuit is powered from a laboratory DC supply, and the current drawn by the circuit must remain within a safe and expected range.

Figure 10.9 shows the simulated current drawn from the supply.

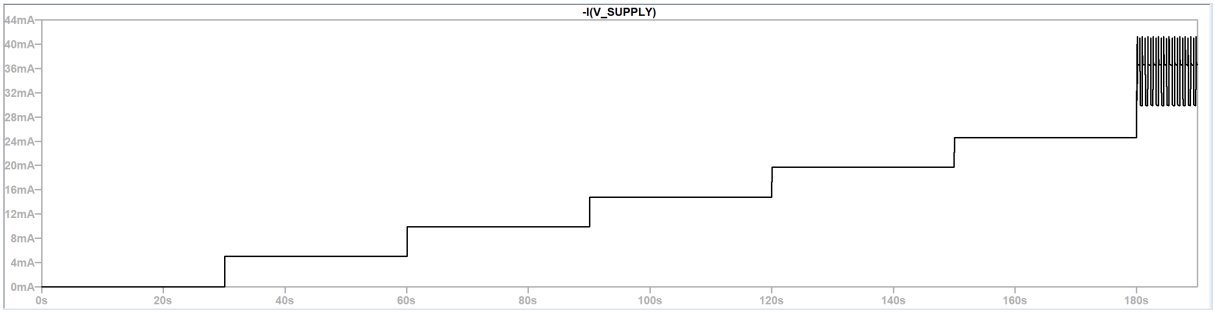


Figure 10.9: Simulated supply current $-I(V_{\mathrm{SUPPLY}})$ during sequential level activation and alarm-stage operation.

The supply current increases in a step-like manner as the modeled water level reaches additional probes. This behavior is expected because each newly activated water level turns on another BJT switching stage and LED branch. Therefore, every new active level adds an additional current path to the circuit.

Before the highest level is reached, the current remains approximately steady after each step. This confirms that the first five indication stages behave as steady loads after activation. The current rises from nearly zero to a few milliamps when the first LED branch turns on, and then continues increasing as D_2, D_3, D_4, and D_5 are activated.

After P_6 is reached near $t = 180\,\mathrm{s}$, the supply current no longer remains constant. Instead, it becomes oscillatory. This oscillation is caused by the astable multivibrator and the blinking alarm LED. As the alarm stage switches between its two states, the instantaneous current drawn from the supply changes periodically.

The final current range is on the order of a few tens of milliamps. In the blinking region, the current is approximately between

$$30\,\mathrm{mA} \quad \text{and} \quad 42\,\mathrm{mA}.$$

This agrees with the current estimate used in Section 8, where the total current was expected to be approximately

$$I_{\mathrm{total}} \approx 30\,\mathrm{mA} - 40\,\mathrm{mA}.$$

The slight extension above $40\,\mathrm{mA}$ occurs during instantaneous peaks of the astable

alarm operation. This is not a problem, because the circuit current is still far below the 1 A current capability of the Agilent E3631A supply and the 1 A average forward-current rating of the 1N5819 diode.

The corresponding simulated power waveform is shown in Figure 10.10.

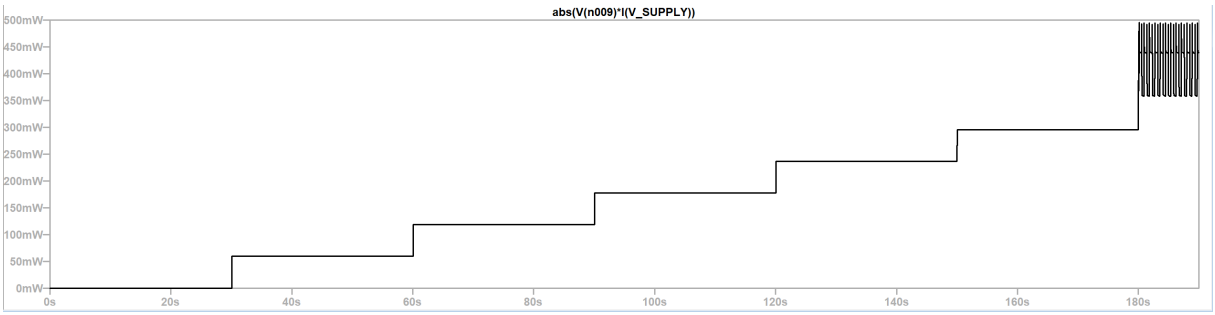


Figure 10.10: Simulated power drawn from the DC supply during sequential level activation and alarm-stage operation.

The plotted expression is

$$|V(n009)I(V_{\mathrm{SUPPLY}})|\,.$$

The absolute value is used because LTspice defines the current through a voltage source according to its own reference direction. When a voltage source delivers power to a circuit, the raw product VI may appear with a negative sign depending on the source orientation. Taking the absolute value gives the magnitude of the power drawn from the supply.

The power waveform follows the same general pattern as the supply-current waveform. At first, the power is nearly zero because no water level has been activated. As the first five LED branches turn on, the power increases in discrete steps. This step-like increase is expected because each active level adds another LED branch and switching path.

After the sixth level is reached, the power waveform becomes oscillatory. This occurs for the same reason as the oscillation in the supply current: the astable multivibrator repeatedly changes the load seen by the supply. When the alarm LED branch conducts, the instantaneous power is higher; when the branch is off, the instantaneous power is lower.

The simulated power remains below approximately

$$0.5\,\mathrm{W}.$$

This agrees with the design estimate in Section 8, where the total power was expected to be approximately

$$0.35\,\mathrm{W} \lesssim P_{\mathrm{total}} \lesssim 0.47\,\mathrm{W}.$$

The peak values in the simulation approach the upper end of this range because the astable alarm stage introduces time-varying current peaks. However, the circuit is still clearly a low-power design. The simulated result supports the conclusion that component heating is not expected to be a major limitation in the laboratory prototype.

Therefore, the supply-current and power-dissipation simulations confirm that the implemented Version 3 circuit operates within a safe low-current and low-power range.

10.9 Overall Interpretation of the Simulation Results

The LTspice simulation results confirm that the implemented Version 3 circuit operates according to the intended design logic. The simulation first verifies that the modeled water level reaches the probes sequentially. Then, the base-node voltage results show that the corresponding BJT stages are activated in the same order. After this internal switching behavior is confirmed, the LED current plots show the actual visual output of the circuit: the first five LEDs operate as steady level indicators, while the sixth LED operates as a blinking alarm indicator.

The simulation results also connect the signal behavior of the circuit with the supply-side behavior. As more water levels are activated, additional LED branches and BJT switching paths draw current from the supply. Therefore, both the supply current and the simulated power consumption increase step by step. After the sixth level is reached, the astable alarm stage begins oscillating, and this causes periodic variation in the LED current, supply current, supply voltage, and power waveform.

Table 10.1 summarizes the main simulation observations and their interpretations.

Observed Quantity	Simulation Result	Interpretation
Water-level trigger signals	The switch-control signals activate at approximately 30 s, 60 s, 90 s, 120 s, 150 s, and 180 s.	The LTspice model represents a rising water level that reaches P_1 to P_6 sequentially.
BJT base-node voltages	The first five base nodes settle near 0.68–0.69 V.	The base-emitter junctions are forward biased, so the BJT switching stages are properly activated.
Input triggering behavior	The first base node rises after the first switch-control signal activates the simulated water path.	The sensing input successfully converts modeled water contact into BJT base drive.
Steady LED currents	D_1 to D_5 turn on sequentially and remain close to the intended low-current range of approximately 5 mA.	The first five levels operate as steady visual water-level indicators.
Alarm LED current	D_6 switches periodically between nearly 0 mA and approximately 4.8 mA.	The highest level activates a blinking alarm instead of a steady LED.
Astable multivibrator nodes	The two astable node voltages alternate periodically between high and low states.	The blinking behavior is produced by the internal oscillation of the astable multivibrator.
Protected supply voltage	V_{CC} remains close to the 12 V source voltage, with a small stepwise decrease caused by the protection diode.	The 1N5819 Schottky diode protects the circuit without excessively reducing the usable supply voltage.
Supply current	The current increases step by step and becomes oscillatory after P_6 is activated.	The supply current follows the activation of the LED branches and the periodic operation of the alarm stage.
Power consumption	The simulated power remains below approximately 0.5 W.	The circuit operates as a low-power design, and heating is not expected to be a major limitation.

Table 10.1: Summary of the main LTspice simulation observations for the implemented Version 3 circuit.

Overall, the simulation verifies that the implemented circuit satisfies the intended functional requirements:

$$\begin{gathered}\text{sequential water-level triggering} \rightarrow \text{BJT switching} \\ \rightarrow \text{steady LED indication for } P_1\text{–}P_5 \\ \rightarrow \text{blinking alarm indication for } P_6\end{gathered} .$$

The first five levels behave as ordinary water-level indicators, while the sixth level produces a visually distinct alarm response. The supply-side results also remain consistent with the design estimates from Section 8. The circuit current stays in the range of a few tens of milliamps, and the total simulated power remains below approximately 0.5 W. Therefore, the LTspice simulation supports the correctness of the implemented Version 3 topology before physical construction and experimental testing.

However, these results should still be interpreted as functional simulation results rather than complete real-world proof. The LTspice model represents water contact using controlled switches and modeled resistance. It does not fully model real water conductivity variation, probe corrosion, electrolysis, unstable surface motion, or long-term practical degradation. For this reason, physical implementation and experimental testing are still required to evaluate the behavior of the circuit under real laboratory conditions.

11 Experimental Implementation and Physical Testing

After the LTspice simulation verified the intended operation of the implemented Version 3 circuit, the circuit was physically constructed on breadboards and tested under real water-contact conditions. The purpose of the physical experiment was to check whether the simulated sequential indication behavior and the highest-level blinking alarm behavior could also be observed in the laboratory implementation.

11.1 Physical Breadboard Implementation

The physical implementation of the conductive-probe-based water level indicator is shown in Figure 11.1. The circuit was supplied from a $12\,\mathrm{V}$ DC source. The gray wire represents the input supply line V_{in}, which is the voltage before the reverse-polarity protection diode. Therefore, the protected internal circuit supply satisfies approximately

$$V_{\mathrm{in}} \approx V_{CC} + V_D,$$

where V_D is the forward voltage drop of the Schottky protection diode.

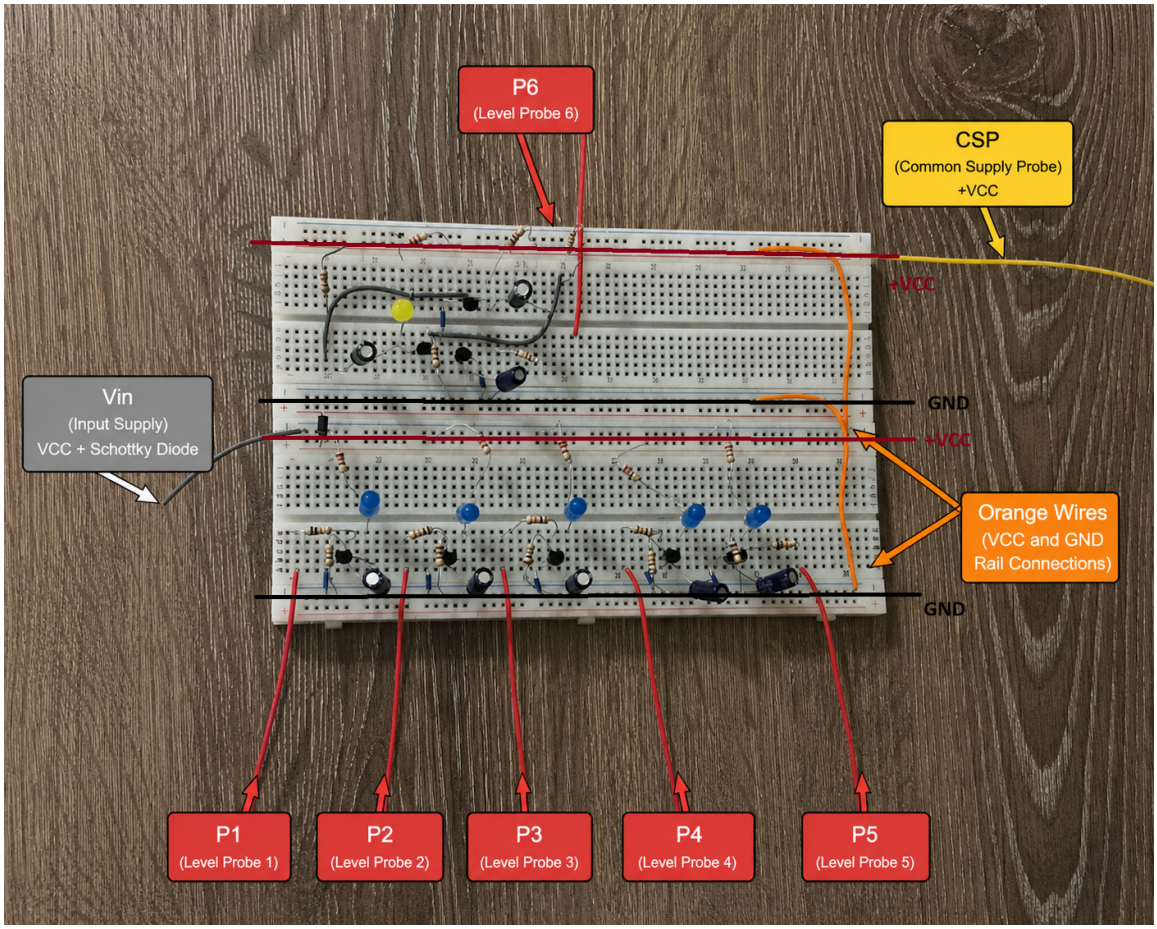


Figure 11.1: Physical breadboard implementation of the conductive-probe-based water level indicator circuit. The gray wire is the input supply V_{in}, the yellow wire is the common supply probe (CSP), the red wires are the level probes P_1 to P_6, and the orange jumpers provide the rail connections for V_{CC} and GND.

In the physical setup, the yellow wire is used as the common supply probe. This probe is connected to the protected positive supply rail and acts as the common electrode that contacts the water. The red wires are the individual level probes. The lower red probes correspond to P_1 through P_5, while the upper red probe corresponds to P_6, the highest-level alarm probe.

When the water reaches a given red level probe, the water forms a conductive path between the common supply probe and that level probe. This creates the base-drive path for the corresponding transistor stage. As a result, the related BJT turns on and the associated LED indicator becomes active. For P_1 through P_5, this produces steady level indication. For P_6, the contact enables the astable multivibrator stage, causing the alarm LED to blink.

The orange jumper wires are used to connect the corresponding V_{CC} and GND rails between different breadboard sections. This was necessary because the physical circuit was distributed across breadboard rails, and all circuit modules needed a common reference ground while sharing the protected supply rail. During construction, special attention was given to keeping the astable multivibrator enable path controlled by the

$Q_{\mathrm{AST_SW}}$ transistor, so that the alarm stage would operate only when the P_6 probe was activated.

Overall, this setup allowed the simulated sensing structure to be tested physically: the common supply probe and the level probes formed the real water-contact interface, while the BJT stages and the astable alarm circuit implemented the electrical response of the system.

11.2 Supply Voltage and Protected Rail Measurement

The first physical measurement was performed to verify the input supply voltage and the protected internal supply rail after the Schottky protection diode. In the design calculations, the forward voltage drop of the $1\mathrm{N}5819$ Schottky diode was approximated as $0.3\,\mathrm{V}$. Therefore, for a $12\,\mathrm{V}$ input supply, the expected protected supply voltage was approximately

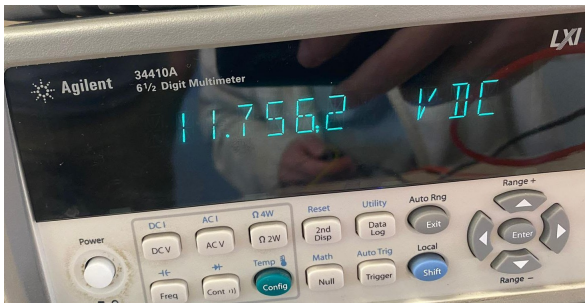
$$V_{CC} \approx V_{\mathrm{in}} - V_D$$

$$V_{CC} \approx 12\,\mathrm{V} - 0.3\,\mathrm{V} = 11.7\,\mathrm{V}.$$

The measured input supply voltage and protected rail voltage are shown in Figure 11.2.



(a) DC source adjusted to approximately $12\,\mathrm{V}$.



(b) Measured protected supply rail after the Schottky diode.

Figure 11.2: Physical measurement of the input supply voltage and the protected internal supply voltage V_{CC}.

From the measurements,

$$V_{\mathrm{in}} = 12.01\,\mathrm{V},$$

and

$$V_{CC} = 11.7562\,\mathrm{V}.$$

Therefore, the measured voltage drop across the Schottky protection diode is

$$V_D = V_{\mathrm{in}} - V_{CC},$$

$$V_D = 12.01\,\mathrm{V} - 11.7562\,\mathrm{V} = 0.2538\,\mathrm{V}.$$

Thus,

$$V_D \approx 0.254\,\mathrm{V}.$$

This result is consistent with the expected low forward-voltage behavior of the $1\mathrm{N}5819$ Schottky diode. The measured protected rail voltage,

$$V_{CC} \approx 11.76\,\mathrm{V},$$

is also close to the value used in the design calculations. Therefore, the reverse-polarity protection stage does not significantly reduce the available supply voltage, and the remaining voltage is sufficient for the LED branches, BJT switching stages, and astable alarm circuit.

11.3 Sequential Supply Current Measurement

After verifying the protected supply rail, the supply current was measured during sequential water-level activation. The current values were read directly from the DC power supply display while the circuit was operating at approximately

$$V_{\mathrm{in}} \approx 12.01\,\mathrm{V}.$$

This measurement was performed from the first activated level P_1 to the highest activated level P_6. Therefore, this measurement set does not include the completely OFF condition. Instead, it shows how the supply current changes as more stages become active.

Table 11.1: Measured supply current during sequential water-level activation from P_1 to P_6.

Condition	Expected LED behavior	Measured current	Approx. power
P_1 reached	D_1 ON	$0.005\,\mathrm{A}$	$0.060\,\mathrm{W}$
P_2 reached	D_1, D_2 ON	$0.010\,\mathrm{A}$	$0.120\,\mathrm{W}$
P_3 reached	D_1, D_2, D_3 ON	$0.015\,\mathrm{A}$	$0.180\,\mathrm{W}$
P_4 reached	D_1 to D_4 ON	$0.021\,\mathrm{A}$	$0.252\,\mathrm{W}$
P_5 reached	D_1 to D_5 ON	$0.026\,\mathrm{A}$	$0.312\,\mathrm{W}$
P_6 reached	Alarm stage active	$0.038\,\mathrm{A}$	$0.456\,\mathrm{W}$

The approximate power values in Table 11.1 were calculated using

$$P \approx V_{\mathrm{in}} I.$$

For example, when the highest level P_6 was reached, the measured supply current was approximately

$$I_{P_6} = 0.038\,\mathrm{A}.$$

Thus, the approximate power drawn from the supply was

$$P_{P_6} \approx (12.01\,\mathrm{V})(0.038\,\mathrm{A}) = 0.456\,\mathrm{W}.$$

The measured current increased step by step as the water level activated additional stages. This is the expected behavior because each newly activated level adds another conducting LED branch. When P_6 was reached, the current increased further because the astable alarm stage was enabled in addition to the ordinary level-indication stages.



(a) P_1: 0.005 A.



(b) P_2: 0.010 A.



(c) P_3: 0.015 A.



(d) P_4: 0.021 A.



(e) P_5: 0.026 A.



(f) P_6: 0.038 A.

Figure 11.3: Measured supply current readings from P_1 activation to P_6 activation.

Although the power supply display gives a relatively coarse current reading, the measured trend is clear. The current rises as the number of active stages increases. This experimentally supports the LTspice result, where the source current also increased in a stepwise manner during sequential level activation. The largest measured current was still below $40\,\mathrm{mA}$, showing that the implemented circuit operates in a low-current range.

11.4 LED Current Measurement

The LED branch current was measured for a representative blue LED stage. In this stage, the LED series resistor was

$$R_{\mathrm{LED}} = 2.2\,\mathrm{k\Omega}.$$

Two measurement methods were used. First, the voltage across the LED series resistor was measured and the current was calculated using Ohm's law. Second, the LED current was measured directly using the ammeter connection.

11.4.1 Current Calculation from Resistor Voltage

The voltage across the LED series resistor was measured as

$$V_{R_{\mathrm{LED}}} = 9.0034\,\mathrm{V}.$$

Since the resistor and LED are in series, the resistor current is equal to the LED current:

$$I_{\mathrm{LED}} = I_{R_{\mathrm{LED}}}.$$

Using Ohm's law,

$$I_{\mathrm{LED}} = \frac{V_{R_{\mathrm{LED}}}}{R_{\mathrm{LED}}}.$$

Substituting the measured value,

$$I_{\mathrm{LED}} = \frac{9.0034\,\mathrm{V}}{2.2 \times 10^{3}\,\Omega} = 4.09 \times 10^{-3}\,\mathrm{A}.$$

Thus,

$$I_{\mathrm{LED}} \approx 4.09\,\mathrm{mA}.$$

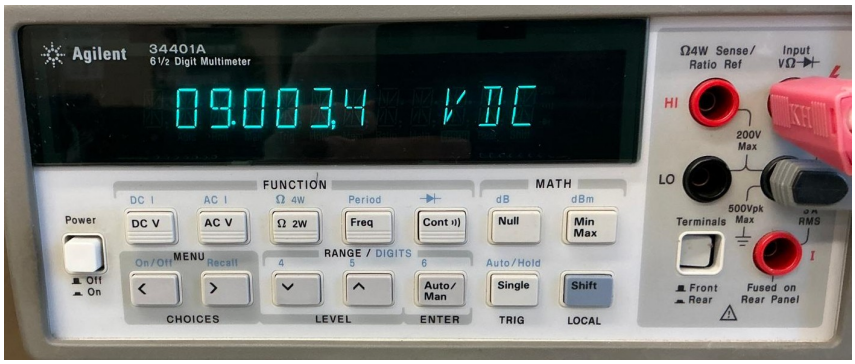


Figure 11.4: Measured voltage across the blue LED series resistor. The measured value was $V_{R_{\mathrm{LED}}} = 9.0034\,\mathrm{V}$.

11.4.2 Direct Ammeter Measurement

The LED current was also measured directly by inserting the ammeter in series with the LED branch. The measured current was

$$I_{\mathrm{LED,measured}} = 3.9089\,\mathrm{mA}.$$

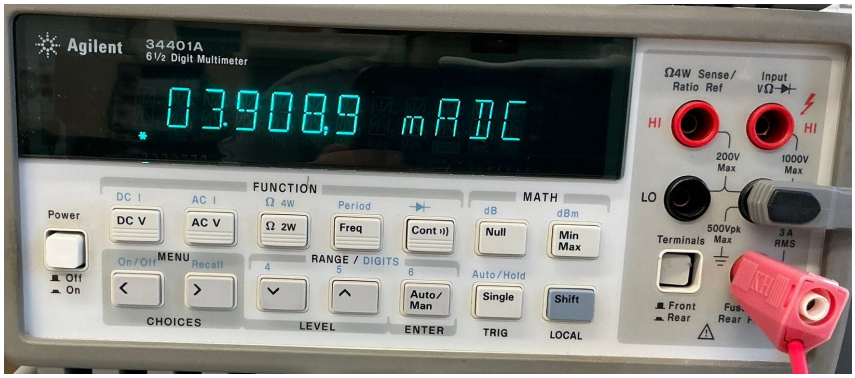


Figure 11.5: Direct ammeter measurement of the blue LED branch current. The measured value was $I_{\mathrm{LED}} = 3.9089\,\mathrm{mA}$.

The two methods produced similar results:

$$I_{\mathrm{LED,calculated}} \approx 4.09\,\mathrm{mA},$$

$$I_{\mathrm{LED,measured}} \approx 3.91\,\mathrm{mA}.$$

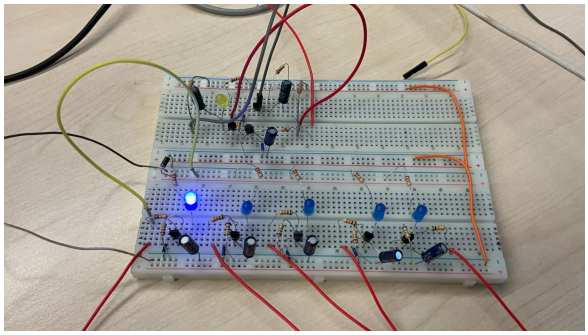
The small difference between these values can be explained by resistor tolerance, breadboard contact resistance, meter insertion effects, and the non-ideal behavior of the LED and BJT switching stage. Both values confirm that the LED current is in the intended low-current range of only a few milliamperes.

Table 11.2: Comparison of calculated and directly measured LED current.

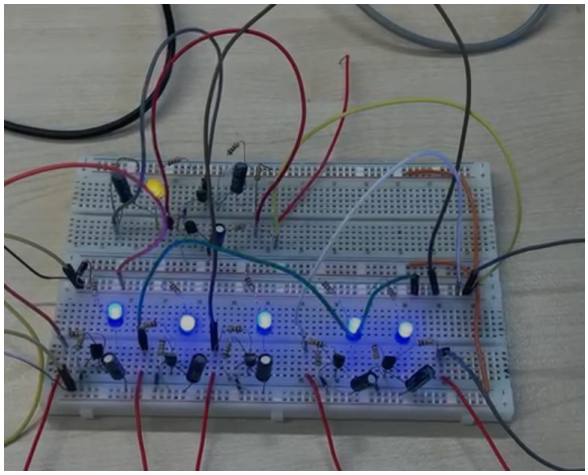
Method	**Measured quantity**	**Result**
Voltage across R_{LED}	$V_{R_{\mathrm{LED}}} = 9.0034\,\mathrm{V}$	$I_{\mathrm{LED}} \approx 4.09\,\mathrm{mA}$
Direct ammeter measurement	Current through LED branch	$I_{\mathrm{LED}} \approx 3.91\,\mathrm{mA}$

11.5 Observed LED Activation

The physical LED activation was also observed directly during the water-level test. Figure 11.6 shows representative operating states of the implemented circuit.



(a) Example of a lower-level LED activation.



(b) Multiple level LEDs active during a higher water-level condition.

Figure 11.6: Observed physical LED activation states of the implemented water level indicator circuit.

The observed behavior matched the intended operation: as the water level was increased, the corresponding LED indicators turned on sequentially. At the highest level, the alarm stage was enabled and the alarm LED produced a blinking output.

11.6 BJT Switching Verification

After measuring the LED current, the switching behavior of a representative BJT stage was verified experimentally. Since the LED stages are implemented as low-side BJT switches, the transistor should operate in two clearly different states:

$$\text{OFF state: } V_{BE} \approx 0, \qquad V_{CE} \text{ high}$$

$$\text{ON state: } V_{BE} \approx 0.6\text{–}0.8\,\mathrm{V}, \qquad V_{CE} \text{ low}.$$

The measured values are summarized in Table 11.3.

Table 11.3: Measured V_{BE} and V_{CE} values for a representative BJT switching stage.

Measurement	LED ON state	LED OFF state
V_{BE}	$0.72946\,\mathrm{V}$	$3.253\,\mathrm{mV}$
V_{CE}	$19.335\,\mathrm{mV}$	$9.6227\,\mathrm{V}$

When the LED was ON, the measured base-emitter voltage was

$$V_{BE,\mathrm{ON}} = 0.72946\,\mathrm{V}.$$

This value is in the expected forward-bias range of a silicon BJT base-emitter junction. Therefore, the measurement confirms that the transistor was driven into conduction.

In the same ON state, the collector-emitter voltage was measured as

$$V_{CE,\mathrm{ON}} = 19.335\,\mathrm{mV}.$$

This very small collector-emitter voltage shows that the BJT provided a low-resistance current path to ground. Therefore, the transistor successfully operated as a low-side switch for the LED branch.

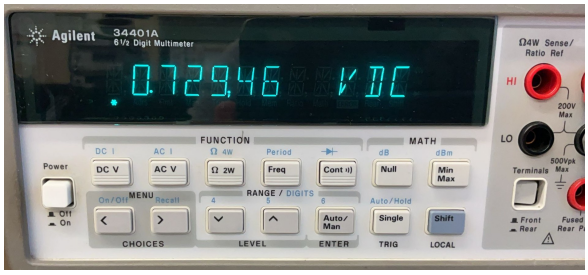
When the LED was turned OFF, the base-emitter voltage decreased to approximately

$$V_{BE,\mathrm{OFF}} = 3.253\,\mathrm{mV}.$$

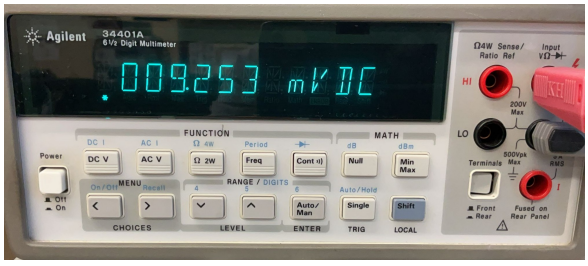
This confirms that the base-emitter junction was no longer forward biased and that the transistor returned to the cutoff state. The measured collector-emitter voltage in the OFF state was

$$V_{CE,\mathrm{OFF}} = 9.6227\,\mathrm{V}.$$

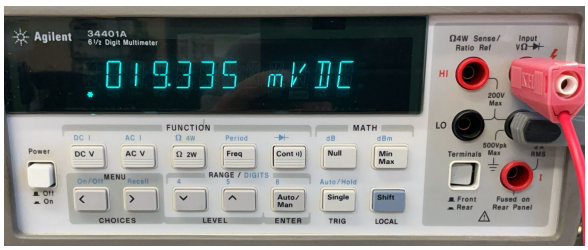
This high V_{CE} value is expected because the transistor no longer pulls the collector node toward ground when it is OFF.



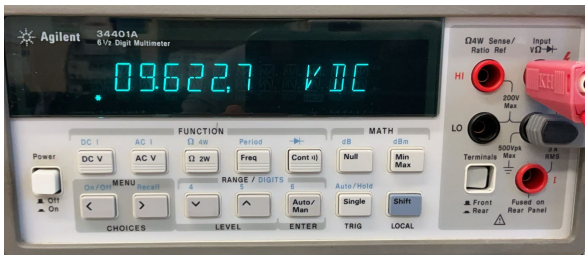
(a) V_{BE} in the LED ON state: $0.72946\,\mathrm{V}$.



(b) V_{BE} in the LED OFF state: $3.253\,\mathrm{mV}$.



(c) V_{CE} in the LED ON state: $19.335\,\mathrm{mV}$.



(d) V_{CE} in the LED OFF state: $9.6227\,\mathrm{V}$.

Figure 11.7: Experimental verification of the ON and OFF states of a representative BJT switching stage.

An important observation was that V_{BE} did not drop instantly when the LED stage was turned OFF. Instead, it decreased gradually. This behavior is caused by the debounce capacitor connected at the base node. When the probe signal is removed, the capacitor discharges through the pull-down resistance, so the base voltage follows an RC decay instead of falling immediately to zero:

$$V_B(t) = V_B(0)e^{-t/(R_{\mathrm{eq}}C_{\mathrm{DB}})}.$$

This gradual decrease is not a fault. It confirms the intended input stabilization behavior of the debounce capacitor. The capacitor helps suppress rapid contact fluctuations caused by water motion or unstable probe contact.

Overall, these measurements confirm the expected BJT switching behavior. In the ON state, the base-emitter junction is forward biased and V_{CE} drops close to zero. In the OFF state, V_{BE} is nearly zero and V_{CE} becomes high. Therefore, the transistor stages successfully operate as low-side LED switches in the physical circuit.

11.7 Astable Alarm Video Verification

The highest-level alarm behavior was also verified using a recorded video of the physical circuit. The purpose of this video evidence was to confirm two important points. First, the P_6 probe successfully enabled the astable multivibrator stage. Second, the alarm LED produced the intended blinking behavior.

A representative frame from the recorded video is shown in Figure 11.8. The full video

is not embedded as a figure in the PDF. Instead, it is included in the project file structure under the figures folder as

`figures/11.17.mp4`.

Therefore, the video can be opened and watched manually from the submitted project files.

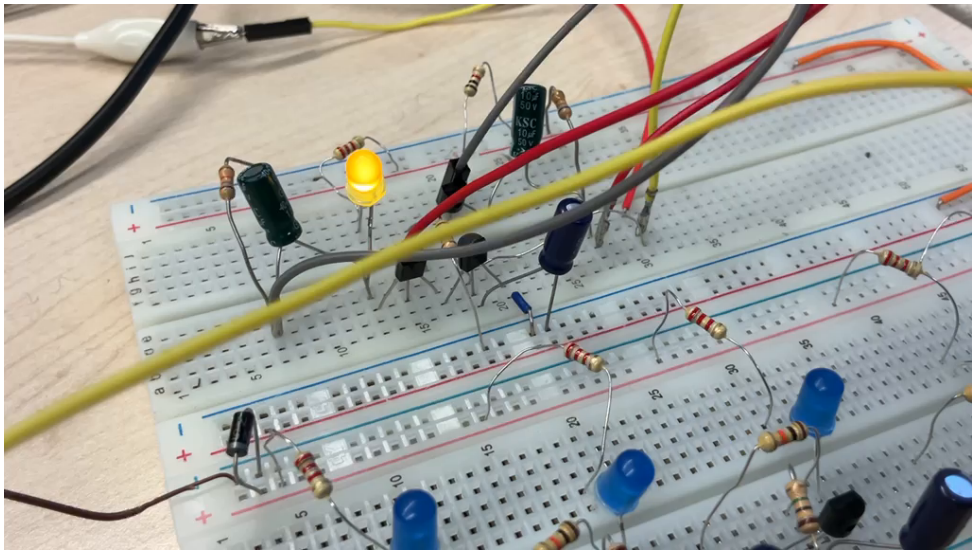


Figure 11.8: Representative frame from the physical alarm test video. The full video file is included separately as `figures/11.17.mp4`.

From the recorded video, the alarm LED was observed to blink 10 times in a time interval of $10\,\mathrm{s}$. Therefore, the measured blinking frequency is

$$f_{\mathrm{blink}} = \frac{N_{\mathrm{blink}}}{t} = \frac{10}{10\,\mathrm{s}} = 1\,\mathrm{Hz}.$$

This measured value is close to the theoretical astable frequency obtained from the selected timing components,

$$f_{\mathrm{theoretical}} \approx 1.06\,\mathrm{Hz}.$$

Thus, the video evidence confirms that the physical astable multivibrator produced the intended approximately $1\,\mathrm{Hz}$ blinking alarm output. It also verifies that the P_6 level successfully activates the alarm stage in the physical implementation.

11.8 Comparison Between Simulation and Physical Experiment

The physical experiment confirmed the main functional behavior predicted by the LTspice simulation. In the simulation, the water-contact process was represented by con-

trolled switches and finite water-path resistances. In the physical implementation, the same behavior was tested using real conductive probes and water contact. Therefore, exact numerical equality between simulation and experiment is not expected, but the main operating sequence should match.

The comparison between the LTspice results and the physical measurements is summarized in Table 11.4.

Table 11.4: Comparison between LTspice simulation and physical experiment.

Feature	**LTspice simulation**	**Physical experiment**
Supply voltage	$12\,\mathrm{V}$ source used	$V_{\mathrm{in}} = 12.01\,\mathrm{V}$ measured
Protected V_{CC} rail	Approximately $11.7\,\mathrm{V}$ expected	$V_{CC} = 11.7562\,\mathrm{V}$ measured
Schottky diode drop	Approximately $0.3\,\mathrm{V}$ assumed	$V_D = 0.2538\,\mathrm{V}$ measured
Sequential activation	P_1 to P_6 activated in order	P_1 to P_6 activated in order
P_1 to P_5 LEDs	Steady cumulative indication	Steady cumulative indication observed
P_6 alarm	Blinking alarm current observed	Blinking alarm LED observed
Blinking frequency	Approximately $1.06\,\mathrm{Hz}$ expected	Approximately $1.00\,\mathrm{Hz}$ measured
Supply current trend	Stepwise increase	$0.005\,\mathrm{A}$ to $0.038\,\mathrm{A}$ measured
LED current	Few-milliampere range expected	$3.91{-}4.09\,\mathrm{mA}$ measured
BJT ON behavior	V_{BE} forward biased, V_{CE} low	$V_{BE} = 0.72946\,\mathrm{V}$, $V_{CE} = 19.335\,\mathrm{mV}$
BJT OFF behavior	V_{BE} near zero, V_{CE} high	$V_{BE} = 3.253\,\mathrm{mV}$, $V_{CE} = 9.6227\,\mathrm{V}$

The measured supply voltage and protected rail voltage support the supply design assumption. The measured Schottky diode voltage drop was

$$V_D = 0.2538\,\mathrm{V},$$

which is close to the $0.3\,\mathrm{V}$ value used in the design calculations. Therefore, the protection diode did not significantly reduce the available circuit supply voltage.

The physical supply-current measurements also followed the same trend as the LTspice source-current result. As the water level activated more stages, the measured current increased from

$$0.005\,\mathrm{A}$$

at P_1 activation to

$$0.038\,\mathrm{A}$$

at P_6 activation. This stepwise increase is expected because each additional water level activates another LED branch, and the P_6 condition also enables the astable alarm stage.

The LED current measurements confirmed that the indicator LEDs operated in the intended low-current range. For the representative blue LED stage with

$$R_{\mathrm{LED}} = 2.2\,\mathrm{k\Omega},$$

the voltage-based calculation gave

$$I_{\mathrm{LED}} \approx 4.09\,\mathrm{mA},$$

while the direct ammeter measurement gave

$$I_{\mathrm{LED}} \approx 3.91\,\mathrm{mA}.$$

Both values are close to the intended few-milliampere LED operating range.

The BJT switching measurements also agree with the intended low-side switching behavior. In the ON state, the measured base-emitter voltage was

$$V_{BE,\mathrm{ON}} = 0.72946\,\mathrm{V},$$

which confirms that the base-emitter junction was forward biased. At the same time, the collector-emitter voltage decreased to

$$V_{CE,\mathrm{ON}} = 19.335\,\mathrm{mV},$$

showing that the transistor provided a low-resistance path to ground. In the OFF state,

$$V_{BE,\mathrm{OFF}} = 3.253\,\mathrm{mV},$$

and

$$V_{CE,\mathrm{OFF}} = 9.6227\,\mathrm{V}.$$

This confirms that the transistor returned to cutoff when the corresponding probe signal was removed.

The highest-level alarm behavior also matched the design expectation. The physical video showed 10 LED blinks in $10\,\mathrm{s}$, giving

$$f_{\text{blink}} = \frac{10}{10\,\text{s}} = 1.00\,\text{Hz}.$$

This is close to the theoretical astable frequency,

$$f_{\text{theoretical}} \approx 1.06\,\text{Hz}.$$

Therefore, the physical astable multivibrator produced the intended approximately $1\,\text{Hz}$ blinking alarm output.

Although the physical results match the simulation functionally, some differences are expected. The LTspice model uses controlled switches and a fixed modeled water resistance, while the real circuit depends on water conductivity, probe contact quality, breadboard contact resistance, component tolerances, and the non-ideal behavior of real BJTs, LEDs, capacitors, and resistors. Therefore, exact numerical agreement is not the main criterion. The important result is that both simulation and physical testing verify the same operating sequence:

water contact $\rightarrow$ BJT switching $\rightarrow$ steady LED indication $\rightarrow$ P_6 astable alarm $\rightarrow$ blinking output.

Overall, the physical experiment supports the LTspice verification. The implemented Version 3 circuit successfully produced sequential water-level indication for P_1 through P_5, activated the astable alarm at P_6, and operated within the expected low-current and low-power range.

11.9 Summary of Physical Testing

The physical testing confirmed the main design objectives. The protected supply rail was measured successfully, the supply current increased with sequential level activation, the LED branch operated in the expected milliampere range, and the measured V_{BE} and V_{CE} values verified that the BJTs operated as low-side switches.

The approximately $1\,\text{Hz}$ blinking frequency confirmed that the astable multivibrator operated as intended. Therefore, the physical implementation supported the LTspice results and verified the main functional behavior of the Version 3 circuit.

The main practical limitation remains the DC conductive sensing method. Since current passes through water and metal probes, long-term use may cause electrolysis and probe corrosion; therefore, a more robust future design would benefit from MOSFET-based sensing or AC-based probe excitation.

12 Proposed MOSFET-Based Improvement

The purpose of this section is to introduce the proposed MOSFET-based Version 4 circuit and explain why it was considered as an improvement direction. This section focuses on the structure and operating principle of the MOSFET-based design. A direct comparison between the implemented BJT-based Version 3 circuit and the proposed MOSFET-based Version 4 circuit is given in the following section.

12.1 Motivation for the MOSFET-Based Version

The implemented Version 3 circuit uses BC547B NPN BJTs as switching devices. This design satisfies the project requirements and was selected for the laboratory implementation. However, the BJT-based sensing method has one important practical limitation: the conductive water path must provide base current to turn on each transistor.

In the implemented BJT circuit, when water reaches a probe, current flows through the water path, the base resistor, and the base-emitter junction of the transistor. This current allows the BJT to switch the corresponding LED branch. Although this works for the modeled water resistance, it also means that DC current passes through the water and the metal probes during operation.

For a short laboratory demonstration, this is acceptable. However, for longer-term practical use, DC current through water can contribute to electrolysis and probe corrosion. Therefore, after identifying this limitation, a MOSFET-based version was considered as a possible improvement direction.

The main idea of Version 4 is to replace the BJT switching stages with N-channel MOSFET switching stages. A MOSFET is mainly controlled by gate-source voltage rather than continuous base current. Therefore, a MOSFET-based sensing stage can reduce the steady current required through the water path.

12.2 Proposed Version 4 Schematic

The proposed MOSFET-based Version 4 circuit is shown in Figure 12.1. In this version, the BJT switching stages are replaced by BS170 N-channel MOSFETs. The overall function of the circuit remains the same: water contact at a probe activates the corresponding switching stage and turns on the related LED indicator.

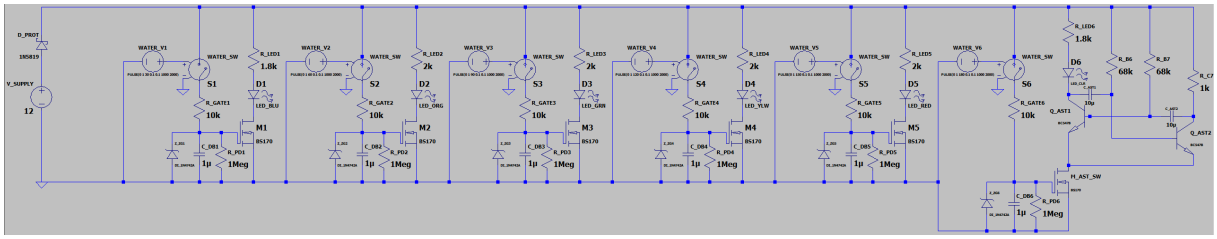


Figure 12.1: Proposed Version 4 MOSFET-based water level indicator circuit using BS170 N-channel MOSFET switching stages.

The schematic keeps the same general conductive-probe sensing philosophy as the implemented circuit. There is still a common supply probe and separate level probes placed at increasing heights. When water reaches a probe, it creates a conductive path that affects the control terminal of the corresponding switching device. The difference is that the controlled terminal is now the MOSFET gate instead of the BJT base.

12.3 Operation of the MOSFET Sensing Stage

In the proposed Version 4 circuit, each BS170 MOSFET is used as a low-side switch. The source terminal is connected to ground, the drain terminal is connected to the LED branch, and the gate terminal is controlled by the conductive probe sensing network.

When the water level is below a probe, the corresponding gate is held near ground by the pull-down resistor. Therefore, the gate-source voltage is approximately

$$V_{GS} \approx 0.$$

In this condition, the MOSFET remains off. Since the drain-source path does not conduct, the LED branch is open and the corresponding LED stays off.

When water reaches a probe, a conductive path is formed between the common supply probe and the level probe. This raises the gate voltage of the corresponding MOSFET. If the gate-source voltage becomes large enough, the MOSFET turns on and completes the LED current path:

$$V_{CC} \rightarrow R_{\text{LED}} \rightarrow D_n \rightarrow \text{MOSFET drain} \rightarrow \text{MOSFET source} \rightarrow \text{ground}.$$

Thus, the external indication behavior remains similar to the implemented version: when the water reaches a level, the corresponding LED turns on. The internal control method, however, changes from base-current drive to gate-voltage control.

12.4 Gate Pull-Down and Input Filtering

The MOSFET gate is a high-impedance node. This is useful because it reduces steady sensing current, but it also creates a practical problem: if the gate is left undefined, it can float. A floating MOSFET gate can accumulate charge or respond to noise, which may cause false switching.

For this reason, the proposed MOSFET design keeps a pull-down resistor at each gate node. The pull-down resistor defines the off-state by connecting the gate to ground when the probe is dry. Therefore, when there is no water contact,

$$V_G \approx 0, \qquad V_{GS} \approx 0,$$

and the MOSFET remains off.

The filtering capacitor is also kept at the input node. Its purpose is to reduce sudden gate-voltage changes caused by unstable water contact, splashing, small waves, or brief probe contact interruptions. Therefore, the same practical stabilization idea used in the BJT version is preserved, but it is now applied to a MOSFET gate node.

The input network can be interpreted as having two practical roles:

pull-down resistor	capacitor
defines the dry-probe state	smooths rapid contact changes

.

12.5 Gate-Source Voltage Protection

One additional design issue appears when MOSFETs are used. The MOSFET gate is insulated from the channel by a thin oxide layer, so excessive gate-source voltage can damage the device. Therefore, the proposed Version 4 design includes gate-source protection using reverse-connected zener diodes.

The purpose of the zener diode is to limit the voltage between the gate and source. If the gate voltage rises beyond the intended safe range, the zener diode provides a clamping path and helps prevent excessive V_{GS}.

This protection is especially important because the sensing input is connected to conductive probes and water. In a practical environment, long wires, accidental contact, electrostatic charge, or unexpected transients may appear at the sensing node. The zener diode makes the MOSFET gate network more robust against these effects.

Therefore, in the proposed MOSFET version, the input stage contains three important supporting elements:

$$\begin{array}{c}\text{gate pull-down}\\ \text{for a defined off-state}\end{array} \rightarrow \begin{array}{c}\text{input capacitor}\\ \text{for contact smoothing}\end{array} \rightarrow \begin{array}{c}\text{zener protection}\\ \text{for limiting } V_{GS}\end{array}.$$

12.6 Expected Function of the MOSFET Version

The expected function of the MOSFET-based circuit is the same at the user-output level. The first water levels would turn on their corresponding LEDs as steady indicators, and the highest level could be connected to the alarm stage to produce a blinking warning indication.

The main expected improvement is not a different visual output, but a better sensing interface. Since the MOSFET gate requires much smaller steady current than a BJT base, the water path is expected to carry less continuous DC current. This makes the MOSFET-based version a more suitable direction for reducing electrolysis and probe corrosion in longer-term use.

However, Version 4 is still a conductive-probe design. It still relies on electrical contact through water, and it still uses a DC-related sensing method. Therefore, it should be understood as an improvement over the implemented design, not as a complete solution to all practical water-sensing limitations.

12.7 Implementation Status of Version 4

The MOSFET-based Version 4 circuit was not physically implemented in this project because the required MOSFET components were not available for the laboratory build. For this reason, Version 3 remains the implemented and simulated main design of the report.

Version 4 is included as a proposed improvement because it follows naturally from the limitations observed in the BJT-based sensing method. It shows how the design could be developed further if suitable MOSFET components were available.

Therefore, the role of Version 4 in this project is not to replace the implemented circuit, but to document a realistic improvement path. The implemented Version 3 design satisfies the given project requirements, while the proposed Version 4 design addresses one of its most important practical weaknesses: continuous sensing current through the water path.

13 Comparison Between Implemented BJT Design and Proposed MOSFET Design

The previous section introduced the MOSFET-based Version 4 circuit as a proposed improvement direction. This section compares the implemented BJT-based Version 3 circuit and the proposed MOSFET-based Version 4 circuit in terms of control mechanism, sensing current, water-conductivity dependence, corrosion risk, protection requirements, and implementation status.

The purpose of this section is not to re-explain the full MOSFET circuit operation. Instead, the aim is to clarify what changes when the switching device is changed from a BJT to a MOSFET, and why this change may be useful for a more practical long-term water-level indicator.

13.1 Control Mechanism

The main difference between the two designs is the control mechanism of the switching device. In the implemented Version 3 design, each sensing stage uses a BC547B NPN BJT. A BJT requires base current to control the collector-emitter current. Therefore, the water path must provide enough current to forward-bias the base-emitter junction and drive the transistor.

In the proposed Version 4 design, each sensing stage uses an N-channel MOSFET. A MOSFET is mainly controlled by gate-source voltage. Ideally, after the gate capacitance is charged, the gate does not require continuous DC current in the same way that a BJT base does.

This difference can be summarized as

$$\begin{array}{c}\text{BJT Version 3}\\ \text{base-current controlled}\end{array} \quad \text{versus} \quad \begin{array}{c}\text{MOSFET Version 4}\\ \text{gate-voltage controlled}\end{array}.$$

This control difference is the reason why the MOSFET version is considered an improvement direction.

13.2 Sensing Current and Water Conductivity

In the BJT-based design, the conductive water path is part of the base-current path. Therefore, the amount of current available to the BJT depends directly on the resistance of the water path. If the water is sufficiently conductive, enough base current is produced and the transistor turns on. If the water resistance is high, the base current decreases and the transistor may not turn on strongly.

For the implemented Version 3 simulation, the water path was modeled as

$$R_{\text{water}} = 100\,\text{k}\Omega.$$

Under this modeled condition, the available base current was sufficient for switching the LED branches. However, real water resistance can vary significantly depending on dissolved ions, probe spacing, probe surface condition, and contamination.

In the MOSFET-based design, the water path mainly affects the gate voltage rather than supplying continuous base current. Because the MOSFET gate has high input impedance, the required steady sensing current is expected to be much smaller. This makes the MOSFET design less dependent on continuous current flow through the water.

However, the MOSFET design is not completely independent of the sensing environment. The gate voltage still depends on the conductive path, leakage effects, noise, and the selected pull-down resistance. Therefore, the MOSFET version reduces the water-current problem, but it does not completely remove all water-sensing limitations.

13.3 Electrolysis and Probe Corrosion

One of the most important practical concerns in conductive water-level sensing is electrolysis and probe corrosion. In the implemented BJT version, DC current flows through the water and probes whenever a level is active. This current is small, but it is still chemically relevant because it passes through metal electrodes and an electrolyte-like medium.

For short-term laboratory testing, this is acceptable. The current levels are low, and the circuit is mainly intended as a discrete analog demonstration. However, for long-term practical use, continuous DC current through water can gradually contribute to electrolysis, corrosion, and changes in probe surface condition.

The proposed MOSFET version is expected to reduce this issue because the gate requires much smaller steady current than a BJT base. Therefore, less continuous current should flow through the water path. This makes the MOSFET version more suitable as a long-term improvement direction.

Nevertheless, the MOSFET version still uses conductive probes and DC-related sensing. Therefore, corrosion risk is reduced but not completely eliminated. A more advanced future design would require AC-based sensing or another method that avoids net DC current through the water.

13.4 Stability and Protection Requirements

Both designs require input stabilization. In the BJT version, the base node must not be left floating when the probe is dry. Therefore, a pull-down resistor is used to hold the base near ground. A debounce capacitor is also used to reduce rapid voltage changes caused by unstable water contact.

The MOSFET version also requires a pull-down resistor and filtering capacitor. In fact, gate stabilization is even more important for a MOSFET because the gate is a high-impedance node. If the MOSFET gate is left floating, it can hold charge or respond to noise, causing false switching.

The MOSFET version also requires additional gate-source protection. Since the MOSFET gate oxide can be damaged by excessive gate-source voltage, zener protection is included in the proposed Version 4 design. This makes the MOSFET input stage more protective, but also slightly more complex than the BJT input stage.

Thus, the MOSFET version improves sensing-current behavior, but it requires more careful gate protection and noise control.

13.5 Practical Implementation Status

The BJT-based Version 3 circuit is the implemented design in this report. It was selected because the required components were available, the design could be built using discrete analog components, and the LTspice simulation verified the intended operating behavior.

The MOSFET-based Version 4 circuit was not physically implemented in this project because the required MOSFET components were not available for the laboratory build. Therefore, its role in this report is different. Version 4 is not the tested implementation; it is a proposed improvement based on the limitations observed in the BJT version.

This distinction is important. The BJT version is the verified and implemented circuit. The MOSFET version is a design proposal that should be tested experimentally in future work before being treated as a final replacement.

13.6 Summary of the Comparison

Table 13.1 summarizes the main differences between the implemented BJT-based Version 3 design and the proposed MOSFET-based Version 4 design.

Criterion	Implemented BJT Version 3	Proposed MOSFET Version 4
Switching device	BC547B NPN BJT	BS170 N-channel MOSFET
Control mechanism	Controlled by base current.	Controlled mainly by gate-source voltage.
Input current requirement	Requires continuous base current when active.	Requires much smaller steady gate current.
Water-path role	Water path supplies BJT base current.	Water path mainly establishes MOSFET gate voltage.
Dependence on water conductivity	Stronger dependence, because low conductivity reduces base current.	Lower dependence on continuous current, but gate voltage still depends on sensing conditions.
Electrolysis and corrosion risk	Higher, because DC sensing current flows through water and probes.	Lower expected risk, because steady sensing current is reduced.
Floating input problem	Base node must be pulled down when the probe is dry.	Gate node must be pulled down carefully because it is high impedance.
Protection requirement	Base resistor and pull-down are sufficient for basic operation.	Gate pull-down and gate-source protection are more important.
Circuit complexity	Simpler and easier to build with available components.	More careful input protection is needed.
Implementation status	Simulated and selected for physical implementation.	Proposed improvement; not physically implemented in this project.
Main role in the project	Main implemented design satisfying the project requirements.	Future improvement direction for reducing sensing current and corrosion risk.

Table 13.1: Comparison between the implemented BJT-based Version 3 design and the proposed MOSFET-based Version 4 design.

The comparison shows that the implemented BJT-based Version 3 circuit is suitable for the current laboratory project because it is simple, available, discrete, and verified in LTspice. It satisfies the required water-level indication behavior and provides a blinking alarm at the highest level.

The proposed MOSFET-based Version 4 circuit is more suitable as a future improvement direction. Its main advantage is the reduction of continuous sensing current through the water path. This can reduce electrolysis and probe corrosion, which are

important practical concerns for long-term use.

Therefore, the two designs should not be interpreted as simply “good” and “bad.” They serve different roles in the project:

$$\begin{array}{c}\text{Version 3}\\ \text{implemented and verified}\end{array} \rightarrow \begin{array}{c}\text{satisfies current}\\ \text{project requirements}\end{array}$$

$$\begin{array}{c}\text{Version 4}\\ \text{proposed improvement}\end{array} \rightarrow \begin{array}{c}\text{better direction for}\\ \text{long-term durability}\end{array}.$$

Thus, Version 3 is the appropriate choice for the present implementation, while Version 4 represents a more practical direction for future development.

14 Real-World Applications

The implemented circuit can be used in situations where a simple, low-cost, and local visual indication of water level is sufficient. Its main purpose is not precision measurement, but threshold-based indication. In other words, the circuit does not measure the exact height of water continuously; instead, it shows which predefined level has been reached.

The output behavior is simple to interpret. The first five LEDs indicate normal water-level thresholds, while the sixth LED gives a blinking indication for the highest level. Therefore, the circuit is most suitable for small-scale monitoring, warning, and educational applications.

14.1 Small Water Tank Level Indication

A direct application of the circuit is small water tank level indication. The probes can be placed at different heights inside a container, and each LED can represent one approximate water-level threshold. As the water rises and reaches higher probes, more LEDs turn on.

This type of indication is useful when the user only needs a qualitative or stepwise estimate of the water level. For example, the LEDs can be interpreted as

$$\underset{\text{low level}}{D_1} \rightarrow \underset{\text{middle level}}{D_3} \rightarrow \underset{\text{nearly full}}{D_5} \rightarrow \underset{\text{critical level}}{D_6}.$$

This makes the circuit appropriate for simple containers, demonstration tanks, and prototype storage systems where exact numerical measurement is not required.

14.2 Overflow Warning

The sixth level can be used as a simple overflow warning. Unlike the first five LEDs, the final LED is designed to blink when the highest probe is reached. This makes the critical level visually different from the normal indication levels.

The distinction can be summarized as

$$\underset{\text{normal levels}}{P_1\text{–}P_5} \rightarrow \underset{\text{indication}}{\text{steady LED}} \qquad \underset{\text{highest level}}{P_6} \rightarrow \underset{\text{warning}}{\text{blinking LED}}.$$

This is useful when a user must notice that the water has reached a maximum allowed level. In a future version, the blinking LED output could also be replaced or extended by another warning device, such as a buzzer or relay driver.

14.3 Educational Circuit Demonstration

The circuit is especially suitable as an educational laboratory project because it combines several basic circuit ideas in one physical system. It demonstrates conductive sensing, transistor switching, LED current limiting, pull-down resistors, debounce capacitors, reverse-polarity protection, and astable multivibrator operation.

Therefore, the circuit is useful not only as a water level indicator, but also as a teaching example. It shows how a physical event,

water touching a probe,

can be converted into an electrical signal and then into a visible output.

This makes the design appropriate for introductory electronics laboratories, circuit theory demonstrations, and projects where the goal is to understand discrete analog behavior rather than hide the sensing process inside a ready-made module or microcontroller.

14.4 Low-Cost Prototype Monitoring

Another possible use is low-cost prototype monitoring. Since the circuit uses common discrete components, it can be built without a microcontroller, ADC, display, or software interface. This makes it useful when the required output is only local visual indication.

The design is also modular. Additional threshold levels can be added by repeating the sensing and switching stage, while fewer levels can be used if the application is simpler. This makes the circuit adaptable as a first prototype before moving to a more advanced sensing system.

14.5 Application Scope

The realistic application scope of this circuit is simple threshold-based indication. It is suitable for educational demonstrations, laboratory prototypes, and small low-cost monitoring tasks.

It should not be treated as a final industrial liquid-level measurement system without further development. Industrial or long-term applications would require additional design work, stronger mechanical construction, more robust sensing methods, and proper safety isolation. These practical and design limitations are discussed in the following sections.

15 Practical Implementation Considerations

The physical behavior of the circuit depends not only on the schematic, but also on the way the circuit is built and tested. Probe placement, water contact, breadboard wiring, component orientation, and power supply setup can all affect the observed result. Therefore, the circuit should be implemented gradually and checked stage by stage.

15.1 Probe Placement and Water Contact

The conductive probes must be placed at different heights inside the container. The common supply probe should extend low enough so that water can form a conductive path between the common probe and any reached level probe.

The probes should be fixed mechanically and should not touch each other directly. If two probes make direct metal-to-metal contact, the circuit may falsely detect water even when the water level has not reached that height.

The intended physical sensing action is

$$\begin{array}{c}\text{water reaches}\\ \text{a level probe}\end{array} \rightarrow \begin{array}{c}\text{conductive path}\\ \text{forms through water}\end{array} \rightarrow \begin{array}{c}\text{corresponding stage}\\ \text{is activated}\end{array}.$$

Real water contact may be unstable near a probe because of small waves, splashing, or partial wetting. The debounce capacitors reduce rapid contact-induced voltage changes, but the probe positions should still be mechanically stable during testing.

15.2 Water Conductivity During Testing

The circuit relies on conductive sensing, so the water sample must be conductive enough to activate the BJT stages. Tap water is usually suitable for a laboratory demonstration because it contains dissolved ions. Very pure or distilled water may have much higher resistance and may not trigger the circuit reliably.

For this reason, the physical test should be performed using the actual water sample intended for the experiment. If an LED is dim, unstable, or does not turn on, the water path may not be providing enough base current. This should be checked before assuming that the LED branch or transistor stage is faulty.

15.3 Breadboard Construction and Debugging

Since the circuit contains repeated sensing stages, wiring errors are possible. Each level stage includes a probe input, base resistor, pull-down resistor, debounce capac-

itor, BJT, LED, and LED resistor. Therefore, the circuit should not be debugged only after the entire system is assembled.

A better construction strategy is

$$\begin{array}{c}\text{build one}\\ \text{sensing stage}\end{array} \rightarrow \begin{array}{c}\text{test its}\\ \text{LED response}\end{array} \rightarrow \begin{array}{c}\text{repeat for}\\ \text{other levels}\end{array}.$$

This makes it easier to identify wiring mistakes. If the full circuit is built at once and does not work, it becomes harder to determine whether the problem is caused by the supply, probes, BJT orientation, LED polarity, the astable stage, or a breadboard connection.

15.4 Component Orientation and Polarity

Several components must be connected with the correct orientation.

The BC547B transistor pinout must be checked before insertion into the breadboard. If the collector, base, and emitter are connected incorrectly, the transistor will not behave as the intended low-side switch.

The LEDs must also be connected with correct polarity. If an LED is reversed, it will not light even if the BJT stage is working correctly.

The 1N5819 Schottky diode must be connected in the correct direction. Under correct supply polarity, it should conduct from the external positive supply toward the protected V_{CC} rail. If it is reversed, the circuit will not receive the intended supply voltage.

The $1\,\mu$F, 50 V debounce capacitors are electrolytic capacitors, so their polarity matters. Since the BJT base node is positive with respect to ground during normal operation, the positive capacitor terminal should be connected to the base node and the negative terminal should be connected to ground.

15.5 Power Supply Setup

The circuit is designed to operate from a 12 V DC laboratory supply. Before powering the circuit, the supply voltage and polarity should be checked. It is also good practice to set a current limit during the first test.

The expected current is only a few tens of milliamps. Therefore, a current limit moderately above the expected operating current can help protect the circuit if a wiring error or short circuit exists.

A safe power-up sequence is

$$\begin{matrix}\text{set } 12\,\text{V} \\ \text{supply voltage}\end{matrix} \rightarrow \begin{matrix}\text{set current} \\ \text{limit}\end{matrix} \rightarrow \begin{matrix}\text{check wiring} \\ \text{and polarity}\end{matrix} \rightarrow \begin{matrix}\text{power the} \\ \text{circuit}\end{matrix}.$$

The reverse-polarity protection diode provides an additional safety layer, but it should not replace careful wiring and supply checking.

15.6 Testing Procedure

The circuit should be tested in the same order as its intended water-level operation. First, the protected supply rail V_{CC} should be measured to confirm that the supply and protection diode are connected correctly.

Then the water level should be raised gradually. When water reaches P_1, only D_1 should turn on. When water reaches P_2, D_1 and D_2 should be on. This sequence should continue up to P_5. When water reaches P_6, the final LED should blink instead of remaining steadily on.

The expected test sequence is

$$\text{below } P_1 \rightarrow \begin{matrix}\text{all LEDs} \\ \text{off}\end{matrix}, \quad \begin{matrix}P_1\text{–}P_5 \\ \text{reached}\end{matrix} \rightarrow \begin{matrix}D_1\text{–}D_5 \\ \text{steady}\end{matrix}, \quad \begin{matrix}P_6 \\ \text{reached}\end{matrix} \rightarrow \begin{matrix}D_6 \\ \text{blinking}\end{matrix}.$$

If a stage does not behave as expected, the corresponding probe connection, base resistor, pull-down resistor, capacitor polarity, transistor orientation, LED polarity, and ground connection should be checked.

15.7 Summary of Practical Considerations

The practical implementation should be treated as a separate validation stage after simulation. The circuit should be built gradually, powered carefully, and tested level by level. The most important practical checks are stable probe placement, sufficient water conductivity, correct component orientation, reliable breadboard wiring, and safe power supply setup.

These considerations do not change the theoretical design, but they strongly affect whether the physical circuit behaves like the LTspice model.

16 Limitations and Possible Improvements

Although the implemented Version 3 circuit satisfies the main project requirements in simulation, it should be understood as a laboratory-level discrete analog prototype rather than a final industrial water-level sensor. The design successfully demonstrates conductive sensing, BJT switching, steady LED indication, and blinking alarm behavior. However, several limitations remain because the circuit relies on DC conductive sensing, real water contact, discrete component tolerances, and simplified LTspice models.

16.1 Dependence on Water Conductivity

The main limitation of the implemented BJT-based design is its dependence on water conductivity. In the LTspice simulation, the water path is represented by a fixed resistance,

$$R_{\text{water}} = 100\,\text{k}\Omega.$$

In a real system, this resistance is not fixed. It depends on dissolved ions, probe spacing, temperature, contamination, and probe surface condition. If the water resistance becomes much larger than the modeled value, the available BJT base current decreases. This can make the LED dim, unstable, or unable to turn on reliably.

The limitation can be summarized as

$$\begin{matrix}\text{higher water} \\ \text{resistance}\end{matrix} \rightarrow \begin{matrix}\text{lower BJT} \\ \text{base current}\end{matrix} \rightarrow \begin{matrix}\text{weaker switching} \\ \text{behavior}\end{matrix}.$$

A possible improvement is to use a higher-input-impedance sensing stage, such as the proposed MOSFET-based Version 4 design, or a comparator-based threshold detector.

16.2 DC Probe Current, Electrolysis, and Corrosion

The circuit uses DC conductive sensing. When water reaches a probe, DC current flows through the water and metal probes. Although this current is small, it can still be important during long-term use because it may contribute to electrolysis and probe corrosion.

For a short laboratory demonstration, this is acceptable. For a long-term installation, however, corrosion can change the effective probe surface and increase or destabilize the contact resistance. Therefore, the limiting issue is not mainly heating, because the simulated power is low. The more important issue is electrochemical degradation at the probes.

The proposed MOSFET version reduces the steady sensing current, but it does not completely eliminate the problem because it still uses conductive probes. A stronger improvement would be to use AC-based probe excitation, which reduces the net DC component through the water path.

16.3 Component Tolerance and Non-Ideal Behavior

The circuit uses real resistors, capacitors, LEDs, transistors, and diodes. Therefore, the physical circuit will not exactly match the calculated values. Resistor tolerance affects LED current and timing resistance. Capacitor tolerance affects debounce behavior and astable blinking frequency. LED forward voltage and brightness also vary with color, manufacturer, and device type.

For this reason, the calculated values should be interpreted as design estimates rather than exact predictions. For example, the expected astable frequency is approximately

$$f \approx 1.06\,\text{Hz},$$

but the real blinking rate may deviate because the timing capacitors and resistors are not ideal. This is acceptable for a visual alarm, but it would not be sufficient for a precision timing application.

16.4 Simulation Model Limitations

The LTspice simulation verifies the functional behavior of the circuit, but it does not fully model real water-contact physics. In the simulation, water contact is represented by voltage-controlled switches and fixed resistive paths. This is useful for checking sequential activation and alarm behavior, but it does not include all physical effects.

The simulation does not fully model:

- changing water conductivity,
- unstable water surface contact,
- bubbles or partial wetting,
- probe corrosion and electrolysis,
- breadboard contact resistance,
- long-wire noise pickup,
- long-term changes in probe surface condition.

Therefore, the simulation should be interpreted as functional verification, not complete real-world proof. Physical testing is still necessary to evaluate the actual behavior of the circuit under laboratory conditions.

16.5 Possible Version 5: AC-Based Probe Excitation

A possible future improvement is to replace DC conductive sensing with AC-based probe excitation. The goal would be to reduce net DC current through the water and probes.

The design motivation is

$$\begin{matrix}\text{DC probe}\\ \text{excitation}\end{matrix} \rightarrow \begin{matrix}\text{net DC current}\\ \text{through water}\end{matrix} \rightarrow \begin{matrix}\text{electrolysis and}\\ \text{corrosion risk}\end{matrix}.$$

With AC excitation, the polarity alternates, so the net DC component can be reduced. Such a design would require extra circuitry, such as an oscillator, coupling network, rectification or envelope detection, filtering, and threshold comparison. This would increase complexity, but it would make the sensing method more suitable for long-term use.

16.6 Other Possible Improvements

Several additional improvements could be considered in future versions:

- using MOSFET sensing stages to reduce steady sensing current,
- using comparators for sharper and more repeatable switching thresholds,
- using more corrosion-resistant probe materials,
- transferring the design from breadboard to PCB,
- adding a buzzer, relay, or pump-control output to the alarm stage,
- testing different water samples and calibrating the sensing thresholds.

These improvements would make the circuit more robust, but they would also increase complexity. Therefore, the present Version 3 design is best understood as a simple educational prototype, while the proposed improvements show possible directions for more practical future development.

16.7 Summary of Limitations and Improvements

The implemented circuit demonstrates the required discrete analog water-level indication behavior, but its main limitations are water-conductivity dependence, DC current through the probes, component tolerance, and idealized simulation modeling. The most important future improvements are MOSFET-based sensing, AC-based probe excitation, comparator-based threshold detection, and improved mechanical construction.

Thus, the current design is suitable for educational demonstration and laboratory prototyping. For long-term real-world use, the most important improvement would be to reduce or eliminate net DC current through the water path.

17 Conclusion

This report presented the design and LTspice simulation of a discrete analog conductive-probe-based water level indicator circuit. The implemented design, referred to as Version 3, uses a $12\,\text{V}$ DC supply, a 1N5819 Schottky diode for reverse-polarity protection, BC547B NPN BJT switching stages, LED indicators, input stabilization networks, and an astable multivibrator alarm stage.

The design satisfies the main functional requirement of detecting multiple water levels and displaying them visually. Although the minimum requirement was three detected levels, the selected design uses six sensing levels. The first five levels are indicated by steady LEDs, while the sixth and highest level activates a blinking LED alarm.

The LTspice simulation verified the intended operating sequence of the Version 3 circuit. The simulated water-contact triggers activated the sensing stages in order, the BJT base nodes reached the expected forward-bias region, the first five LED currents became steady after activation, and the sixth LED current became periodic after the alarm stage was enabled. The astable multivibrator waveforms confirmed that the blinking behavior was produced by internal oscillation.

The supply-side results were consistent with the design estimates. The protected supply rail remained close to the $12\,\text{V}$ input, the supply current stayed in the range of a few tens of milliamps, and the simulated power consumption remained below approximately $0.5\,\text{W}$. The component selection and datasheet-based verification also showed that the selected components operate within suitable voltage, current, and power limits for the simulated design.

The main limitation of the implemented circuit is its dependence on DC conductive sensing. The BJT stages require sensing current through the water path, so real performance depends on water conductivity, probe contact quality, and long-term probe condition. For this reason, the MOSFET-based Version 4 and AC-based Version 5 directions were discussed as possible future improvements for reducing sensing current and improving long-term durability.

Overall, the implemented Version 3 circuit satisfies the intended discrete analog water-level indication behavior in simulation. It detects multiple water-level thresholds, provides steady LED indication for the lower levels, produces a blinking alarm at the highest level, and remains within safe simulated operating ranges. Physical implementation and experimental testing are still required to evaluate the behavior of the circuit with real probes, real water contact, component tolerances, and practical laboratory conditions.

References

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3. RS Components / Keysight Technologies, *E3631A/32A/33A/34A Programmable DC Power Supply Specifications*. Used to verify the 0 to $+25$ V, 0 to 1 A output capability of the Agilent E3631A laboratory power supply. Available at: `https://ca.rs-online.com/m/d/69e02f6fa6915a2ac4203f14c300830f.pdf`

4. Analog Devices, *LTspice Circuit Simulator*. Used for transient simulation of the implemented Version 3 water level indicator circuit.

The datasheet excerpts used directly in the component verification process are included in Appendix F.

Appendix

Detailed Calculations, Datasheet Excerpts, and Full-Size Schematics

This part contains the detailed derivations, supporting calculations, component datasheet references, and full schematic figures used to support the main design report.

A Detailed LED Resistor Calculations

This appendix gives the detailed LED current-limiting resistor calculations used in Section 8. In the main report, only the selected values and final current estimates are presented. Here, the arithmetic steps are shown explicitly.

The LED current target is

$$I_{\mathrm{LED}} \approx 5\,\mathrm{mA}.$$

For each LED branch, the current path is

$$V_{CC,\mathrm{eff}} \rightarrow R_{\mathrm{LED}} \rightarrow \text{LED} \rightarrow \text{BJT collector-emitter path} \rightarrow \text{ground}.$$

Using Kirchhoff's voltage law, the voltage across the LED resistor is

$$V_R = V_{CC,\mathrm{eff}} - V_F - V_{CE(sat)}.$$

Therefore, by Ohm's law,

$$I_{\mathrm{LED}} = \frac{V_R}{R_{\mathrm{LED}}} = \frac{V_{CC,\mathrm{eff}} - V_F - V_{CE(sat)}}{R_{\mathrm{LED}}}.$$

Solving for the resistor gives

$$R_{\mathrm{LED}} = \frac{V_{CC,\mathrm{eff}} - V_F - V_{CE(sat)}}{I_{\mathrm{LED}}}.$$

The effective circuit supply is

$$V_{CC,\mathrm{eff}} \approx 11.7\,\mathrm{V},$$

and the transistor saturation voltage is approximated as

$$V_{CE(sat)} \approx 0.2\,\mathrm{V}.$$

A.1 Blue LED

For the blue LED, the assumed forward voltage is

$$V_F \approx 3.2\,\mathrm{V}.$$

The required LED resistor is

$$R_{\text{LED}} = \frac{11.7 - 3.2 - 0.2}{0.005}.$$

The numerator is

$$11.7 - 3.2 - 0.2 = 8.3\,\text{V}.$$

Therefore,

$$R_{\text{LED}} = \frac{8.3}{0.005} = 1660\,\Omega.$$

The selected standard value is

$$R_{\text{LED}} = 1.8\,\text{k}\Omega.$$

With this selected value, the actual estimated current becomes

$$I_{\text{LED}} = \frac{8.3}{1800} = 0.00461\,\text{A}.$$

Thus,

$$I_{\text{LED}} \approx 4.61\,\text{mA}.$$

This is slightly below $5\,\text{mA}$, but it is still suitable for visible low-current indication.

A.2 Orange LED

For the orange LED, the assumed forward voltage is

$$V_F \approx 2.0\,\text{V}.$$

The required resistor is

$$R_{\text{LED}} = \frac{11.7 - 2.0 - 0.2}{0.005}.$$

The numerator is

$$11.7 - 2.0 - 0.2 = 9.5\,\text{V}.$$

Therefore,

$$R_{\text{LED}} = \frac{9.5}{0.005} = 1900\,\Omega.$$

The selected standard value is

$$R_{\text{LED}} = 2.0\,\text{k}\Omega.$$

With this selected value,

$$I_{\text{LED}} = \frac{9.5}{2000} = 0.00475\,\text{A}.$$

Thus,

$$I_{\text{LED}} \approx 4.75\,\text{mA}.$$

A.3 Green LED

For the green LED, the assumed forward voltage is

$$V_F \approx 2.1\,\text{V}.$$

The required resistor is

$$R_{\text{LED}} = \frac{11.7 - 2.1 - 0.2}{0.005}.$$

The numerator is

$$11.7 - 2.1 - 0.2 = 9.4\,\text{V}.$$

Therefore,

$$R_{\text{LED}} = \frac{9.4}{0.005} = 1880\,\Omega.$$

The selected value is

$$R_{\text{LED}} = 2.0\,\text{k}\Omega.$$

With this selected value,

$$I_{\text{LED}} = \frac{9.4}{2000} = 0.00470\,\text{A}.$$

Thus,

$$I_{\text{LED}} \approx 4.70\,\text{mA}.$$

A.4 Yellow LED

For the yellow LED, the assumed forward voltage is

$$V_F \approx 2.0\,\text{V}.$$

This is the same forward-voltage assumption as the orange LED. Therefore,

$$R_{\text{LED}} \approx 1900\,\Omega.$$

The selected standard value is

$$R_{\text{LED}} = 2.0\,\text{k}\Omega.$$

The resulting current is

$$I_{\text{LED}} = \frac{9.5}{2000} = 0.00475\,\text{A}.$$

Thus,

$$I_{\text{LED}} \approx 4.75\,\text{mA}.$$

A.5 Red LED

For the red LED, the assumed forward voltage is

$$V_F \approx 1.9\,\text{V}.$$

The required resistor is

$$R_{\text{LED}} = \frac{11.7 - 1.9 - 0.2}{0.005}.$$

The numerator is

$$11.7 - 1.9 - 0.2 = 9.6\,\text{V}.$$

Therefore,

$$R_{\text{LED}} = \frac{9.6}{0.005} = 1920\,\Omega.$$

The selected value is

$$R_{\text{LED}} = 2.0\,\text{k}\Omega.$$

With this selected value,

$$I_{\text{LED}} = \frac{9.6}{2000} = 0.00480\,\text{A}.$$

Thus,

$$I_{\text{LED}} \approx 4.80\,\text{mA}.$$

A.6 Clear LED

The clear LED is treated as a high-forward-voltage LED in the calculation. Therefore,

$$V_F \approx 3.2\,\text{V}.$$

This gives the same calculation as the blue LED:

$$R_{\text{LED}} \approx 1660\,\Omega.$$

The selected value is

$$R_{\text{LED}} = 1.8\,\text{k}\Omega.$$

The resulting current is

$$I_{\text{LED}} = \frac{8.3}{1800} = 0.00461\,\text{A}.$$

Thus,

$$I_{\mathrm{LED}} \approx 4.61\,\mathrm{mA}.$$

If the clear LED is actually a clear package with a lower-forward-voltage internal die, its real current may be higher than this estimate. Therefore, the clear LED current should be checked during physical testing.

A.7 Summary of LED Resistor Calculations

LED Color	Assumed V_F	Selected R_{LED}	Estimated I_{LED}
Blue	$3.2\,\mathrm{V}$	$1.8\,\mathrm{k}\Omega$	$4.61\,\mathrm{mA}$
Orange	$2.0\,\mathrm{V}$	$2.0\,\mathrm{k}\Omega$	$4.75\,\mathrm{mA}$
Green	$2.1\,\mathrm{V}$	$2.0\,\mathrm{k}\Omega$	$4.70\,\mathrm{mA}$
Yellow	$2.0\,\mathrm{V}$	$2.0\,\mathrm{k}\Omega$	$4.75\,\mathrm{mA}$
Red	$1.9\,\mathrm{V}$	$2.0\,\mathrm{k}\Omega$	$4.80\,\mathrm{mA}$
Clear	$3.2\,\mathrm{V}$	$1.8\,\mathrm{k}\Omega$	$4.61\,\mathrm{mA}$

Table A.1: Detailed LED resistor calculation summary.

The calculated currents are close to the intended $5\,\mathrm{mA}$ target. The exact brightness of each LED may still differ because perceived brightness depends on LED color, efficiency, lens structure, and manufacturer.

A.8 LED Resistor Tolerance Effect

The resistors used in the circuit have 5% tolerance. For a $2.0\,\mathrm{k}\Omega$ resistor,

$$R_{\min} = 0.95(2.0\,\mathrm{k}\Omega) = 1.90\,\mathrm{k}\Omega,$$

and

$$R_{\max} = 1.05(2.0\,\mathrm{k}\Omega) = 2.10\,\mathrm{k}\Omega.$$

For the orange or yellow LED case, the resistor voltage is approximately

$$V_R = 9.5\,\mathrm{V}.$$

The maximum current occurs when resistance is minimum:

$$I_{\max} = \frac{9.5}{1.90\,\text{k}\Omega} = 5.00\,\text{mA}.$$

The minimum current occurs when resistance is maximum:

$$I_{\min} = \frac{9.5}{2.10\,\text{k}\Omega} = 4.52\,\text{mA}.$$

Therefore, due to resistor tolerance alone,

$$4.52\,\text{mA} \leq I_{\text{LED}} \leq 5.00\,\text{mA}.$$

This variation is acceptable for a simple LED indicator circuit.

B Detailed BJT Base-Current and Gain Calculations

This appendix gives the detailed base-current and gain calculations used to evaluate the BC547B switching stages. The purpose is to verify whether the conductive water path can provide enough base current to turn on the BJT and drive the LED branch.

B.1 Base Resistor Purpose

Each BJT base has a base resistor:

$$R_{\mathrm{BASE}} = 10\,\mathrm{k}\Omega.$$

The base-emitter junction of a BJT behaves similarly to a diode. Once it is forward biased, it can conduct significant current. Without a base resistor, excessive base current could flow and damage the transistor or disturb the circuit.

Therefore, the base resistor limits the base current and makes the switching behavior more controlled.

B.2 Base Current Calculation

When water reaches a probe, the base-current path is

$$V_{CC,\mathrm{eff}} \rightarrow \text{common supply probe} \rightarrow \begin{array}{c}\text{water path} \rightarrow P_n \rightarrow R_{\mathrm{BASE}} \\ \rightarrow \text{base-emitter junction} \rightarrow \text{ground}\end{array}.$$

The simulated water resistance is

$$R_{\mathrm{water}} = 100\,\mathrm{k}\Omega.$$

The BJT base-emitter voltage is approximated as

$$V_{BE} \approx 0.7\,\mathrm{V}.$$

Therefore, the voltage available across the water resistance and base resistor is

$$V_{\mathrm{available}} = V_{CC,\mathrm{eff}} - V_{BE}.$$

Using

$$V_{CC,\text{eff}} = 11.7\,\text{V}, \qquad V_{BE} = 0.7\,\text{V},$$

we get

$$V_{\text{available}} = 11.7 - 0.7 = 11.0\,\text{V}.$$

The total series resistance in the base path is

$$R_{\text{total}} = R_{\text{water}} + R_{\text{BASE}}.$$

Using

$$R_{\text{water}} = 100\,\text{k}\Omega, \qquad R_{\text{BASE}} = 10\,\text{k}\Omega,$$

we obtain

$$R_{\text{total}} = 110\,\text{k}\Omega.$$

Therefore,

$$I_B = \frac{V_{\text{available}}}{R_{\text{total}}} = \frac{11.0\,\text{V}}{110\,\text{k}\Omega} = 0.0001\,\text{A}$$

Thus,

$$I_B = 100\,\mu\text{A} = 0.1\,\text{mA}.$$

B.3 Required Current Gain

The LED current is approximately

$$I_{\text{LED}} \approx 5\,\text{mA}.$$

Since the LED current flows through the collector-emitter path of the BJT,

$$I_C \approx I_{\text{LED}}.$$

Therefore,

$$I_C \approx 5\,\text{mA}.$$

The required effective current gain is

$$\beta_{\text{required}} = \frac{I_C}{I_B}.$$

Substituting

$$I_C = 5\,\text{mA}, \qquad I_B = 100\,\mu\text{A},$$

we get

$$\beta_{\text{required}} = \frac{5\,\text{mA}}{100\,\mu\text{A}}.$$

Since

$$5\,\text{mA} = 5000\,\mu\text{A},$$

$$\beta_{\text{required}} = \frac{5000}{100} = 50.$$

Thus, the transistor must provide an effective current gain of approximately

$$\beta_{\text{required}} \approx 50.$$

This is lower than the reference h_{FE} value used from the BC547B datasheet, so the transistor is expected to switch the LED branch under the modeled water resistance.

B.4 Forced-Beta Saturation Check

For a conservative saturated-switch design, designers often use

$$\beta_{\text{forced}} \approx 10.$$

This means the transistor is driven as if its gain were only 10, even if the datasheet gain is larger. The purpose is to ensure strong saturation.

For

$$I_C = 5\,\text{mA},$$

the base current required for forced-β switching is

$$I_{B,\text{forced}} = \frac{I_C}{\beta_{\text{forced}}}.$$

Substituting,

$$I_{B,\text{forced}} = \frac{5\,\text{mA}}{10} = 0.5\,\text{mA}.$$

But the available base current through the modeled water path is

$$I_B = 0.1\,\text{mA}.$$

Therefore,

$$0.1\,\text{mA} < 0.5\,\text{mA}.$$

This means that the transistor can switch the LED under the modeled condition, but it should not be described as a strongly overdriven saturated switch for every possible water condition.

B.5 Effect of Water Resistance

The base current formula is

$$I_B = \frac{V_{CC,\text{eff}} - V_{BE}}{R_{\text{water}} + R_{\text{BASE}}}.$$

The numerator is approximately fixed:

$$V_{CC,\text{eff}} - V_{BE} = 11.7 - 0.7 = 11.0\,\text{V}.$$

Therefore, the main variable is R_{water}. If water resistance increases, base current decreases.

For example, if

$$R_{\text{water}} = 500\,\text{k}\Omega,$$

then

$$I_B = \frac{11.0}{500\,\text{k}\Omega + 10\,\text{k}\Omega} = \frac{11.0}{510\,\text{k}\Omega}.$$

Thus,

$$I_B = 21.6\,\mu\text{A}.$$

The required gain becomes

$$\beta_{\text{required}} = \frac{5\,\text{mA}}{21.6\,\mu\text{A}}.$$

Since

$$5\,\text{mA} = 5000\,\mu\text{A},$$

$$\beta_{\text{required}} = \frac{5000}{21.6} \approx 231.$$

This is much higher than the previous required gain of 50. Therefore, if the water is less conductive, the BJT may not turn on as strongly.

Therefore, the implemented BJT stage should be interpreted as a functional switching stage under the modeled water-resistance condition, not as a transistor stage guaranteed to be deeply saturated for all possible water samples. The use of $V_{CE(sat)} \approx 0.2\,\text{V}$ in the LED and power calculations is an approximate design assumption. A more exact verification would require measuring or simulating the actual collector-emitter voltage V_{CE} of the transistor during operation.

B.6 Can the Base Current Be Increased to $0.5\,\text{mA}$?

To obtain

$$I_B = 0.5\,\text{mA},$$

the required total resistance would be

$$R_{\text{total}} = \frac{V_{CC,\text{eff}} - V_{BE}}{I_B}.$$

Substituting,

$$R_{\text{total}} = \frac{11.0\,\text{V}}{0.5\,\text{mA}}.$$

Since

$$0.5\,\text{mA} = 0.0005\,\text{A},$$

$$R_{\text{total}} = \frac{11.0}{0.0005} = 22000\,\Omega.$$

Therefore,

$$R_{\text{total}} \approx 22\,\text{k}\Omega.$$

However, the modeled water resistance alone is already

$$R_{\text{water}} = 100\,\text{k}\Omega.$$

Thus, with the modeled water resistance, $0.5\,\text{mA}$ of base current cannot be obtained directly through the water path. This explains why the MOSFET-based Version 4 is considered a better improvement direction for reducing sensing-current dependence.

C Detailed RC Time-Constant Derivations

This appendix gives the detailed derivation of the input stabilization time constants. Each sensing node includes a pull-down resistor and a debounce capacitor. These elements reduce floating-node behavior and unstable triggering near the water-contact boundary.

C.1 Pull-Down Resistor Current

Each base node has a pull-down resistor

$$R_{\mathrm{PD}} = 1\,\mathrm{M}\Omega.$$

When the transistor is on, the base voltage is approximately

$$V_B \approx 0.7\,\mathrm{V}.$$

The current through the pull-down resistor is

$$I_{\mathrm{PD}} = \frac{V_B}{R_{\mathrm{PD}}}.$$

Substituting,

$$I_{\mathrm{PD}} = \frac{0.7\,\mathrm{V}}{1\,\mathrm{M}\Omega}.$$

Since

$$1\,\mathrm{M}\Omega = 1000000\,\Omega,$$

$$I_{\mathrm{PD}} = 0.0000007\,\mathrm{A}.$$

Thus,

$$I_{\mathrm{PD}} = 0.7\,\mu\mathrm{A}.$$

This is much smaller than the base current:

$$0.7\,\mu\mathrm{A} \ll 100\,\mu\mathrm{A}.$$

Therefore, the pull-down resistor does not significantly steal current from the base when water contact exists.

C.2 Turn-On Time Constant

When water reaches a probe, the debounce capacitor charges mainly through the water resistance and the base resistor. The approximate turn-on time constant is

$$\tau_{\text{on}} \approx (R_{\text{water}} + R_{\text{BASE}})C_{\text{DB}}.$$

Using

$$R_{\text{water}} = 100\,\text{k}\Omega,$$

$$R_{\text{BASE}} = 10\,\text{k}\Omega,$$

and

$$C_{\text{DB}} = 1\,\mu\text{F},$$

we get

$$\tau_{\text{on}} \approx (100\,\text{k}\Omega + 10\,\text{k}\Omega)(1\,\mu\text{F}).$$

Therefore,

$$\tau_{\text{on}} \approx (110\,\text{k}\Omega)(1\,\mu\text{F}).$$

Since

$$1\,\text{k}\Omega \cdot 1\,\mu\text{F} = 1\,\text{ms},$$

$$110\,\text{k}\Omega \cdot 1\,\mu\text{F} = 110\,\text{ms}.$$

Thus,

$$\tau_{\text{on}} \approx 0.11\,\text{s}.$$

This means that the base node does not respond to extremely short contact distur-

bances instantly. Instead, the voltage rises over a short but finite time.

C.3 More Exact Turn-On Time Constant

The simplified turn-on estimate ignores the pull-down resistor during charging. A more exact approximation includes it in parallel with the charging path:

$$\tau_{\text{on,exact}} \approx \left((R_{\text{water}} + R_{\text{BASE}}) \parallel R_{\text{PD}} \right) C_{\text{DB}}.$$

First,

$$R_{\text{water}} + R_{\text{BASE}} = 110\,\text{k}\Omega.$$

The equivalent resistance is

$$R_{\text{eq}} = \frac{(110\,\text{k}\Omega)(1\,\text{M}\Omega)}{110\,\text{k}\Omega + 1\,\text{M}\Omega}.$$

Using

$$1\,\text{M}\Omega = 1000\,\text{k}\Omega,$$

$$R_{\text{eq}} = \frac{(110)(1000)}{110 + 1000}\,\text{k}\Omega.$$

$$R_{\text{eq}} = \frac{110000}{1110}\,\text{k}\Omega.$$

Thus,

$$R_{\text{eq}} \approx 99.1\,\text{k}\Omega.$$

Therefore,

$$\tau_{\text{on,exact}} \approx (99.1\,\text{k}\Omega)(1\,\mu\text{F}) \approx 0.099\,\text{s}.$$

This is close to the simpler value $0.11\,\text{s}$. Therefore, the simplified estimate is acceptable for design explanation.

C.4 Turn-Off Time Constant

When water contact disappears, the capacitor discharges mainly through the pull-down resistor. Therefore,

$$\tau_{\text{off}} \approx R_{\text{PD}} C_{\text{DB}}.$$

Using

$$R_{\text{PD}} = 1\,\text{M}\Omega, \qquad C_{\text{DB}} = 1\,\mu\text{F},$$

we obtain

$$\tau_{\text{off}} \approx (1\,\text{M}\Omega)(1\,\mu\text{F}).$$

Since

$$1\,\text{M}\Omega \cdot 1\,\mu\text{F} = 1\,\text{s},$$

$$\tau_{\text{off}} \approx 1\,\text{s}.$$

This means that after a brief loss of water contact, the LED does not immediately turn off. This reduces flickering caused by small waves or unstable contact.

C.5 Low-Pass-Like Interpretation for Contact-Induced Fluctuations

The pull-down resistor and debounce capacitor can be interpreted as producing a low-pass-like behavior at the input node. However, this interpretation must be stated carefully. The circuit is not filtering the DC supply voltage. Instead, it is smoothing rapid base-node voltage variations caused by unstable water contact.

When the water surface moves near a probe, the electrical connection between the water and the probe may appear and disappear rapidly. This causes the base-node voltage to change rapidly. The capacitor opposes sudden voltage changes, so short disturbances are reduced.

For a simple RC low-pass estimate, the cutoff frequency can be written as

$$f_c = \frac{1}{2\pi RC}.$$

Using

$$R = 1\,\mathrm{M\Omega}, \qquad C = 1\,\mu\mathrm{F},$$

we get

$$f_c = \frac{1}{2\pi(1\,\mathrm{M\Omega})(1\,\mu\mathrm{F})}.$$

Since

$$(1\,\mathrm{M\Omega})(1\,\mu\mathrm{F}) = 1\,\mathrm{s},$$

$$f_c = \frac{1}{2\pi}.$$

Therefore,

$$f_c \approx 0.159\,\mathrm{Hz}.$$

This value gives a qualitative idea of why rapid disturbances are suppressed. However, the real input behavior is not a perfect linear RC filter because the BJT base-emitter junction clamps the base voltage near the forward-bias region, and the water contact is modeled as a switched resistive path. Therefore, the term “low-pass” should be understood as a practical description of input stabilization, not as an exact small-signal filter model.

D Detailed Astable Frequency and Tolerance Calculation

This appendix gives the detailed frequency calculation for the astable multivibrator alarm stage.

D.1 Nominal Frequency Calculation

For a symmetric BJT astable multivibrator, each half-period is approximately

$$t_{\frac{1}{2}} \approx 0.693RC.$$

A full period consists of two half-periods:

$$T \approx 2(0.693RC).$$

Therefore,

$$T \approx 1.386RC.$$

Frequency is the reciprocal of period:

$$f = \frac{1}{T}.$$

Thus,

$$f \approx \frac{1}{1.386RC}.$$

The selected timing values are

$$R = 68\,\mathrm{k}\Omega, \qquad C = 10\,\mu\mathrm{F}.$$

First calculate RC:

$$RC = (68\,\mathrm{k}\Omega)(10\,\mu\mathrm{F}).$$

Using

$$68\,\mathrm{k}\Omega = 68000\,\Omega,$$

and

$$10\,\mu\mathrm{F} = 10 \times 10^{-6}\,\mathrm{F},$$

$$RC = 68000(10 \times 10^{-6}).$$

Therefore,

$$RC = 0.68\,\mathrm{s}.$$

The period is

$$T \approx 1.386(0.68).$$

Thus,

$$T \approx 0.942\,\text{s}.$$

The frequency is

$$f \approx \frac{1}{0.942}.$$

Therefore,

$$f \approx 1.06\,\text{Hz}.$$

This means the alarm LED is expected to blink approximately once per second.

D.2 Tolerance Calculation

The blinking frequency depends on the timing resistor and capacitor values. The $68\,\text{k}\Omega$ resistors have 5% tolerance:

$$R_{\min} = 0.95(68\,\text{k}\Omega) = 64.6\,\text{k}\Omega,$$

$$R_{\max} = 1.05(68\,\text{k}\Omega) = 71.4\,\text{k}\Omega.$$

Assuming the $10\,\mu\text{F}$ capacitor may vary by approximately 20%,

$$C_{\min} = 0.80(10\,\mu\text{F}) = 8\,\mu\text{F},$$

$$C_{\max} = 1.20(10\,\mu\text{F}) = 12\,\mu\text{F}.$$

The minimum period occurs when both R and C are minimum:

$$T_{\min} \approx 1.386(64.6\,\text{k}\Omega)(8\,\mu\text{F}).$$

First,

$$64.6\,\text{k}\Omega \cdot 8\,\mu\text{F} = 0.5168\,\text{s}.$$

Therefore,

$$T_{\min} \approx 1.386(0.5168) = 0.716\,\text{s}.$$

The maximum frequency is

$$f_{\max} = \frac{1}{T_{\min}} = \frac{1}{0.716}.$$

Thus,

$$f_{\max} \approx 1.40\,\text{Hz}.$$

The maximum period occurs when both R and C are maximum:

$$T_{\max} \approx 1.386(71.4\,\text{k}\Omega)(12\,\mu\text{F}).$$

First,

$$71.4\,\text{k}\Omega \cdot 12\,\mu\text{F} = 0.8568\,\text{s}.$$

Therefore,

$$T_{\max} \approx 1.386(0.8568) = 1.188\,\text{s}.$$

The minimum frequency is

$$f_{\min} = \frac{1}{T_{\max}} = \frac{1}{1.188}.$$

Thus,

$$f_{\min} \approx 0.84\,\text{Hz}.$$

Therefore, the expected blinking frequency range is

$$0.84\,\text{Hz} \lesssim f \lesssim 1.40\,\text{Hz}.$$

This range is still acceptable for a human-visible blinking alarm.

E Detailed Current and Power Dissipation Calculations

This appendix provides the detailed current and power dissipation calculations summarized in Section 8.

E.1 Estimated Total Current Consumption

When the first five LEDs are on, their total current is approximately

$$I_{D1-D5} \approx 5(5\,\text{mA}) = 25\,\text{mA}.$$

The sixth LED blinks. Its instantaneous current is approximately

$$I_{D6,\text{instant}} \approx 5\,\text{mA}.$$

Assuming approximately 50% duty cycle,

$$I_{D6,\text{avg}} \approx 0.5(5\,\text{mA}) = 2.5\,\text{mA}.$$

The base current per sensing transistor is approximately

$$I_B \approx 100\,\mu\text{A}.$$

If six sensing paths are active,

$$I_{B,\text{total}} \approx 6(100\,\mu\text{A}) = 600\,\mu\text{A} = 0.6\,\text{mA}.$$

Therefore, the LED and sensing-base current is approximately

$$I_{\text{avg}} \approx 25\,\text{mA} + 2.5\,\text{mA} + 0.6\,\text{mA}.$$

Thus,

$$I_{\text{avg}} \approx 28.1\,\text{mA}.$$

The astable multivibrator also draws current. Therefore, a practical estimate for total current is

$$I_{\text{total}} \approx 30\,\text{mA} - 40\,\text{mA}.$$

E.2 Estimated Total Power from the Supply

The total input power after the protection diode can be estimated by

$$P_{\text{total}} \approx V_{CC,\text{eff}} I_{\text{total}}.$$

Using

$$V_{CC,\text{eff}} \approx 11.7\,\text{V},$$

and the lower current estimate

$$I_{\text{total,min}} = 30\,\text{mA} = 0.030\,\text{A},$$

we get

$$P_{\text{total,min}} \approx (11.7)(0.030) = 0.351\,\text{W}.$$

Using the upper current estimate

$$I_{\text{total,max}} = 40\,\text{mA} = 0.040\,\text{A},$$

we get

$$P_{\text{total,max}} \approx (11.7)(0.040) = 0.468\,\text{W}.$$

Therefore,

$$0.35\,\text{W} \lesssim P_{\text{total}} \lesssim 0.47\,\text{W}.$$

E.3 LED Resistor Power Dissipation

For a $2.0\,\text{k}\Omega$ LED resistor carrying

$$I_{\text{LED}} = 4.75\,\text{mA},$$

the power is

$$P_R = I^2 R.$$

Substituting,

$$P_R = (0.00475)^2(2000).$$

Thus,

$$P_R \approx 0.045\,\mathrm{W} = 45\,\mathrm{mW}.$$

For a $1.8\,\mathrm{k}\Omega$ LED resistor carrying

$$I_{\mathrm{LED}} = 4.61\,\mathrm{mA},$$

$$P_R = (0.00461)^2(1800).$$

Thus,

$$P_R \approx 0.038\,\mathrm{W} = 38\,\mathrm{mW}.$$

Both are safely below a typical $0.25\,\mathrm{W}$ resistor rating.

E.4 LED Power Dissipation

LED power is

$$P_{\mathrm{LED}} = V_F I_{\mathrm{LED}}.$$

For a blue LED,

$$V_F \approx 3.2\,\mathrm{V}, \qquad I_{\mathrm{LED}} \approx 4.61\,\mathrm{mA}.$$

Therefore,

$$P_{\mathrm{LED}} \approx (3.2)(0.00461) = 0.0148\,\mathrm{W}.$$

Thus,

$$P_{\mathrm{LED}} \approx 14.8\,\mathrm{mW}.$$

For a red LED,

$$V_F \approx 1.9\,\text{V}, \qquad I_{\text{LED}} \approx 4.80\,\text{mA}.$$

Therefore,

$$P_{\text{LED}} \approx (1.9)(0.00480) = 0.00912\,\text{W}.$$

Thus,

$$P_{\text{LED}} \approx 9.12\,\text{mW}.$$

E.5 BJT Power Dissipation

When a BJT is on,

$$P_Q \approx V_{CE(sat)} I_C.$$

Using

$$V_{CE(sat)} \approx 0.2\,\text{V}, \qquad I_C \approx 5\,\text{mA},$$

we get

$$P_Q \approx (0.2)(0.005) = 0.001\,\text{W}.$$

Thus,

$$P_Q \approx 1\,\text{mW}.$$

This is very small for the BC547B.

E.6 Base Resistor Power Dissipation

The base resistor carries approximately

$$I_B = 100\,\mu\text{A} = 0.0001\,\text{A}.$$

With

$$R_{\text{BASE}} = 10\,\text{k}\Omega = 10000\,\Omega,$$

the power is

$$P_{R_{\mathrm{BASE}}} = I_B^2 R_{\mathrm{BASE}}.$$

Therefore,

$$P_{R_{\mathrm{BASE}}} = (0.0001)^2(10000) = 0.0001\,\mathrm{W}.$$

Thus,

$$P_{R_{\mathrm{BASE}}} = 0.1\,\mathrm{mW}.$$

E.7 Pull-Down Resistor Power Dissipation

The pull-down resistor power is

$$P_{R_{\mathrm{PD}}} = \frac{V_B^2}{R_{\mathrm{PD}}}.$$

Using

$$V_B \approx 0.7\,\mathrm{V}, \qquad R_{\mathrm{PD}} = 1\,\mathrm{M}\Omega,$$

$$P_{R_{\mathrm{PD}}} = \frac{(0.7)^2}{1000000}.$$

Thus,

$$P_{R_{\mathrm{PD}}} = 0.00000049\,\mathrm{W}.$$

Therefore,

$$P_{R_{\mathrm{PD}}} = 0.49\,\mu\mathrm{W}.$$

This is negligible.

E.8 Water Path Power Dissipation

The modeled water path carries approximately the base current:

$$I_{\text{water}} \approx I_B \approx 100\,\mu\text{A}.$$

With

$$R_{\text{water}} = 100\,\text{k}\Omega,$$

the modeled water-path power is

$$P_{\text{water}} = I_{\text{water}}^2 R_{\text{water}}.$$

Substituting,

$$P_{\text{water}} = (0.0001)^2(100000).$$

Thus,

$$P_{\text{water}} = 0.001\,\text{W} = 1\,\text{mW}.$$

This is thermally small, but chemically relevant because the current is DC and flows through water and metal probes.

E.9 Protection Diode Power Dissipation

The protection diode carries the total circuit current. Its power is

$$P_D = V_D I_{\text{total}}.$$

Using

$$V_D \approx 0.3\,\text{V}, \qquad I_{\text{total}} \approx 40\,\text{mA} = 0.040\,\text{A},$$

$$P_D \approx (0.3)(0.040) = 0.012\,\text{W}.$$

Therefore,

$$P_D \approx 12\,\text{mW}.$$

E.10 Astable Timing Resistor Power Dissipation

A conservative estimate assumes nearly the full effective supply across one $68\,\mathrm{k}\Omega$ timing resistor:

$$P = \frac{V^2}{R}.$$

Using

$$V = 11.7\,\mathrm{V}, \qquad R = 68\,\mathrm{k}\Omega,$$

$$P = \frac{(11.7)^2}{68000}.$$

Since

$$(11.7)^2 = 136.89,$$

$$P = \frac{136.89}{68000} \approx 0.0020\,\mathrm{W}.$$

Thus,

$$P \approx 2.0\,\mathrm{mW}.$$

There are two $68\,\mathrm{k}\Omega$ timing resistors, so

$$P_{68k,\mathrm{total}} \approx 2(2.0\,\mathrm{mW}) = 4.0\,\mathrm{mW}.$$

E.11 Astable Collector Resistor Power Dissipation

For the $1\,\mathrm{k}\Omega$ astable collector resistor, the instantaneous current is estimated as

$$I_{RC} \approx \frac{V_{CC,\mathrm{eff}} - V_{CE(sat)}}{R_C}.$$

Using

$$V_{CC,\mathrm{eff}} = 11.7\,\mathrm{V}, \qquad V_{CE(sat)} = 0.2\,\mathrm{V}, \qquad R_C = 1\,\mathrm{k}\Omega,$$

$$I_{RC} \approx \frac{11.7 - 0.2}{1000} = 0.0115\,\text{A}.$$

Thus,

$$I_{RC} = 11.5\,\text{mA}.$$

The instantaneous resistor power is

$$P_{RC,\text{instant}} = I_{RC}^2 R_C.$$

Therefore,

$$P_{RC,\text{instant}} = (0.0115)^2(1000).$$

Thus,

$$P_{RC,\text{instant}} \approx 0.132\,\text{W}.$$

Assuming approximately 50% duty cycle,

$$P_{RC,\text{avg}} \approx 0.5(0.132\,\text{W}) = 0.066\,\text{W}.$$

Therefore,

$$P_{RC,\text{avg}} \approx 66\,\text{mW}.$$

This is below a $0.25\,\text{W}$ resistor rating, but it is not negligible compared with many other resistors in the circuit.

E.12 Summary of Component Power Dissipations

Component	Estimated Power Dissipation	Interpretation
$2.0\,\text{k}\Omega$ LED resistor	$45\,\text{mW}$ each	Safe for a typical $0.25\,\text{W}$ resistor.
$1.8\,\text{k}\Omega$ LED resistor	$38\,\text{mW}$ each	Safe for a typical $0.25\,\text{W}$ resistor.
Blue / clear LED	$14.8\,\text{mW}$ each when on	Low-power indicator operation.
Red LED	$9.1\,\text{mW}$ when on	Low-power indicator operation.
BC547B transistor	$1\,\text{mW}$ each when on	Very small transistor dissipation.
$10\,\text{k}\Omega$ base resistor	$0.1\,\text{mW}$ each	Negligible power dissipation.
$1\,\text{M}\Omega$ pull-down resistor	$0.49\,\mu\text{W}$ each	Extremely small.
Modeled water path	$1\,\text{mW}$ each active path	Thermally small, but chemically relevant.
1N5819 protection diode	$12\,\text{mW}$	Safe for expected current.
$68\,\text{k}\Omega$ astable timing resistor	$2\,\text{mW}$ each	Safe for a typical $0.25\,\text{W}$ resistor.
$1\,\text{k}\Omega$ astable collector resistor	$66\,\text{mW}$ average	Safe, but not negligible.

Table E.1: Detailed estimated power dissipation values.

E.13 Summed Component Power Check

At the highest water level, the first five LEDs are on continuously and the sixth LED blinks. Assuming 50% duty cycle for D_6, the LED resistor powers are

$$P_{RLED,\text{total}} \approx 38 + 45 + 44 + 45 + 46 + 0.5(38) \quad \text{mW}.$$

Therefore,

$$P_{RLED,\text{total}} \approx 38 + 45 + 44 + 45 + 46 + 19 = 237\,\text{mW}.$$

The LED powers are approximately

$$P_{LED,\text{total}} \approx 14.8 + 9.5 + 9.9 + 9.5 + 9.1 + 0.5(14.8) \quad \text{mW}.$$

Therefore,

$$P_{LED,\text{total}} \approx 14.8 + 9.5 + 9.9 + 9.5 + 9.1 + 7.4 = 60.2\,\text{mW}.$$

For practical reporting,

$$P_{LED,\text{total}} \approx 60\,\text{mW}.$$

The transistor power for the LED switching stages is approximately

$$P_{Q,\text{total}} \approx 5(1\,\text{mW}) + 0.5(1\,\text{mW}) = 5.5\,\text{mW}.$$

The six base resistors dissipate approximately

$$P_{RBASE,\text{total}} = 6(0.1\,\text{mW}) = 0.6\,\text{mW}.$$

The six pull-down resistors dissipate approximately

$$P_{RPD,\text{total}} = 6(0.49\,\mu\text{W}) = 2.94\,\mu\text{W}.$$

Since

$$2.94\,\mu\text{W} = 0.00294\,\text{mW},$$

this is negligible compared with the LED branch powers.

The six modeled water paths dissipate approximately

$$P_{\text{water,total}} = 6(1\,\text{mW}) = 6\,\text{mW}.$$

The protection diode dissipates approximately

$$P_D \approx 12\,\text{mW}.$$

The two $68\,\text{k}\Omega$ astable timing resistors dissipate approximately

$$P_{68k,\text{total}} \approx 4\,\text{mW}.$$

The $1\,\text{k}\Omega$ astable collector resistor dissipates approximately

$$P_{RC,\text{avg}} \approx 66\,\text{mW}.$$

Therefore, the summed component power is approximately

$$P_{\text{sum}} \approx 237 + 60 + 5.5 + 0.6 + 0.003 + 6 + 12 + 4 + 66 \quad \text{mW}.$$

Thus,

$$P_{\text{sum}} \approx 391.1\,\text{mW}.$$

Therefore,

$$P_{\text{sum}} \approx 0.39\,\text{W}.$$

This lies inside the supply-side estimate

$$0.35\,\text{W} \lesssim P_{\text{total}} \lesssim 0.47\,\text{W}.$$

Therefore, the supply-side estimate and the component-sum estimate are consistent.

F Datasheet Excerpts

This appendix contains the datasheet excerpts used in the component selection and verification process. The main report refers to these excerpts when discussing voltage ratings, current ratings, transistor pin configuration, transistor gain, saturation voltage, diode reverse voltage, diode current rating, and laboratory power supply capability.

F.1 BC547B NPN BJT Datasheet Excerpt

BC547B

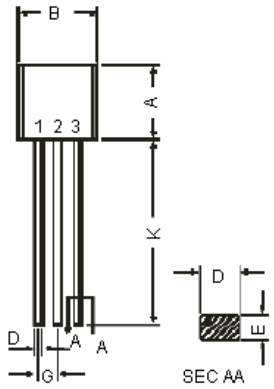
General Purpose Transistor



Features:

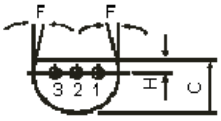
- NPN general purpose transistors, especially suited for use in driver stages of audio amplifiers, low noise input stages of tape recorders, HI-FI amplifiers, signal processing circuits of television receivers.

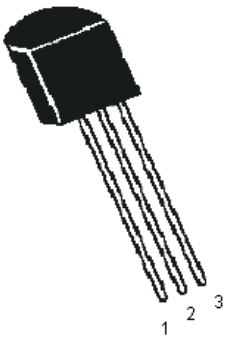
TO-92 Plastic Package



Dimensions	Minimum	Maximum
A	4.32	5.33
B	4.45	5.20
C	3.18	4.19
D	0.41	0.55
E	0.35	0.50
F	5°	
G	1.14	1.40
H		1.53
K	12.70	-

Dimensions : Millimetres





Pin Configuration:
1. Collector
2. Base
3. Emitter

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Figure F.1: BC547B datasheet excerpt used for pin configuration, collector-emitter voltage rating, collector current rating, DC current gain, and saturation-voltage reference.

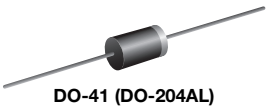
F.2 1N5819 Schottky Diode Datasheet Excerpt

VISHAY
www.vishay.com

1N5817, 1N5818, 1N5819

Vishay General Semiconductor

Schottky Barrier Plastic Rectifier



DO-41 (DO-204AL)

FEATURES

- Guardring for overvoltage protection
- Very small conduction losses
- Extremely fast switching
- Low forward voltage drop
- High frequency operation
- Solder dip 275 °C max. 10 s, per JESD 22-B106
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	1.0 A
V_{RRM}	20 V, 30 V, 40 V
I_{FSM}	25 A
V_F	0.45 V, 0.55 V, 0.60 V
T_J max.	125 °C
Package	DO-41 (DO-204AL)
Circuit configuration	Single

TYPICAL APPLICATIONS

For use in low voltage high frequency inverters, freewheeling, DC/DC converters, and polarity protection applications.

MECHANICAL DATA

Case: DO-41 (DO-204AL)
Molding compound meets UL 94 V-0 flammability rating
Base P/N-E3 - RoHS-compliant, commercial grade

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
E3 suffix meets JESD 201 class 1A whisker test

Polarity: color band denotes the cathode end

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER	SYMBOL	1N5817	1N5818	1N5819	UNIT
Maximum repetitive peak reverse voltage	V_{RRM}	20	30	40	V
Maximum RMS voltage	V_{RMS}	14	21	28	V
Maximum DC blocking voltage	V_{DC}	20	30	40	V
Maximum non-repetitive peak reverse voltage	V_{RSM}	24	36	48	V
Maximum average forward rectified current at 0.375" (9.5 mm) lead length at T_L = 90 °C	$I_{F(AV)}$	1.0			A
Peak forward surge current, 8.3 ms single half sine-wave superimposed on rated load	I_{FSM}	25			A
Voltage rate of change (rated V_R)	dV/dt	10 000			V/µs
Operating junction and storage temperature range	T_J, T_{STG}	-65 to +125			°C

ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)

PARAMETER	TEST CONDITIONS		SYMBOL	1N5817	1N5818	1N5819	UNIT
Maximum instantaneous forward voltage	1.0		V_F (1)	0.450	0.550	0.600	V
Maximum instantaneous forward voltage	3.1		V_F (1)	0.750	0.875	0.900	V
Maximum average reverse current at rated DC blocking voltage		T_A = 25 °C	I_R (1)	1.0			mA
		T_A = 100 °C		10			
Typical junction capacitance	4.0 V, 1.0 MHz		C_J	125	110		pF

Note
(1) Pulse test: 300 µs pulse width, 1 % duty cycle

For technical questions within your region: DiodesAmericas@vishay.com, DiodesAsia@vishay.com, DiodesEurope@vishay.com
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Figure F.2: 1N5819 datasheet excerpt used for reverse-voltage rating, average forward-current rating, and Schottky diode forward-voltage behavior.

F.3 Agilent E3631A Power Supply Datasheet Excerpt

Agilent Technologies

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E3631A 80W Triple Output Power Supply, 6V, 5A & ±25V, 1A

Product Status: Currently Orderable | Currently Supported

Overview

Key Specifications

Output Ratings

- Output 1: 0 to 6 V, 0 to 5 A
- Output 2: 0 to +25 V, 0 to 1 A
- Output 3: 0 to -25 V, 0 to 1 A

Programming Accuracy at 25°C ±5°C

- Voltage: 0.05% + 20 mV, 0.05% + 20 mV, 0.1% + 5 mV
- + Current: 0.15% + 4 mA, 0.15% + 4 mA, 0.2% + 10 mA

Ripple & Noise (20 Hz to 20 MHz)

- Normal Mode Voltage: <350 µVrms/ 2 mV p-p, <350 µV rms/2 mV p-p, <350 µV rms/2 mV p-p
- Normal-mode current: <500 µA rms, <500 µA rms, <2 mA rms
- Common-mode current: <1.5 µA rms, <1.5 µA rms, <1.5 µA rms

Readback Accuracy at 25°C ±5°C

- Voltage: 0.05% + 10 mV, 0.05% + 10 mV, 0.1% + 5 mV
- Current: 0.15% + 4 mA, 0.15% + 4 mA, 0.2% + 10 mA

Description

Agilent basic DC power supplies offer essential features for a tight budget. The triple output, 80 W, GPIB, E3631A provides small, compact size for bench use; low output ripple and noise; built-in measurements and basic programmable features with GPIB and RS232 interface. This clean and reliable supply is designed for general purpose applications; and combines convenient benchtop capabilities with flexible system features in a compact, multi-output package.

Figure F.3: Agilent E3631A datasheet excerpt used to verify the available 0 to $+25$ V, 0 to 1 A output capability.

G Full-Size Schematics

This appendix provides full-size schematic figures for the design versions discussed in the main report. These figures are included here because the schematics may be too detailed to read comfortably in the main text.

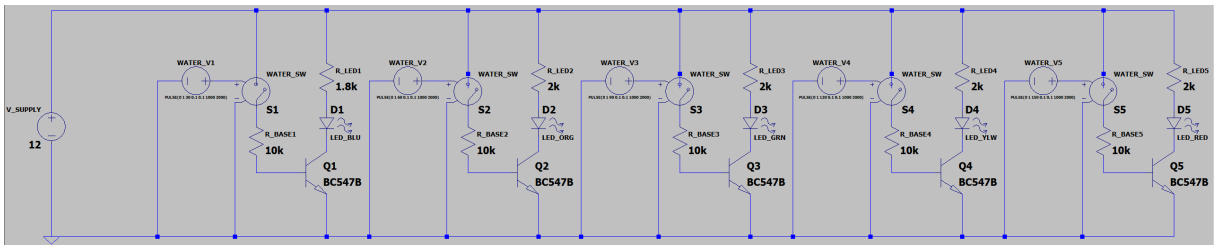


Figure G.1: Full-size Version 1 schematic.

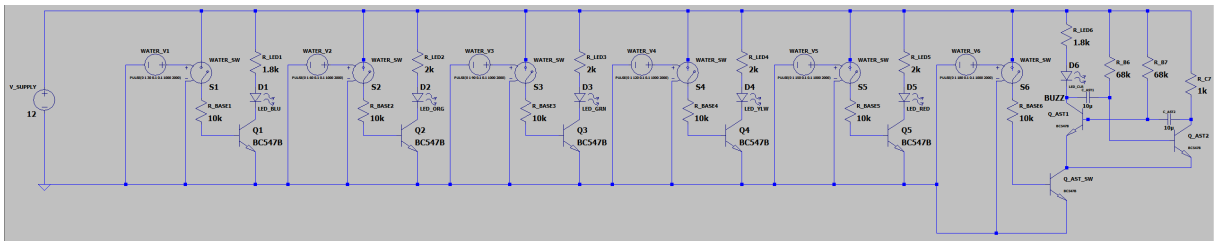


Figure G.2: Full-size Version 2 schematic.

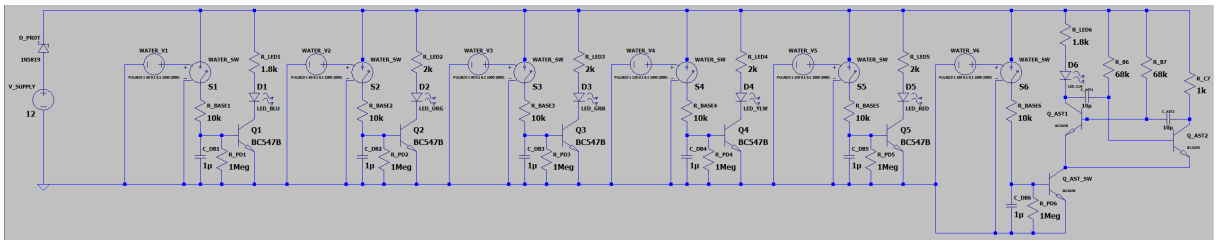


Figure G.3: Full-size implemented Version 3 schematic.

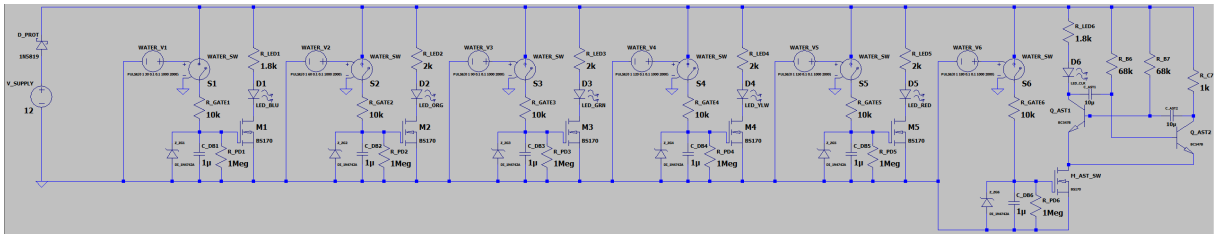


Figure G.4: Full-size proposed MOSFET-based Version 4 schematic.